

I N D E X

FAQs for Investors Relation Conferences & Analyst Meet 2025

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Gas Marketing

1. What is the global LNG demand-supply outlook for next five years? Please highlight key events, along with their likelihood that could potentially alter this outlook. Additionally, kindly provide the outlook on impact of US Tariff on Spot LNG (price & import)?

LNG supply growth was limited in 2024, with just 9.1 MMtpa of new liquefaction capacity reaching commercial start. Demand growth for importers outside Europe outpaced new supply capacity growth in 2024, keeping the global market tight and limiting European LNG imports to 100 MMt, a decline of 23 MMt YOY. In 2025 LNG supply is forecasted to grow by 7% YOY (29 MMt) from a combination of new projects coming online and existing and commissioning projects ramping up. Following the cessation of Russian gas through Ukraine and the forecast strong withdrawals from European storage in Q1, Europe will need this increased LNG to refill storage over the summer. Furthermore, although new capacity is coming online in 2025, much of this is in H2 2025, meaning the global LNG market is expected to stay tight in H1. However, as per IEA's report on "Indian Gas Market report 2030" approximately 202MMT of new capacity will be commissioned between 2024 and 2030 which should ease the global LNG market.

Despite global LNG prices falling from record highs this past winter, security of supply concerns and high spot LNG prices have continued to incentivize strong contracting activities and investment in liquefaction projects, which could increase the risk of oversupplied market conditions emerging at the end of the decade.

The new US tariffs and the Chinese retaliation have put downward pressure on global markets — including gas and LNG markets — based on the assumption that a reduction in global trade will slow global economic growth. Several significant LNG markets in Asia could face sharply negative economic consequences as a result of the new US tariffs.

Pre-FID US LNG projects could face additional challenges as they seek to finalize sales and purchase agreements (SPAs) owing to a combination of higher costs, economic uncertainty, and heightened trade risk for buyer-nations. Meanwhile, SPAs with Chinese buyers are significant for a couple of pre-FID US LNG projects under development, which might create some uncertainty.

China's tariff on US LNG imports had already reduced Chinese imports of US LNG to negligible, with Chinese off takers swapping their cargoes into other markets — so this step is unlikely to have any immediate additional effect on the LNG markets.

2. How GAIL's Natural Gas Trading business manages volatility in the energy prices?

Majority of RLNG sales in GAIL's portfolio occur on a back-to-back basis, hence, energy price volatility is passed on to customers, and GAIL trading profits are unaffected by index price changes. GAIL is also using hedging activity to limit index volatility risk in various sale and purchase transactions including Basis Risk, as well as chasing on opportune sales timings depending on the volatility of various indexes.

To manage upstream contracts, GAIL employs a variety of strategies, including destination swaps, time swaps, foreign sales, and so on. This entails trading crude-linked and HH-linked gas quantities with a variety of counterparties, as well as cargo diversion and optimization to meet

market demands. For example, depending on the transaction dynamics, we shift part of the volumes coming to India towards the foreign market to maintain our margins. We are using innovative strategies including simultaneous FOB-DES transactions and DES-DES transactions to combat unpredictable prices, wherein GAIL floats an enquiry for simultaneous FOB sale and DES buy and DES sale DES buy, respectively.

The ramp-up of flexible quantities from the United States, Australia, and Russia are bolstering the natural gas trading market. Aggregators and traders are able to manage the increase in LNG quantities by purchasing and selling LNG over a variety of contract durations, allowing them to optimize their portfolios. GAIL also conducts business with such counterparties.

3. What are the recent steps taken for Short Term & Long-Term sourcing of LNG and highlights about future plans?

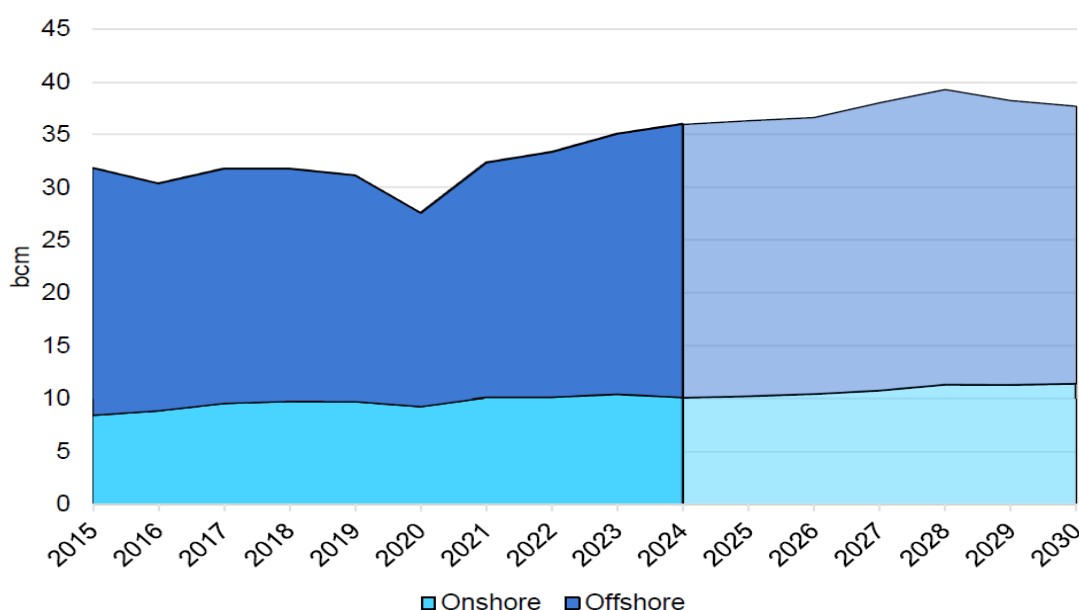
GAIL has recently concluded Long Term deals with Vitol & ADNOC totaling to ~1.50 MMTPA wherein supplies shall commence from 2026. Further, GAIL has also concluded a mid-term transaction for sourcing of ~0.8 MMTPA for 5 years.

Also, GAIL is in discussions with various global LNG producers/traders for future potential LNG tie-ups.

4. What is the Year-wise (for next five years) & the Source-wise GAIL's estimate of change in the gas availability from domestic and overseas imports?

As per IEA's "India's Gas Market Report 2030", India's Domestic Natural Gas production stood at 97.2 MMSCMD (36 bcm) and will range between 94.5 MMSCMD (35 bcm) to 108 MMSCMD (40 bcm). The same may be observed from below graphical representation which illustrates India's Natural Gas production from year 2015 till 2024 and projected Natural Gas production from 2025 till 2030.

Annual natural gas production in India, 2015-2030



Offshore production is set to increase due to additional supplies from ONGC's deepwater KG-D5 project between 2025 and 2030. However, offshore (and overall) gas supply growth will be

tempered by plateauing output from the KG-D6 fields and declining production from legacy assets like ONGC's Mumbai offshore fields.

Change in Gas availability from overseas imports is as per the table below:

Particulars	CY-26	CY-27	CY-28	CY-29	CY-30	Remarks
Ras Gas - 2003	15.8	15.8	12.6	-	-	Assumed ~7.3 MMSCMD towards Make Good and Restoration Quantities accumulated till CY 2024 shall be supplied in CY 2028
Gorgon	1.6	1.7	1.7	1.7	1.6	
RG-2015*	1.0	1.0	0.4	-	-	
US Volume	20.9	20.9	20.9	20.9	20.9	
SMTS Volume	10.3	10.3	10.3	10.3	10.3	
RG-2015 High Sea Sale	-	-	-	-	-	
Mitsui Purchase	1.4	1.4	1.4	1.4	1.4	
Midterm/1	2.8	2.8	2.8	2.8	0.7	Cargo size assumed as 85 MMSCM
Ras Gas 2003 renew			3.2	15.8	15.8	
EMAPPL	-	-	1.8	1.9	1.9	
Vitol	0.5	3.6	3.6	3.6	3.6	
ADNOC	0.8	1.9	1.9	1.9	1.9	
LNG - Total Available (MMSCMD)	55.0	59.4	60.7	60.3	58.0	

5. Is GAIL planning to source more volume of LNG in coming years? What is the kind of formula/slope/index the company is looking to procure fresh volume? How much volume the company is looking to source?

In order to cater to the increasing downstream demand in India and supply natural gas to various sectors, GAIL endeavors to enter into medium- and long-term LNG/RLNG contracts on various indices either on back-to-back basis or based on market and customer requirements.

Further, GAIL is committed to meet increasing downstream demand and accordingly the corresponding sourcing of volumes. Crude linked price is more acceptable in Indian market.

Additionally, based on sudden increase in downstream demand from key sectors like power/fertilizer/steel etc., GAIL can also source LNG cargoes on spot basis to meet such demand. By 2030, GAIL intends to source 7-8 MMTPA of additional LNG volumes out of which approx. 2.25 has already been sourced.

6. Kindly provide outlook on how does GAIL plan to increase its share in Gas Marketing of India?

Robust Pipeline Infrastructure: GAIL's Natural Gas Pipeline Infrastructure connects various gas sources to different gas markets to meet the existing/ future natural gas demand of various Powers, Fertilizer, CGD and other industries in the Country. GAIL has ~16,421 kms operational Natural Gas pipeline in India (as on 31st December 2024) and is in the process of laying another ~3193 kms pipeline across the country which would increase the accessibility and reach of natural gas to potential untapped markets, thereby enhancing the market share.

LNG terminal: GAIL intends to increase the capacity of KLL's Dabhol LNG terminal from 5 MMTPA at present to 6.3 MMTPA and subsequently to 15 MMTPA. Further, Breakwater project at KL Dabhol has already been completed which has enabled KLL to operate throughout the year as an all-weather terminal. This will enable GAIL to bring additional volume to LNG terminal as per evolving market dynamics and to respond in flexible manner.

Fertilizer Sector: GAIL is actively pursuing customers for extension of their Long-term Contracts with GAIL beyond April'2028. Further, GAIL has also taken up this matter with Department of Fertilizers (DoF) for facilitating the same.

Power Sector: GAIL has been consistently supplying Gas to Power Sector customers especially during the crunch period viz through NVVN Scheme and Section-11 of the Electricity Act, 2003 as and when implemented by CEA. Further, to increase the share of Gas in power plants, GAIL has been actively participating in tenders and bilateral discussions by giving competitive price offers.

Steel Sector: GAIL has been pursuing Steel sector customers for replacing coking coal with natural gas as a cleaner and efficient alternative. With the laying of NG pipelines such as JHBDPL, MNJPL, etc., Eastern Part of India which has the predominant presence of Steel sector has access to the Natural Gas. Further, GAIL is actively taking up with customers for long term tie-ups by provide competitive offers.

CGD: Out of 307 GAs in India, about 72 GAs are being operated by GAIL, its Subsidiaries and Joint Ventures. By constantly expanding the above CGD Network, GAIL is able to increase its presence in the PNG & CNG segments across the country. CGDs are expected to grow at 10% CAGR and presents a good potential for future growth. GAIL is actively offering competitive short/long term offers to these CGD customers to increase its market share.

7. Projected year-wise, sector wise volume details in respect of Gas transmission for next 5 years for GAIL? Please segregate the data in GAIL's own volumes and shippers' volumes also.

NATURAL GAS TRANSMISSION:

GAIL's sector wise transmission volumes for FY23-24, FY24-25 and projected transmission volumes till FY29-30 is as per the table below:

GAIL's Transmission volume (MMSCMD) – Sector wise							
Sector	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30
CGD	25.16	32.39	38.32	42.16	46.78	52.29	58.63
Fertilizer Industry	51.89	50.28	49.48	50.08	50.68	51.28	51.88
Others	18.31	16.70	17.14	19.34	21.91	24.95	28.48
Refinery	8.46	11.60	14.08	11.76	12.29	12.84	13.50
Power Industry	9.61	9.77	11.24	16.57	18.28	20.11	21.80
IC	7.03	7.23	7.93	8.40	8.90	8.90	8.90
Grand Total	120.46	127.32	138.19	148.31	158.83	170.37	183.18

GAIL's Natural Gas Transmission volume Projection (MMSCMD) - Pipeline wise					
Pipeline wise Gas Trans (MMSCMD)	2025-2026	2026-2027	2027-2028	2028-2029	2029-2030
GAIL's Integrated Natural Gas Pipeline (Z-1)	33.37	35.71	37.69	39.83	43.06
GAIL's Integrated Natural Gas Pipeline (Z-2)	9.03	9.66	10.20	10.78	11.65

GAIL's Natural Gas Transmission volume Projection (MMSCMD) - Pipeline wise					
Pipeline wise Gas Trans (MMSCMD)	2025-2026	2026-2027	2027-2028	2028-2029	2029-2030
GAIL's Integrated Natural Gas Pipeline (Z-3)	21.35	22.84	24.11	25.48	27.55
GAIL's Integrated Natural Gas Pipeline (Z-4)	40.68	43.53	45.95	48.55	52.49
GAIL's Integrated Natural Gas Pipeline (Z-5)	16.22	17.36	18.32	19.36	20.93
GAIL's Integrated Natural Gas Pipeline (Z-6)	2.35	2.52	2.66	2.81	3.04
GAIL's Integrated Natural Gas Pipeline (Z-7)	1.60	1.71	1.80	1.90	2.06
GAIL's Integrated Natural Gas Pipeline (Z-8)	0.03	0.03	0.03	0.03	0.04
GAIL's Integrated Natural Gas Pipeline (Z-9)	0.08	0.08	0.09	0.09	0.10
GAIL's Integrated Natural Gas Pipeline (Z-10)	0.21	0.23	0.24	0.25	0.28
Agartala Regional Network	0.69	0.69	0.69	0.69	0.69
K.G. Basin: Kakinada-Tatipaka-Kovvur	4.24	4.54	4.79	5.06	5.47
Kochi - Koottanad - Mangalore - Bangalore: Zone 1	2.60	2.78	2.94	3.10	3.35
Kochi - Koottanad - Mangalore - Bangalore: Zone 2	1.59	1.70	1.79	1.89	2.05
Cauvery: Narimanam and Kuthalam	0.74	0.74	0.74	0.74	0.74
Cauvery: Ramnad	0.92	0.92	0.92	0.92	0.92
North Gujarat: Kadi-Kalol	0.02	0.02	0.02	0.02	0.02
North Gujarat: Kalol - Ramol	0.02	0.02	0.02	0.02	0.02
North Gujarat:Shobhasan	0.02	0.02	0.02	0.02	0.02
North Gujarat: Paliyad	0.02	0.02	0.02	0.02	0.02
South Gujarat: Motwan	0.01	0.01	0.01	0.01	0.01
South Gujarat: GGCL and UPL	1.00	1.00	1.00	1.00	1.00
Rajasthan: RRVUNL,Ramgarh - ONGC/OIL Pipeline	0.53	0.53	0.53	0.53	0.53
Rajasthan: RRVUNL,Ramgarh - Focus Pipeline	0.07	0.07	0.07	0.07	0.07
Other Dedicated	0.28	0.28	0.28	0.28	0.28
MNJPL	0.35	1.07	1.73	4.32	4.20
SAPL	0.15	0.22	2.16	2.57	2.58
TOTAL	138.19	148.31	158.83	170.37	183.18

GAIL's sector wise transmission volumes for FY23-24, FY24-25 and projected transmission volumes for FY25-26 is as per the table below:

GAIL's Transmission volume (MMSCMD) – Sector wise			
Sector	2023-24	2024-25	2025-26
CGD	25.16	32.39	38.32
Fertilizer Industry	51.89	50.28	49.48
Others	18.31	16.70	17.14
Refinery	8.46	11.60	14.08
Power Industry	9.61	9.77	11.24
IC	7.03	7.23	7.93
Grand Total	120.46	127.32	138.19

NATURAL GAS MARKETING:

GAIL's Gas Marketing Volume Projection (Sector wise)						
Domestic Gas						Remarks/Assumptions
Sector	FY-26	FY-27	FY-28	FY-29	FY-30	
CGD	19.0	19.4	19.9	20.5	21.1	Assumed - Domestic gas availability at current level & Incremental Volume from CBG assumed as CGD consumption
Fertilizer Industry	6.0	6.0	6.0	6.0	6.0	
Others	2.6	2.6	2.6	2.6	2.6	
Power Industry	6.5	6.5	6.5	6.5	6.5	
IC	1.4	1.4	1.4	1.4	1.4	
Only Marketing	4.2	4.2	4.2	4.2	4.2	

GAIL's Gas Marketing Volume Projection (Sector wise)						
Domestic Gas						
Sector	FY-26	FY-27	FY-28	FY-29	FY-30	Remarks/Assumptions
Sub-Total (A)	39.7	40.1	40.6	41.2	41.8	
RLNG						
Sector	FY-26	FY-27	FY-28	FY-29	FY-30	
CGD	11.7	14.0	16.8	20.2	24.2	20% CAGR assumed from current level
Fertilizer Industry	29.7	30.3	30.9	31.5	32.1	2% CAGR - renewals / EPMC / HPHT to RLNG etc
Others	7.6	9.1	10.9	13.1	15.7	20 % CAGR assumed considering new customers along JHBDPL, MNJPL, KKMBPL, SAPL
Power Industry	3.5	3.9	4.3	4.7	5.2	10% CAGR assumed
IC	6.5	7.0	7.5	7.5	7.5	Increased consumption considering GMPL & Usar Plant commissioning
International Sale	9.3	9.3	9.3	9.3	9.3	
Sub-Total (B)	68.3	73.6	79.7	86.3	94.0	
Total (A+B)	108.0	113.7	120.3	127.5	135.8	

8. Detail of Various Gas LNG Contracts for next 3 years w.r.t.

a. Expected Year-wise volumes – Supply Contract wise and date of start of Gas supply.

Reply (a)

Supplier / Project	Country of Supply	Supplies from	LNG Quantity (MMTPA)
Qatar Energy (earlier RasGas)	Qatar	Since 2004 (Contract through PLL)	4.5 (GAIL's share out of 7.5 MMTPA)
Qatar Energy (earlier RasGas)	Qatar	Jan 2016	0.3 (GAIL's share out of 1 MMTPA)
Gorgon-Exxon	Portfolio	Q1'2017 (Contract through PLL)	0.43 (GAIL's share out of 1.44 MMTPA)
Gorgon – EMAPPL	Portfolio	2028	0.36 (GAIL's share out of 1.2 MMTPA)
Sabine Pass	USA	March 2018	3.5 MMTPA
Cove Point	USA	April 2018	2.3 MMTPA
Gazprom	Portfolio	June, 2018	2.85 MMTPA w.e.f. 2023
Vitol	Portfolio	2026	1 MMTPA
ADNOC	Abu Dhabi	2026	~ 0.5 MMTPA
Qatar Energy Trading	Portfolio	2025	~0.75 MMTPA

b. Year wise back-to-back tied-up volume out of Point a. Year wise tied up volumes with fixed price or formula but not on the procurement price out of point a

Reply for (b)

Particulars	CY-25	CY-26	CY-27
LT-RLNG Sale @ PLL Price (back-to-back)	22.0	21.9	21.8
Portfolio RLNG Sale @ PLL Price to Fertilizers	15.3	15.3	15.2
Portfolio RLNG Sale @ HH Indexed Formula	16.5	17.3	14.9
Portfolio RLNG Sale @ JCC Indexed Formula	2.1	1.8	0.3
Portfolio RLNG Sale @ Hybrid Formula	1.2	0.8	0.6
PF-RLNG @ Others / Fixed Price	0.6	1.1	2.9
Internal Consumption at UPPC, Pata	4.6	4.6	4.6
Pipeline IC	1.5	1.5	1.5
Total (CV@9450)	63.8	64.4	61.8

Note:

- Internal Consumption requirement for UPPC, Pata: ~ 4.5 - 5.0 MMSCMD.
- Internal Consumption requirement for pipeline SUG: ~1.5 MMSCMD (~0.62 MMSCMD HPHT volume tied-up till May'26 & ~0.2 MMSCMD till 2028)
The ToP for long term contracts is in the range of 80 - 90% and for CGD sector customers 60 -80%. Accordingly, drawl by customers are largely upto Top level.

c. Expected Year-wise Prices with assumptions**Reply for b.**

Expected Year Wise HH Price

Particulars	UoM	FY 2025-26	FY 2026-27	FY 2028-29
HH	\$/MBG	3.79	3.97	3.70

9. Details of source-wise cargos in FY24-25.

Details for FY 24-25		
S. No.	Source Country	Volume (MMTPA)
1	Angola	0.92
2	Cameroon	0.55
3	Equitorial Guinea	0.40
4	Indonesia	0.06
5	Mozambique	0.13
6	Nigeria	0.32
7	Oman	0.59
8	Qatar	1.69
9	Russia	0.07
10	UAE	1.57
11	USA	2.88
	Grand Total	9.18

10. Status of Gas swapping agreement (Destination swap, Time swap, Own ship, charter cargo etc.) for US Gas including the contracts with SHELL, Gunvor etc. for CY 2025, 2026 & 2027.

- For CY 2025, GAIL has entered into 01 Destination Swap Contract for 12 cargoes, 04 FOB-DES Contracts for 29 cargoes and 04 DES-DES contracts for 07 cargoes
- For CY 2026, GAIL has entered into 01 Destination Swap Contracts for 24 cargoes and 01 FOB-DES Contract for 12 Cargoes

CY	Destination Swap	Time Swaps	FOB-DES/DES-DES Swap
2025	12	6 cargoes (GAIL Sells)	36

2026	24	6 cargoes (GAIL Buys)	12
2027	-	6 cargoes (GAIL Buys)	-

11. What is the possibility of having any special scheme for supplying gas to stalled power plants? What are the incentives required to run such schemes? Will the Government allow/any strategy for more gas-based Power Plants in coming time instead of Imported coal?

- GAIL remains continuously engaged with Ministry of Power (MoP) and various power generators in order to meet their gas requirements.
- In 2015-17, the Ministry of Power developed a power pooling arrangement to help run the country's stranded gas-based power plants
- In 2023, Ministry of Power has developed a scheme to run NTPC & other gas-based power plants during peak summer period of April and May 2023.
- GAIL also signed a Long-term Gas sale agreement with NTPC for supply of 500 MMSCM RLNG for a period of 4.5 years starting Jan'2024.
- Further, in 2024 to meet the power demand in the crunch period, MoP introduced NVVN scheme to procure power from gas-based power plants. Based on the schemes mentioned above, GAIL has managed to sign contracts with few gas-based power plants participating in NVVN scheme.
- Subsequently, to ensure maximum power generation from Gas-based power plants during peak summer demand, Section 11 of the Electricity Act, 2003 was implemented by CEA during May-June 2024. Under the above direction, GAIL consistently supplied gas to power sector customers.
- As per the reports, it is understood that in 2025 likely peak power demand during the summer crunch period will be ~ 270 GW surpassing previous year demand of ~ 250 GW. To meet the peak power demand, GAIL envisages that Gas based power plants may be required to run at full capacity during summer crunch period.
- As was done in the past, GAIL is ready to meet any additional gas requirement (over and above the contracted volume) through sourcing of spot LNG cargoes.

12. What is the likely sustainable margin (\$/MMBTU) & Tentative year wise Spread in Gas Marketing segment for next 3 years?

Out of the total RLNG sales, ~ 60-70 % of volume is being sold on a back-to-back index to various customers, including fertilizer, power, CGD, and other industrial customers, and a standard marketing margin (MM) is applicable. GAIL shall earn trading margin in relation to other RLNG volumes which will be determined by then prevailing market conditions and volume available for catering such sales.

13. What steps have been taken for chartering ships for LNG transportation?

GAIL currently has a fleet of 05 LNG Vessels, which in the year FY25 has transported 47% of the US LNG Volume. Balance volumes are via 3rd party sales and optimization contracts. GAIL has also awarded another contract for long term charter hiring of LNG Vessel.

GAIL Bhuwan is on charter hire up to the period 2026. **Grace Emilia** is on charter hire upto a period between 2027-2028. **GAIL Urja** is on charter hire upto a period between 2030-2032. **GAIL Sagar** which came on charter in Feb 2025 is on GAIL's Charter till Year 2038. Further, GAIL has **Maran Gas Pericles** on Charter from Oct 24 till October 26.

Further, from time to time depending on market and trading needs, GAIL charters vessels on spot and short-term basis. **In FY 25, GAIL chartered 05 Vessels on Spot Charter** and continues to look for opportunities.

Gradually, GAIL has built a sizable fleet of LNG Vessels with state-of-the-art technology and design.

14. What is GAIL's Strategy on sale of US LNG Volumes? How much of it is placed under Long Term formula to customers in India?

In India, GAIL has already tied up some of the US LNG volumes with Fertilizer units, power sector, CGDs and other sector customers. GAIL also has earmarked some of US LNG for internal consumption at the Pata Petrochemical plant in UP. Further, GAIL has also sold part volume in the International market. Overall, about 80% of the US LNG volume has been tied up with domestic customers/ internal consumption and balance is sold in the International market.

15. Status and physical progress of all the fertilizer plants, CGD entities and Refineries which are amongst the anchor customers along JHBDPL. What is their present Gas offtake?

- A. Fertilizer:** Major Fertilizer plants along the JHBDPL, i.e., HURL-Gorakhpur; HURL-Barauni, HURL-Sindri and Matix-Durgapur are offtaking ~8 MMSCMD gas.
- B. Refineries** along JHBDPL, i.e., IOCL-Paradip and IOCL-Barauni are off-taking ~2 MMSCMD gas.
- C. CGDs** - 51 GAs were awarded along JHBDPL, out of which 38 are connected and 13 are in the process of getting connected.

16. Kindly provide unit economics in terms of affordability of delivered Gas price (in \$/MMBTU) and cost of Gas in per unit generation for major sectors including Fertilizers, Gas based power plants, CNG, PNG, Steel sector, Aluminum sector etc.

Sector	Affordability Range (delivered at plant)
Power	9-10 \$/MMBtu
Steel	6-8 \$/MMBtu
CGD	10-12 \$/MMBtu
Petrochemicals	7-8 \$/MMBtu
Fertilizer (for urea plants)	Gas cost is passed through subsidy
Other sectors (Ceramics, glass etc)	10-13 \$/MMBtu

17. Expected GAIL's LNG Volumes at PLL, Dabhol, Shell, Kochi, Dhamra & others in next 5 years.

GAIL has directly imported 141 cargoes during FY 2024-25 and has utilized its full capacities in Dahej, Dabhol & Dhamra terminals. GAIL is confident to utilize full capacities during upcoming years.

GAIL has booked 2.5 MMTPA regas capacities at Dahej and the same is being utilized fully. Dabhol terminal is not operating at full capacities due to non-availability of independent heating system. Further, considering completion of Breakwater, GAIL is in process to schedule cargoes at Dabhol for full year for CY 2025.

Further, GAIL has booked capacity of ~ 1.5 MMTPA at Dhamra regas terminal which was commissioned in May 2023 for commercial operations. It is being utilized as per its contractual obligations.

18. What is Sector-wise Gas Supply by GAIL for FY 2023-24, 24-25? Kindly brief about sector wise consumption in country for aforementioned years?

FY2023-24:

Sector	GAIL (published figures)			India			Difference (Total)
	Domestic	RLNG	Total	Domestic	RLNG	Total	
Power	6.13	3.18	10.58	17.77	7.04	24.81	14.23
Fertilizers	7.41	33.15	39.28	8.28	49.23	57.5	18.22
CGD	20.16	5.32	25.48	27.43	9.43	36.86	11.38
Others incl. IC	7.18	8.42	15.6	44.89	23.8	68.69	53.09
Total	40.88	50.07	90.94	98.37	89.5	187.87	96.93

FY2024-25:

Sector	GAIL			India			Difference (Total)
	Domestic	RLNG	Total	Domestic	RLNG	Total	
Power	7.27	3.57	10.84	16.76	7.56	24.32	13.48
Fertilizers	6.21	30.94	37.15	7.88	48.23	56.11	18.96
CGD	18.97	8.17	27.14	27.68	13.72	41.40	14.26
Others incl. IC	7.12	10.13	17.25	44.18	28.91	73.09	55.84
Total	39.57	52.81	92.38	96.50	98.42	194.92	102.54

Source of India's consumption: PPAC Sectoral Consumption Data FY2024-25

19. Outlook on LNG re-gasification plants in India in terms of future utilization and margins, given significant new re-gasification capacity is coming online in next few years. Will there be open access for capacity booking at regasification terminals in future? Should Regasification charges be brought under regulatory regime?

- India currently has ~52.5 MMTPA Regas Capacity. Further, India's regas capacity is expected to be increased to ~68 MMTPA in next few years. However, if the gas demand in India doesn't materialize, the utilization of the upcoming regasification terminals may be further limited.

- In respect of open access at regasification terminals, some LNG Registration Rules notified by the Central Government in October 2012 have underscored that in order to protect consumer interests, liquefied natural gas terminals must be subject to “commercial transparency” and committed to provide additional capacity to any consumer at “publicized tariffs”.
- Recently, PNGRB has framed draft regulation in this regard and a decision on the same is awaited.

20. Role of GGSPL in GAIL’s overseas volume trading.

GAIL Global Singapore Pte. Ltd. (GGSPL) is a wholly owned subsidiary of GAIL (India) Limited, established in 2004. It was originally set up as an investment arm (investment in Egypt – NATGAS) and commencing 2012 has evolved into a fully functional trading company, focusing on the international liquefied natural gas (LNG).

The company’s strategic location in Singapore allows it to participate effectively in spot and short-term LNG trades, wherein the company procures spot LNG Cargos for its parent company (GAIL) and participates independently in LNG tender purchase enquires of many natural gas players in India. GGSPL also independently participates in GAIL DES- FOB LNG tenders wherein GAIL optimizes its LNG volumes from USA.

GGSPL has in place, Master Sales Purchase Agreements with more than 70 counterparties which includes producers, buyers, traders, and portfolio players of LNG. This enables a wider participation of LNG tenders floated by GAIL or GGSPL. GGSPL had marketed a portion of GAIL HH (0.5MMTPA) in the international LNG market and said supplies commenced in 2018 through Feb 2023.

Over the years, GGSPL has expanded its reach and strengthened its trading capabilities and regularly participates in third party LNG trades. During the financial year 2024-25, GGSPL has traded in total 27 LNG Cargos- out of which 15 Cargos have been procured for GAIL and remaining 12 Cargos were traded independently by GGSPL. For the year 2024-25, GGSPL has contributed about 1.74 MMSCMD sale of Natural Gas for GAIL MOU targets.

21. Brief details of Regasification contract with PLL & Adani Ports for LNG regasification? Renewal Status of Qatar Gas contract of 4.8 MMTPA.

Dahej currently has 17.5 MMTPA of Regasification Capacity. GAIL has booked regasification capacity of 2.5 MMTPA in PLL’s Dahej terminal vide Regasification Agreement dtd 06.09.2012. The regas tariff for CY 2025 is Rs. 62.06 and the same is being escalated @ 5% on calendar year basis.

In addition, GAIL has also sourced 4.5 MMTPA of RLNG from PLL on long term basis, which PLL has sourced from its upstream supplier Qatar Energy.

22. Natural Gas demand is growing rapidly. What are the major avenues of demand growth? Kindly provide a sector wise outlook.

As per PNGRB’s report titled “India's Natural Gas demand projection for 2030-2040”, the historical sectorial trend from FY2015-16 till FY2023-24 is as below:

Sector	FY 15-16	FY 16-17	FY 17-18	FY 18-19	FY 19-20	FY 20-21	FY 21-22	FY 22-23	FY 23-24	CAGR
Power	29.8	31.8	33.0	32.9	30.3	29.7	27.8	22.3	24.8	-2.2%
CGD	14.9	20.1	23.5	25.2	29.7	25.3	33.4	33.0	36.9	12.0%
Refinery	13.9	14.7	17.9	19.3	21.3	21.7	14.6	10.7	16.0	1.8%
I/C for P/L System	1.1	1.3	1.4	1.5	1.4	1.2	1.3	4.8	5.3	
Agriculture (Tea Plantation)	0.0	0.0	0.5	0.5	0.6	0.5	0.4	0.4	0.4	-4.6%
Industrial	1.1	1.9	2.4	2.6	1.6	1.2	2.3	1.9	2.9	12.7%
Manufacturing	0.4	0.3	0.3	0.4	0.4	0.3	0.3	0.5	1.1	14.0%
Other/Misc	11.7	10.8	8.8	9.3	1.5	12.5	25.0	23.4	29.9	12.4%
Non-Energy Sectors										
Fertilizer	44.1	42.3	40.2	41.1	44.0	48.7	49.5	53.2	57.5	3.4%
Petrochemical	10.2	11.4	11.0	9.3	9.8	8.4	7.9	5.4	7.3	-4.1%
LPG Shrinkage	2.1	2.1	2.2	2.4	2.3	2.5	2.9	2.7	2.7	3.6%
Sponge Iron/Steel	1.5	2.4	3.5	3.1	1.6	1.8	3.1	2.6	3.2	10.1%
Total	130.8	139.1	144.8	147.5	144.3	153.8	168.5	160.8	187.9	4.6%

The sector wise demand outlook for year 2030 and 2040 is outlined as under:

			2030		2040	
Sector	Sector Consumption 2024	Sector Consumption 2025	GtG	GtB	GtG	GtB
CGD	36.9	41.4	87.1	126.1	216.4	270.8
Power	25.2	24.3	35.7	40	43.5	52.8
Refinery	22	15.9	43.4	50.9	52.4	57.8
Fertilizer	58	56.1	65.3	69.3	72.9	80.5
Steel	3.2	3.7	4.3	5.1	6.4	9.3
LNG Transport	0	0	3.9	6.6	26.3	65.7
Others (Tea Plantation, Industries, LPG Shrinkage)	42	53.6	57.3	66.6	76.9	93.3
Total	187	194.9	297	365	495	630

*GtG stands for Good to Go scenario which is based on moderate basis.

*GtB stands for Good to Best Scenario which is based on accelerated progress.

Power Sector

- Natural gas for balancing for renewable energy & peak load shaving in power sector- *Peak shaving involves briefly reducing power consumption to prevent spikes. This is achieved by either scaling down production or sourcing additional electricity from local power sources.*
- Power Pooling & encouraging emission-based merit order dispatch (M.o.D.) in the power sector- *Power pooling is used to balance electrical load over a larger network (electrical grid) than a single utility.*
- Natural gas for Trigenation- *also known as Combined Cooling, Heat and Power (CCHP), is a system that simultaneously produces electricity, heat, and cooling from a single source.*

Steel Sector

- Natural gas for pellet plants & blast furnace plants in iron & steel sector- *In pellet plants, natural gas can be used as a fuel source for heating and induration processes, helping to create the iron ore pellets. It can also be used as a reducing gas in direct reduction processes, producing sponge iron, a steelmaking precursor.*
- Substitution of coal in existing Blast Furnace plants- *Globally, injection of Pulverized Coal (PCI), Natural Gas & Coke Oven Gas etc. is used to optimize coking coal consumption in BF.*
- Addition of new gas based DRI plants either through brownfield or greenfield expansion with increase in demand of Steel

Fertilizer Sector

- Existing Fertilizer plants planning to expand their capacity shall require additional gas for increased production.

Others

- LNG for transportation purposes.
- Natural gas as sustainable energy source in small scale industries.
- Mandating use of natural gas for upcoming industrial corridors.

23. Kindly provide a point of view regarding changing pricing dynamics due to geopolitical scenarios as traditionally the growth of Natural Gas demand has been volatile.

- The pricing of domestic natural gas in India is regulated by the government through a formula-based approach.
- The pricing mechanism considers factors such as international benchmarks, production costs, and market dynamics.
- The government periodically revises the price based on these factors to ensure a fair and competitive pricing structure.
- The pricing of domestic natural gas is crucial in balancing the interests of both producers and consumers, promoting investment in the sector, and ensuring the affordability and sustainability of natural gas as an energy source in India.

Thus, the natural gas market is quite price sensitive. Therefore, understanding this is essential for policymakers and industry stakeholders to make informed decisions and develop effective pricing strategies. There are several factors to be considered as-

- The price of domestic natural gas relative to alternative fuels, such as coal, diesel, and liquefied petroleum gas (LPG), significantly impacts its price sensitivity
- Higher infrastructure costs may affect the price sensitivity and competitiveness of natural gas
- Global market dynamics, including LNG prices, supply-demand imbalances, and geopolitical factors, can impact the price sensitivity of imported natural gas in the Indian market.
- Government policies, subsidies, and incentives play a crucial role in mitigating the price sensitivity of natural gas.

- The price elasticity of demand measures the responsiveness of quantity demanded to changes in price. Natural gas demand elasticity in different sectors, such as industrial, power generation, and residential, influences the price sensitivity.
- Elastic demand implies a higher price sensitivity, indicating that even small price changes can have a significant impact on demand.
- Consumer behavior and preferences also affect the price sensitivity of domestic natural gas.

24. Kindly provide an outlook on bringing Natural gas under GST regime. What are the expected benefits? What are the major hurdles in bringing Natural gas under GST? What is the potential impact on GAIL?

View on GST on Natural Gas

GST on five petroleum products viz petroleum crude, High speed Diesel (HSD), motor spirit (commonly known as petrol), natural gas and aviation turbine fuel supply of petroleum crude, high speed diesel, motor spirit (commonly known as petrol), natural gas and aviation turbine fuel, shall be levied with effect from such date as may be notified by the Government on the recommendations of the Council and therefore legacy taxes continue to be applicable on these products.

Expected benefit to gas sector

Natural Gas is primarily an industrial input used as fuel/feedstock by industries like power, fertilizers, petrochemical, LPG and City Gas Distribution (CGD). Natural Gas is an environment – friendly fuel unlike other fuels viz. Crude, Coal and other alternate fuels which pollute the environment due to higher carbon emission. With the increasing level of pollution, thrust of the government is to increase the use of greener fuel. All other polluting fuels are being kept under the ambit of the GST and the environment friendly fuel is kept outside the GST. Bringing Gas under GST will have a significant role in limiting CO₂ emissions, global warming and climate change.

Currently, VAT rates on Natural Gas vary from Nil to 20% across different States, with some States levying an additional cess on taxable turnover. Such differential tax treatment and variation in rates act as a barrier in free and smooth trade of Natural Gas across the country and further aggravate the competitiveness of industries using Natural Gas as an input. Such distortion in tax treatment having effect on price for consumers situated in different states can be removed only by bringing Natural Gas under GST.

Inclusion of Natural Gas under GST will also promote the Natural Gas based industries as seamless credit of GST will be available. Natural gas is one of the major industrial inputs used as fuels in industries and non-inclusion of Natural Gas under GST lead to absorption of VAT paid on procurement of Natural Gas in their cost of production which is rendering large, medium and small industries using Natural Gas economically unviable and uncompetitive.

Key hurdles in inclusion of Natural Gas under GST

1. Petroleum products contribute a significant amount of revenue to the states.
2. Levying VAT on these products, the state governments have control over their revenues. Bringing petroleum under GST will take away that control as the Centre controls fixing the rate of GST.
3. Central Govt. is open to the possibility, but a majority of the states are against the inclusion of petroleum products under the GST.
4. Many states are still against the inclusion of petroleum products under GST.
5. Under GST, even under the 28% tax slab, states will earn a lot less compared to what they currently earn from petrol and diesel sales as value-added tax (VAT).

Impact on GAIL

1. Inclusion of GST will have positive impact on overall gas industry including GAIL.
2. There will be significant reduction in the landed cost of Natural Gas to the customer which will increase the overall demand of Natural Gas. This will have favorable impact on the GAIL's top line.
3. Substantial reduction in cost and time of tax compliances and litigations.

Changes during FY 2024-25 in taxation (VAT / CST etc.) having impact on Natural Gas prices

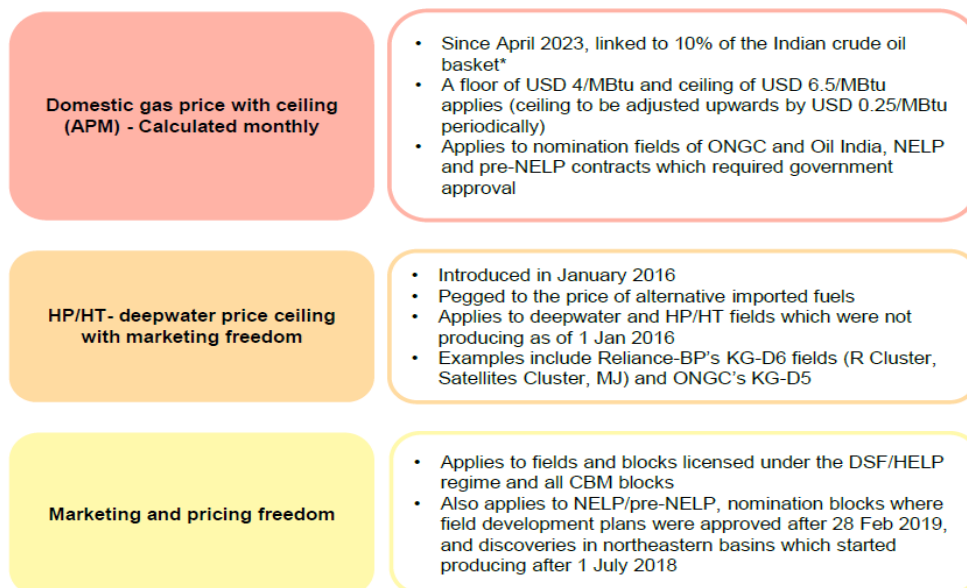
In order to promote the use of Natural Gas, some of the states have reduced the VAT rate on Natural Gas, PNG and CNG during the FY 2024-25. Details of such reductions are as under:

State	Product	Old Rate	New Rate
Andhra Pradesh	NG	24.50%	5.00%
	CNG	24.50%	5.00%
	PNG	5.00%	5.00%
Assam	NG	14.50%	14.50%
	CNG	14.50%	5.00%
	PNG	14.50%	14.50%
Bihar	NG	20.00%	20.00%
	CNG	20.00%	12.50%
	PNG	20.00%	12.50% (Domestic & Commercial PNG) & 5% (Industrial PNG)
Chhattisgarh	NG	25.00%	14.50%
	CNG	25.00%	14.50%
	PNG	25.00%	14.50%

25. How does Natural gas pricing dynamics work in India? Govt. of India has set a target to increase share of Natural gas from 6% to 15%. What do you think is the realistic percentage that may actually be achieved?

The 15% Natural gas share target is a policy and marketing driven goal, dependent in infrastructure and sectoral adoption. Pricing dynamics, which involve regulated and market-linked mechanisms, are only one part of the equation.

A brief on the domestic pricing mechanism may be referred from below: -



Source: IEA's India Gas Market report 2030

In addition to aforementioned type of domestic Gases, Natural Gas Corporation (ONGC) is offsetting the natural decline in output from ageing gas fields by drilling new wells. This Gas has been termed as New Wells Gas and is sold at 10% of Indian Crude Basket + 20% premium i.e. 12% of the Indian crude Basket (source: The Economic Times)

26. Does GAIL serve a large chunk of its customers through spot contracts? Further, what percentage of long-term Gas portfolio is tied up on back-to-back basis?

GAIL sells almost ~95% of volumes through term contracts. GAIL doesn't serve large chunk of its customers through spot contracts.

27. Given that domestic Natural Gas production is not increasing, can it be assumed that the growth in demand will be met with Imports?

Yes. Negotiation with various suppliers to source with favorable terms & conditions can help meet the growth in demand through imports.

India's domestic gas production, which met 50% of demand in 2023, is expected to grow only moderately through 2030. After nearly a decade of decline and stagnation, India's domestic gas production has seen a resurgence. In 2023, total net gas production reached 35 bcm/94.5MMSCMD, meeting about half of the country's gas demand. This growth is primarily driven by the deepwater fields in the Krishna- Godavari basin, which now account for nearly 25% of India's total production.

Between 2024 and 2030, only moderate growth is expected, supported by increasing onshore production from coal bed methane (CBM) and discovered small fields (DSF). Offshore production will also rise with additional supplies from ONGC's deepwater KG-D5 project. However, overall growth will be tempered by plateauing output from the KG-D6 fields and declining production from legacy assets like ONGC's Mumbai offshore fields, leaving production in 2030 (at just under 38 bcm/ 102.6 MMSCMD) only around 8% higher than 2023 levels. Source: IEA India Gas Market Report 2030.

Further, As per PNGRB's India Natural Gas Demand projection report published in April 2025, considering moderate growth (primarily as is scenario) , Indias NG demand by 2030 is projected

to be 297 MMSCMD. Therefore, in 2030, there is a visible gap of ~ 200 MMSCMD between demand and supply of natural gas which will have to be met with imports.

28. Can LNG trucking be expected to add a substantial increase in Natural Gas demand of India?

As per PNGRB's report titled " India's Natural Gas demand projection for 2030-2040", considering moderate growth (primarily as is scenario) , India's natural gas demand for LNG trucking is expected to be 3.9 MMSCMD by year 2030 and 26.3 MMSCMD by year 2040.

29. Kindly provide the status of breakwater construction at Dabhol Terminal. By when Dabhol terminal is expected run at full capacity of 5 MMTPA? Kindly brief about the capacity expansion plans of Dabhol terminal, expected timelines and estimated capex for the same?

Status:

Breakwater construction is 100% complete. 5 Mmpta utilization shall be possible after the commissioning of ambient heating system.

Expansion would be done in 2 phases. Phase-1: 5-6.3 MMPTA and Phase-2: 6.3 to 15 MMPTA. Timeline for Phase-1 is 42 months and for Phase-2 is 48 months from the start date.

30. India's country wise LNG Import Data

In year 2023, India imported 22.1 MMT of LNG. Country wise details are as per below table:

Country	LNG Imported (MMT)	% Share
Qatar	10.90	49%
United Arab Emirates	3.04	14%
United States	3.18	14%
Angola	0.77	3%
Oman	0.81	4%
Nigeria	0.75	3%
Russian Federation	0.42	2%
Cameroon	0.41	2%
Algeria	0.35	2%
Australia	0.35	2%
Mozambique	0.28	1%
Trinidad and Tobago	0.27	1%
Egypt, Arab Rep.	0.20	1%
Equatorial Guinea	0.22	1%
Guinea	0.07	0.3%
Belgium	0.06	0.3%
China	0.05	0.2%
Singapore	0.01	0.05%
TOTAL	22.14	100%

Source: World Integrated Trade Solution

Petrochemical Marketing

1. Kindly provide Global outlook for Petrochemical segment. Kindly mention top ten Petrochem producers & consumer countries with quantities for FY2024-25. Kindly provide a brief of upcoming capacities globally and demand outlook till 2030. Further, kindly brief about India's per capita Polymer consumption as compared to world's average, USA and China?

Global Polymer industry (PE & PP) is growing at a CAGR of ~4-5% with current market size of ~230 Million MT. Current Global Capacity is ~266 Million MT.

Top 10 Countries by Polymer production capacity (PE & PP) is as below:

Country	Capacity (Million MT)
China	75
USA	40
Saudi	22
Russia	16
India	13.5
Iran	13
South Korea	12.6
Kuwait	12
Thailand	6.1
UAE	4.8

Top 10 Countries by Polymer consumption (PE & PP) is as below:

Country	Demand (Million MT)
China (mainland)	81
United States	31
India	15
Germany	10.5
Turkey	6.3
Brazil	5.8
Japan	5.5
Vietnam	5.2
Indonesia	4.6

Expected Global Polymer Capacity (PE&PP) vis-à-vis demand by 2030 will be approximately 286 Million MT & 278 Million MT. Average Per Capita Polymer consumption of India stood at 15 Kg as compared to world Average of 39 Kg. For USA & China, Average Polymer consumption is of 93 Kg & 83 kg respectively.

2. **Kindly provide outlook on Top 10 PE consumers of GAIL with quantities in FY2024-25.**

Top 10 PE consumers of GAIL are as below:

Top 10 Customers	Sales (MT) FY2025
NILKAMAL LIMITED	18,218
FASTRACK DEALCOMM PVT LTD	16,541
SIDDHI VINAYAK INDUSTRIES	15,896
JJ PLASTALLOY PVT. LTD.	13,689
ICON POLYMERS	13,214
NEOTERIC COMPOUNDS	13,170
TIRUBALA POLYMERS	12,918
UNIPOL INDIA PRIVATE LIMITED	12,869
SRM PRODUCTS PRIVATE LIMITED	12,826
GLOBAL PLASTCHEM PRIVATE LIMITED	12,784

3. **Kindly provide India's Polymer Production and consumption for last 5 years. Kindly provide producer wise capacity at present and upcoming/scheduled. Kindly brief about top polymer consuming sectors in India.**

India's PE/PP Production and Consumption for last 5 Years in KTA is as below :

Year	Polyethylene (PE – HD/LL only)		Polypropylene	
	Production	Consumption	Production	Consumption
2020-21	4,830	5,459	5,232	5,417
2021-22	4,845	5,675	5,709	6,158
2022-23	4,079	6,083	5,183	6,457
2023-24	5,272	7,351	5,909	7,024
2024-25	5,870	7,262	6,623	7,615

Producer wise Existing capacity :

Manufacturers	Production Capacity (Million MT)	
	PE	PP
HMEL	1.25	1
Reliance	2.2	3.0
Haldia Petrochemicals	0.72	0.35
OPaL	1.06	0.34
IOCL	0.65	1.28
GAIL	0.81	X
BCPL	0.22	0.06
MRPL	X	0.45
NAYRA		0.45
Total	6.91	6.93

Major upcoming Polyethylene plants:

1. HRRL Barmer – 1,000 KTA by 2026.
2. IOCL Paradeep – 1,100 KTA by 2030.
3. BPCL Bina – 1,200 KTA by 2030.

Major upcoming Polypropylene plants:

1. GAIL USAR & PATA – 550 KTA by 2026 & 2025 respectively.
2. IOCL Barauni & Vadodara – 700 KTA by 2026.
3. IOCL Panipat – 500 KTA by 2026.
4. HRRL Barmer – 980 KTA by 2027.
5. NRL Assam – 360 KTA by 2028.
6. Nayra Ph-II Vadinar – 450 KTA by 2028.
7. IOCL Paradeep – 530 KTA by 2030.
8. BPCL Bina & Kochi – 1000 KTA by 2030.
9. Petronet LNG – 500 KTA by 2030.

Top PE & PP consuming sectors in India are Packaging (68%), Infrastructure (12%) , Agriculture (10%) , Electrical & Electronics (4%), Household (2%) and Automotive (4%).

4.Outlook on oversupply of polymer in India (since PLL and others have also announced Petrochemical plants). Petrochemicals margins are shrinking due to oversupply at global level. How does GAIL see it?

By 2030, PP will definitely be in surplus however, PE will remain in deficit, if past CAGR of 8% is maintained. As per Margins are concerned, its cyclical in nature and feedstock cost has a bigger role in final margins. Even in turbulent situations, Plants running on Naphtha and Ethane are able to be in profit.

5.Present status on Petrochemical Plant operational activities? At which capacity is being operated? Has any specific volume been tied up by GAIL for petrochemicals?

In FY25, Petrochemical plant has operated on 102% of its name plate capacity (production of 827 TMT against a capacity of 810 TMT). No, specific volume has been tied up by GAIL for Petrochemicals.

6.Brief on Demand and Supply scenario of Polymers in India with actual of past 3 years and estimation for future 3 years? Kindly brief about GAIL’s Market share (production and consumption).

Figures in KTA

(KTA)	Polyethylene(PE – HD/LL only)		Polypropylene	
Year	Production	Consumption	Production	Consumption
2022-23	4,079	6,083	5,183	6,457
2023-24	5,272	7,351	5,909	7,024
2024-25	5,870	7,262	6,623	7,615
2025-26 (E)	6,500	7,840	7,900	8,220
2026-27 (E)	6,800	8,470	8,500	8,880
2027-28 (E)	6,800	9,140	9,300	9,590

Since, GAILs Capacity is limited, expected market share in PE for FY 2025-26 is ~10.3% and ~13.1% including BCPL.

7. Present Market Share and outlook on product wise market share of GAIL Petrochemicals in coming years.

Present Market share of GAIL is 11.7%. Market share in HDPE is ~12% whereas in LLDPE, its ~11% . Since our capacity is limited and market is growing, hence this share will further go down in coming years.

8. What is the expected production from GAIL's Petrochemical segment in FY 25-26 and onwards?

Plant is expected to operate at 100% of Capacity.

9. What are the strategies to improve/safeguard the margins in PC segment in context of input cost & to improve market share?

Cheaper feedstock will help in safeguarding the margins in PC segment. Apart from this, GAIL's Marketing team is constantly focusing on those demand pockets/areas and Product grades where realization is higher.

Market share can only improve, if GAIL increases its capacity, as market is growing and our production is constant. However, in FY 2025-26 we will enter new markets through diversification in PP & PTA

10. What is the profitability of GAIL's competitors in Petrochemical segment along with operational reasons for this?

EBITDA of our competitors vary between 14% - 18%.

Reason: Majority of Plants in India are Naphtha Based and they are integrated with refineries as a part of their Oil to Chemical Value Chain.

11. Is GAIL making any direct sales to customers? What is the %age of direct sale to customer vis-à-vis sale through consignment stockiest (CS) during previous three years? Kindly bifurcate CS Sales in Actual User and Trader.

Yes,

Average Direct Sales in past 3 years is in the range of 1-2%.

For FY 2024-25, Actual User Sales is ~54% of total Sales portfolio and for Traders is ~46%.

12. Which are the major polymer companies whose products are imported by India?

Borouge, Chevron Philips, Exxon Mobil, Qatar energy & Dow Chemicals are top exporters of Polyethylene in India.

13. Kindly provide outlook on Region wise marketing of Polymer in previous three years for GAIL. Do you see any anticipated shift in the market structure in view of other plant commissioned/to be commissioned?

Region wise Market Presence in last 3 FY is as below:

Region	2022-23	2023-24	2024-25
North	41%	52.7%	49%
South	18%	17.9%	18.5%
East	5%	4.3%	4.5%
West	36%	25.1%	28%

With upcoming HRRLs Plant in North and HMELs plant operating at 100% Load, Regional Profile may change marginally with anticipated shift from North to West & South.

14. Kindly brief about Production and Imports of Petchem in India during previous five years and future prospects? Further, does GAIL have any plan on import of Petchem products presently or in near future.

Production vs Imports during past years for **PE Segment** is as below:

	Capacity (KTA)	Production (KTA)	Imports (KTA)
2020-21	5,660	4,830	960
2021-22	5,660	4,845	1,002
2022-23	5,660	4,079	2,000
2023-24	6,910	5,272	2,115
2024-25	6,910	5,870	1,700 (E)

Due to Global oversupply situation coupled with slowdown in China (largest importer of polymer in world), substantial influx of imports are expected in India for at least 2-3 years.

However, with capacity rationalization and incremental demand, they are expected to come down marginally but India is expected to be in net-deficit.

In view of surplus in the market (Domestic Supply + Imports), GAIL has no plan to import PE in near future.

15. Probable Impact of US tariff on PC Import / Export in India?

Non-Significant impact estimated

16. Is PC export lucrative and what are the target markets for Export of Polymers? What is the exports target, if any, for next two years? What is the net realization in the present exports and expected realization for expected numbers?

In current market situation, if delta between Price & Cost is within manageable limits, Exports can be lucrative. Net realization for FY 2024-25 is ~ USD 3.8 Million. Target for FY 2025-26 is ~ 40,000 MT and hence expected net realization will be ~ USD 41 Million.

17. With reference to the changing Scenario in crude oil and Naphtha price globally, how competitive are the gas-based crackers considering the elevated levels of Gas prices and no domestic gas allocation for PC?

At current gas Prices (above USD 10.4 in April 2025), Gas based crackers are operating in losses. Petrochemicals Topline is dependent on international prices with high degree of Price sensitivity. Hence, Bottomline can be positive by virtue of cheaper feedstock and economies of scale. However, Naphtha based crackers are able to operate at comparatively better margins.

18. How do you see new capacity being added at USAR/PATA will contribute and GAIL's strategies in this regard?

Currently, GAIL has no presence in PP. This expansion will help us diversify in PP segment with a market share of ~7%.

19. What does the Indian market look like regarding consumption of petrochemicals (PE, PP & PTA)? How does GAIL Plan to enter PTA market? Which are the major producers and consumers of PTA in India? How much is the deficit between India's PTA production and demand?

Consumption of PE, PP & PTA looks positive as all these markets are growing at 7-8% with demand in each of the market is expected to cross 8 Million MT in FY 2025-26.

GAIL has already started Pre-marketing activity for PTA and have reached out to majority of Customers in Domestic Tariff area and some strategic global customers as well.

Major Producers of PTA in India are Reliance, IOCL and MCPI whereas Institutional Customers like – Bhilosa, Bombay dyeing, Wellknown, Indorama, SRF, Polyplex, Uflex, Filatex, etc. account for major demand.

Current deficit in PTA is ~1.3 Million MT, however by next year IOCL Paradeep and GMPL will start its Production, then this deficit will decrease to 0.2 Million MT.,

Pricing Related

1. Kindly provide brief on present Tariffs on GAIL's NG Transmission network. (Kindly mention pipeline wise tariff in \$/MMBTU along with tariff of upcoming pipelines (including bid out pipelines).

Present Tariffs for GAIL's NG Pipeline Network is as below:

Sl. No.	Pipeline Network	Zone	Tariff (Rs/MMBTU-GCV)
1	National Gas Grid System^ (Unified Tariff)	1	42.04
		2	80.08
		3	106.77
2	GAIL's Integrated Natural Gas Pipeline Network~	1	29.67
		2	59.34
		3	74.17
		4	81.59
		5	85.29
		6	87.14
		7	88.07
		8	88.53
		9	88.77
		10	88.88
		11	88.94
		12	88.97
		13	88.98
3	Agartala Regional NG Pipeline	N/A	2.06
4	K. G. Basin	N/A	8.40
5	Kochi – Kootanad – Mangalore/ Bengaluru	1	42.78
		2	42.82
6	Cauvery – Narimanam + Kuthalam	N/A	43.82
	Cauvery – Ramnad	N/A	9.63
7	Gujarat – Motwan	N/A	2.77
	Gujarat – Kadi Kalol	N/A	61.61
	Gujarat – Kalol Ramol	N/A	1.90
	Gujarat – Mehsana	N/A	14.82
	Gujarat – Paliyad	N/A	4.56
	Gujarat – Ex Hazira GGCL	N/A	1.10

^Customers Connected to NGGS shall be levied Unified Tariff. NGGS Includes - GAIL's Integrated NGPL & KG Basin PL, PIL's East West NGPL, RGPL's Shahdol-Phulpur PL, IOCL's Dadri-Panipat PL, GSPL's High Pressure Gujarat Gas Grid & Low-Pressure Gujarat Gas Grid, GIGL's Mehsana-Bhatinda & Bhatinda-Jammu-Srinagar, GITL's Mallavaram-Bhilwara, ONGC's UTNGPL, GGCL's Hazira Ankleshwar NGPL.

2. Brief Details on Integrated Tariff implemented from 01.04.2023 and status of revision of Integrated Tariff along with its impact.

As per NGPL Tariff regulation amendment dated 19.11.2022, PNGRB vide Tariff Order dated 22.03.2023 approved the Tariff for **GAIL's Integrated Pipeline Network as Rs.58.61/MMBTU.**

Further, PNGRB vide tariff order dated 24.03.2023, approved zonal tariffs for 13 zones of GAIL's Integrated Pipeline Network, applicable with effect from 01st April 2023. The tariff notified for each zone is provided in reply to question no.2 above.

Presently, the following pipelines are part of Integrated Natural Gas Pipeline Tariff proposal:

- a. Jagdishpur-Haldia-Bokaro-Dhamra Pipeline (JHBDPL)
- b. Dadri-Bawana-Nangal Pipeline (DBNPL)
- c. Chainsa Jhajjar Hissar NG Pipeline (CJHPL)
- d. Dabhol Bangalore NG Pipeline (DBPL)
- e. Dahej Uran Panvel-Dabhol NG Pipeline (DUPL-DPPL)
- f. HVJ Integrated Pipeline Network (HVJ)
- g. South Gujarat Main Network (SG-Main)
- h. Mumbai Uran – Thal – Usar Network (UTU)
- i. Mumbai Trombay Network (Trombay)

GAIL had submitted the revised tariff Rs. 78.72/MMBtu for GAIL's Integrated NGPL in August 2024 in response to PNGRB's direction to submit the revised tariff submission.

Post tariff submission by GAIL, PNGRB had raised various queries on the revised tariff submission which were suitably replied by GAIL time to time. On 17th April 2025, PNGRB webhosted Public Consultation Document (PCD) on revised tariff submission of GAIL's Integrated PL Network soliciting comments from all stakeholders within 15 days with a timeline up to 1st May 2025 for providing comments.

GAIL is anticipating positive impact on tariff revenue mainly on account of true up of actual fuel cost for FY 2022-23 and FY 2023-24 and lower capacity determination of GAIL's Integrated NGPL Network in compare to capacity considered by PNGRB in previous tariff order.

3. How will revision of GAIL's Integrated Pipeline Network Tariff affect its downstream customers? How much tariff revision is GAIL practically expecting against the submitted tariff?

GAIL is anticipating increase in pipeline tariff for Integrated Pipeline Network mainly on account of the true up of actual fuel cost for FY 2022-23 and FY 2023-24 and lower capacity determination of GAIL's Integrated NGPL network in compare to capacity considered by PNGRB in previous tariff order.

Increase in tariff of GAIL's Integrated Pipeline Network may increase the cost of natural gas for its downstream customers.

GAIL is expecting tariff revision in the range of Rs.67-Rs.70/MMBTU against submitted Tariff.

4. Kindly provide brief Details on Unified Tariff and timelines.

Unified Tariff is a weighted average of notified tariff of the constituent pipelines of NGGS and subject to inter-entity settlement based on the unified tariff collected from consumers and

revenue entitlement of each pipeline entity based on the notified tariff for each pipeline. Thus, Unified Tariff is revenue neutral for all the pipeline entities whose pipelines are part of NGGS.

In Nov-2020, PNGRB notified the NGPL Tariff Second Amendment Regulations (Unified Tariff Regulations). However, its date of coming in to force was put in abeyance and to be notified later.

Further, by notification dated 19.11.2022, PNGRB notified that these regulations will come into force with effect from 01.04.2023 along with some amendments including increase in number of zones for Unified Tariff from two (2) to three (3).

An industry committee was formed by PNGRB comprising of representatives from pipeline entities whose pipelines are part of the National Gas Grid System facilitated by a representative from the Board to develop the settlement mechanism. (Letter dated 02.12.2022).

PNGRB vide Tariff Order dated 29.03.2023 has issued the applicable Unified Tariff. The same has been implemented by GAIL w.e.f. 1st April 2023. Further, the PNGRB has revised the applicable unified tariff and notified contractual path from time to time. The last unified tariff order was issued by the PNGRB on 28.06.2024 will supplementary tariff orders issued on 17.12.2024 and 15.04.2025.

5. View on domestic natural gas prices in India, for gas supplied by ONGC (APM gas) and future deep-water gas. Sector-wise Chronology for allocation of Domestic Gas in India?

Based on the recommendations of Kirit Parikh Committee, MoP&NG vide notification number L-12015/1/2022-GP-II (E-44157) dated 7th April 2023 has notified amendment to New Domestic Gas Pricing Guidelines, 2014. As per the notification:

1. Domestic Natural Gas Price (APM price) shall be 10% of the Indian Crude Basket Price as defined by Petroleum Planning and Analysis Cell (PPAC) from time to time. The present price for the month of April 2025, notified by PPAC is USD 7.26/MMBTU-GCV.
2. For the gas produced by ONGC & OIL from their nomination fields, the APM price shall be subject to a floor and a ceiling. The initial floor and ceiling prices shall be \$4/MMBTU and \$6.5/MMBTU respectively. The ceiling for previous two financial years (FY 2023-24 and 2024-25) was \$6.5/MMBTU and price ceiling for FY 2025-26 is increased by \$0.25/MMBTU to \$6.75/MMBTU.
3. Gas produced from new well or well intervention in the nomination fields of ONGC& OIL, where APM prices are subject to floor and ceiling, would be allowed a premium of 20% on these APM prices.
4. Subject to the provisions of respective Production Sharing Contracts (PSCs), this APM price, so declared by PPAC, would also apply where PSCs of NELP Blocks or Pre NELP blocks provide for Government's approval of prices. Floor and ceiling shall not apply to such APM price for gas produced from fields under these regimes.

Price ceiling for Gas Discoveries in Deepwater, Ultra Deepwater Areas and High Pressure High Temperature Zones is declared by PPAC based on the following guidelines:

- The ceiling price shall be the, lowest of the -
 - Fuel oil import landed price

- Weighted average import landed price of substitute fuels (0.3 x price of imported coal + 0.4 x price of imported fuel oil + 0.3 x price of imported naphtha) and
 - LNG import landed price.
- The ceiling will be calculated once in six months. The price data used shall be the trailing four quarters data with one quarter lag.
- The policy guidelines would be applicable to future and existing discoveries which are yet to commence commercial production as on 1.1.2016.
- Will be applicable to existing discoveries where arbitration/ litigation exist directly on gas price, only after conclusion/ withdrawal of such arbitration/litigation.

The present ceiling price notified for the period of 1st April 2025 to 30th September 2025 is \$10.04/MMBTU-GCV.

6. Kindly brief on formula for Domestic Gas Pricing by PPAC.

MoP&NG vide number L-12015/1/2022-GP-II (E-44157) dated 7th April 2023 has notified amendment to New Domestic Gas Pricing Guidelines, 2014. As per the notification Domestic Natural Gas Price (APM price) shall be 10% of the Indian Crude Basket Price as defined by Petroleum Planning and Analysis Cell (PPAC) from time to time. The Indian basket of Crude Oil represents a derived basket comprising of Sour grade (Oman & Dubai average) and Sweet grade (Brent Dated) of Crude oil processed in Indian refineries in the ratio of 78.50: 21.50.

For the gas produced by ONGC & OIL from their nomination fields, the APM price shall be subject to a floor and a ceiling. The initial floor and ceiling prices shall be \$4/MMBTU and \$6.5/MMBTU respectively. The ceiling price of \$6.50/MMBTU was applicable for previous two financial years (FY 2023-24 and 2024-25) and increased by \$0.25/MMBTU for next financial year i.e. FY2025-26 (\$6.75/MMBTU). The price ceiling is to be increased by \$0.25/MMBTU each year.

7. How much marketing margin is charged by GAIL on various types of Gas?

Marketing margin currently being charged by GAIL for natural gas is as tabulated below:

Gas Type	Marketing Margin
APM	Rs. 200/MSCM linked with NCV at 10,000 kcal/SCM as per MoP&NG order
Domestic gas other than APM fields: for supply to LPG & Fertilizer (Urea) producers in the country	Rs. 200/MSCM linked with NCV at 10,000 kcal/SCM as per MoP&NG order
for supply to other than LPG & Fertilizer (Urea) producers in the country	Rs. 18.32/MMBTU - NCV basis and equivalent as per agreement*
Imported Gas	Rs. 18.32/MMBTU-GCV as per agreement#

*MM is increased @5% every year depending upon the contract.

MM is increased @5% every 1st January.

8. How much is the Regasification charges at different terminals as on 01.04.2025? Periodicity of revision in charges at each terminal. Is there policy/contractual term validity to regulate the regasification charges as its increasing year on year.

Regasification charges of different terminals as re follows:

Regas Charges at Dahej & Dabhol – Rs.66.06/MMBTU (GCV) for CY 2025 (increased by @5% w.e.f. 1st January every year)

Regas Charges at Kochi- Rs.93.81/MMBTU (GCV) for FY 2025-26 (increased by @5% w.e.f. 1st April every year).

Regas Charges at Dhamra- Rs.66.35/MMBTU (GCV) for CY 2025 (increased by @5% w.e.f. 1st January every year)

Regas Charges at Hazira- Rs.59.00/MMBTU (GCV) for CY 2025 up to specified number of slots.

There is no regulation for Regasification Charges as on date. Regasification Charges are being fixed based on contractual agreement between parties.

9. What is the gas sales price from various sources along with assumptions and Pricing formula?

Applicable Gas prices as on 01.04.2025 from various sources are stated below;

Particulars	Price (\$/MMBTU)	Remarks
PLL Long Term Ras Gas (2003) (GCV)	10.7536	Note - 1
Domestic Gas Price (GCV)	7.26	Note - 2
APM/NAPM (GCV) For rest of India, applicable for nominated fields of ONGC/OIL	6.75	Note -3
APM/NAPM (GCV) For NE customers, applicable for nominated fields of ONGC/OIL	4.05	Note -3
New Well/Well Intervention Gas by ONGC/OIL	8.71	Note -4
Focus Energy (GCV)	7.26	
RAVVA 1 (GCV)	10.115	
RAVVA Satellite (GCV)	10.115	
Other Gases	As per the contracts	
Notes:		
1. PLL Long term price for April'25 Ex-terminal excluding Regas charges & sales-tax/VAT.		
2. Domestic Gas price notified by PPAC for the month of April 2025 based on the revised pricing methodology of Domestic Gas price equivalent to 10% of Indian Crude Basket.		
3. PPAC notified Domestic Gas price subject to a floor of \$4.00/MMBTU-GCV and ceiling of \$6.75/MMBTU-GCV for the FY 2025-2026.		
4. As per revised New Domestic Gas Pricing Guidelines 2014, Gas produced from new well or well intervention in the nomination fields of ONGC& OIL, where APM prices are subject to floor and ceiling, would be allowed a premium of 20% on these APM prices		

10. Kindly provide outlook on pricing of Polyethylene (till 2030).

Polymer price is going to remain under pressure till 2027-28, however prices are expected to recover gradually thereafter. Price of LLDPE in South East Asia region may drop 900-950 USD/MT till 2027-28 and after that price may recover upto 1,200-1,250 USD/MT till 2030. Similarly, price of HDPE may drop to 900-950 USD/MT till 2027-28 and after that price may recover upto 1,175-1,225 USD/MT by 2030.

11. Methodology of Petrochemical and LHC (LPG, Propane, Pentane & Naphtha) pricing in GAIL?

Petrochemical - GAIL's Polymer price of various grades are being worked out based upon the **Import Parity Pricing** Mechanism and prevailing market conditions.

LPG - With effect from 1st April 2002, Liquefied Petroleum Gas (LPG) prices has been deregulated and is now based on the **import parity prices fixed by the Oil Marketing Companies** and IOCL is the nodal agency. However, the pricing mechanism is provisional and is yet to be finalized by the MoP&NG. Being the Nodal agency, IOCL shares the LPG prices on a monthly basis but pricing methodology is not shared with GAIL.

Propane /Pentane/Naphtha– Based on the international price of the commodity and the prevailing market conditions.

12. Pricing methodology in detail that's used by IOCL for declaring LHC prices in its circulars?

LPG - With effect from 1st April 2002, Liquefied Petroleum Gas (LPG) prices has been deregulated and is now based on the import parity prices fixed by the Oil Marketing Companies and IOCL is the nodal agency. However, the pricing mechanism is provisional and is yet to be finalized by the MoP&NG. Being the Nodal agency, IOCL shares the LPG prices on a monthly basis but pricing methodology is not shared with GAIL.

13. Impact of change in PNGRB regulation/proposed changes on NG transmission tariff? What are key major changes comparing with previous one and its Impact?

PNGRB has webhosted draft amendments to the Tariff Regulation. The key changes are summarized as under:

S.No.	Proposed Amendments	Impact on GAIL
1	For the "CNG and PNG-Domestic" volumes, applicable unified zonal tariff shall be the tariff for first zone of unified tariff.	Will reduce tariff cost for GAIL's CGD
2	Definition of Isolated Network and Tariff floor for Isolated Networks Proposed: Isolated Network means natural gas pipeline as defined under the Petroleum and Natural Gas Regulatory Board (Authorizing Entities for Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008 which is neither part of 'national gas grid system' nor connected to any 'liquefied natural gas terminal' as defined in Petroleum and	Draft amendments if notified, will help to improve revenue realization from Isolated pipeline networks where operating cost is more than the tariff revenue at present.

	Natural Gas Regulatory Board Act and provided in the Schedule E. The tariff for the isolated natural gas pipeline authorized under regulation 17 of the Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008 listed under Schedule E of Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008 shall be Tariff determined in line with Regulation 4 of Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008 subject to a floor tariff based on minimum of the following: Tariff for first zone for Unified Tariff notified immediately after notification of this amendment; or Tariff determined in line with Regulation 4 of Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008 considering replacing Volume Divisor as per Clause 6 of Schedule A of Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008 with actual volume flow for the period commencing from 20.11.2008. Provided that, in case the isolated natural gas pipeline becomes part of the National Gas Grid system, the tariff for the remaining life shall be determined based on the Regulation 4 of the Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008 by considering its net fixed assets effective from the date one year after the date of it becoming part of National Gas Grid system	
3	Inclusion of Interest free security deposits received in Project cash flows Any interest free security deposit obtained by an natural gas pipeline entity for providing services, facilities, infrastructure, etc. to the shipper with respect to the pipeline the tariff of which is being determined in terms of the regulatory framework, shall be deducted from the Capital Expenditure claimed for the relevant services, facilities, infrastructure, etc. during the relevant year(s) for the purpose of tariff determination, until the said interest free security deposit has been refunded by it to the Shipper.	There will not be any significant impact on GAIL

	Provided that interest earned by the transporting entity on such deposits received will not be considered as other/miscellaneous income of the entity	
4	<i>Long term Contracts for Fuel Purchase</i> Entities to procure their gas volume for fuel requirements under the long-term contract(s) on least cost basis either domestically or internationally for a period of at least 5 years from the end of the period for which SUG volumes are already contracted or within three months after the notification of this amendment, whichever is later. However, if the entity has surplus gas available in their portfolio which can be allocated to SUG at a price lower than the above, then such cheaper gas price will be allocated for SUG. The entity shall provide the details of the gas procured to PNGRB on immediate basis after the procurement or allocation, as the case may be.	Draft amendments if notified, will help to improve pipeline tariff revenues based on fuel cost estimates close to actuals instead of fuel cost estimates based on historical prices. It will further help to reduce the major tariff variations/shocks at future date and ensure consistency in tariff revenue.
5	Sharing of revenue on actual volumes above normative volumes level. In case where the actual volume exceeds normative volume, after the adjustment(s) provided as per clause 10 of Schedule A, in any of the financial year(s), the 50% of such excess volumes shall be shared in the tariff by the pipeline entity. Further, the revenue (net of tax) earned by the pipeline entity on such balance of excess 50% volumes above shall be kept in a separate Natural Gas Pipeline Development Reserve by the entity including the interest earned on the same (net of income tax calculated at the applicable income tax rate, if any), which shall be utilized for laying, building or expanding a natural gas pipeline infrastructure of the entity. This is to further clarify that the amount kept in Natural Gas Pipeline Development Reserve shall not be used for any other purpose like, payment of dividends, other expenses, etc. In this regard, annual statement shall be provided by the transporter duly certified by an Independent Chartered Accountant by 30th June of the following financial year.	Draft amendments if notified, will help to create development fund which can be used for laying, building or expanding a natural gas pipeline infrastructure of the entity
6	Two Unified Tariff Zones second tariff zone for unified tariff being the remaining length of the national gas grid system	As unified tariff is revenue neutral for pipeline operations, there will be no impact on transmission tariff revenue of GAIL.

		However, there will be some increase in internal consumption cost for Petrochemical and LHC Segment.
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14. Any upcoming tariff revision/key change in the Non-Integrated Pipelines?

GAIL has submitted revised tariff for the following pipelines for which revised tariff is expected:

- (a) GAIL Integrated Pipeline Network
- (b) KKB MPL Pipeline Network
- (c) K G Basin NGPL

15. Details of de-allocation of Domestic Gas to GAIL's compressor in last three years and any expectation of re-allocation of domestic APM gas/other specific gas in near future?

Details of de-allocation of Domestic Gas allocated for SUG of GAIL's compressors in last three years is as per following:

De-allocation of Domestic Gas allocated for GAIL's compressor (w.e.f.)	De-allocated Volume (MMSCMD)	Remaining allocation(MMSCMD)
16-08-2022	0.45	1.10
01-02-2023	0.20	0.90
16-03-2023	0.30	0.60
01-06-2023	0.20	0.40
01-10-2023	0.20	0.20
16-12-2023	0.20	Nil

The probability for any re-allocation of domestic APM gas/other specific gas in near future is remote considering the fact that SUG for NG transmission pipelines is not a priority sector for allocation of domestic gas.

Regulatory Affairs

1. Which issues need to be taken up with PNGRB/GoI for increasing the Share of Natural Gas in energy basket of India? Status of various issues taken up and pending with PNGRB and the expected timelines.

Policy incentives promoting gas over polluting fuels, inclusion of gas in GST, rationalization of VAT till gas is outside GST, exempting excise duty on CNG, incentivize DPNG in areas being served by LPG would be some key measures to promote gas. Consideration of cost-overrun in tariff of pipelines for delays beyond the entity's control remains a key issue under deliberation.

2. What is the process of pipeline tariff review?

A Review Petition is required to be filed with PNGRB under section 13(1)(h) of the PNGRB Act. Subsequently, the petition is heard by a PNGRB Bench comprising 2 Members (Chairperson plus one member or two members if Chairperson's post is vacant) and Member-Legal. Post hearing, a revised tariff order is issued.

Methodology for determination of natural gas pipeline tariff (as per Clause 2, Schedule A of Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008): The unit rate of natural gas pipeline tariff to be charged for a period shall be the calculated based on the "Discounted Cash flow" (DCF) methodology considering the reasonable rate of return.

Determination of initial unit natural gas pipeline tariff and review of unit natural gas pipeline tariff (as per Clause 9, Schedule A of Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008):

The unit natural gas pipeline tariff shall be determined for the natural gas pipeline over its economic life and levelized during:

- (a) the period applicable for the initial unit natural gas pipeline tariff;
- (b) the **period of five consecutive years** after the end of the initial unit natural gas pipeline tariff
- (c) the period between any two consecutive tariff reviews

3. Latest status on tariff revision of Integrated P/L and other pipelines (separately). Kindly provide a table mentioning GAIL's pipeline wise tariff (including bid out pipelines).

Pipeline	Category	Existing Tariff Rs. /MMBtu	Tariff submitted Rs. /MMBtu	Status
MNJPL	Bid-out	Rs.60- Rs.139.85	-	-
SAPL	Bid-out	Rs.15- Rs.188.68	-	-
GJPL	Bid-out	Rs.1.00 – Rs.234	-	-
Integrated	Cost-plus	58.61	78.72	PCD stage
KG-Basin	Cost-plus	8.40	In process	
Cauvery Basin: NKM	Cost-plus	43.82	In process	
Cauvery Basin: RMD	Cost-plus	9.63	In process	
Gujarat Regional				

Pipeline	Category	Existing Tariff Rs. /MMBtu	Tariff submitted Rs. /MMBtu	Status
Kalol-Ramol	Cost-plus	1.9	In process	
Mehsana	Cost-plus	14.82	In process	
Paliyad	Cost-plus	4.56	In process	
Ex-Hazira GGCL	Cost-plus	1.1	In process	
Kadi-Kalol	Cost-plus	61.61	In process	
Motwan	Cost-plus	2.77	In process	
KKBMPL	Cost-plus	42.79	176.85	
Agartala	Cost-plus	2.06	-	-

4. Any new amendments or proposed changes in regulatory sphere / PNGRB regulation?

PNGRB has come out with draft amendments in tariff regulations inter-alia proposing for reduction of Unified Zonal tariff from three to two zones, applicability of first zone unified tariff for "CNG and PNG-Domestic" segment of City Gas Distribution (CGD) Networks, revising tariff of cost-plus isolated networks by volume-based assumptions and reimbursing fuel cost based on contractual LNG pricing.

5. What are the regulatory requirements, if any pipeline asset has to be transferred to Infrastructure Investment Trust (InvIT) to monetize on existing pipelines? How regulator controls the assets under InvIT?

It would depend upon the exact model adopted for InvIT. In any case, it would require amendment in authorization in favour of the entity to whom the pipeline is transferred. Earlier also, in respect of East West Pipeline, the InvIT mode of monetization of pipeline asset has happened.

6. Can CGD authorization by central government to GAIL alongside JHBDPL be transferred to GAIL Gas? How regulator and other market players can challenge it? Is there any potential benefit/loss for GAIL in doing so?

On 06.04.2018, PNGRB notified the amended CGD Authorization Regulations wherein PNGRB may accept transfer of CGD authorization by the entity to its wholly owned subsidiary subject to the condition that the entity submits a Corporate Guarantee to PNGRB, inter-alia, agreeing to fulfil all the technical/commercial obligations of the wholly owned subsidiary in the event of failure/default of the wholly owned subsidiary to fulfil the same.

This Regulation is in place and has not been challenged by any market player.

7. After completion of economic life, a Pipelines gets over, can PNGRB legally appoint a new operator for the pipeline on new terms and condition? How will the tariff of the pipeline be determined in these cases?

At present, neither there is any explicit provision nor PNGRB has announced any such draft for changing the operator. However, as per the extant PNGRB Regulations, extension of the period of economic life may be considered by PNGRB for a block of ten years at a time depending on the satisfactory compliance of the service obligations and on such terms and conditions to the

existing operator, as it may deem fit at that point in time. Further, Tariffs will continue to be determined as per extant regulations irrespective of pipeline operator.

8. Can charges of regasification terminal be covered under regulation? When and how?

There are some LNG Registration Rules notified by the Central Government in 2012 which inter-alia specifies that in order to protect consumer interests, liquefied natural gas terminals must be subject to commercial transparency and committed to provide additional capacity to any consumer at publicized tariffs. However, the enabling regulations in this regard are not in place.

10 What is the designated IRR for pipeline projects? Is 12% post tax returns reasonable for GAIL or it can earn more returns on pipelines?

Cost Plus Pipelines: As per extant regulations, returns are **fixed** at 12% post-tax for cost-plus pipelines.

Bid-out Pipelines: There is no ceiling for bid-out pipelines.

11 Has GAIL approached PNGRB for authorization of Natural Gas pipelines which were previously authorized to other entities and have not been laid by them? If yes, kindly mention the pipeline name, length and estimated capex.

Based on MoP&NG directive to GAIL under section 42 for developing the Mallavaram-Bhopal-Vijaipur-Bhilwara Pipeline, GAIL is taking up with PNGRB for authorization of the same. Route details, estimated capex etc. are under finalization.

Business Development (BD) /Project Development (PD)

1. Kindly provide the outlook for investment in PDH-PP plant at USAR and PP Plant at Pata. Kindly provide year wise expected returns. Further, kindly provide the details of IPA Project at USAR. By when the aforementioned projects are scheduled to be completed?

PDHPP Project:

GAIL's board of directors in its 400th meeting held on 13th May 2019 had approved setting up 500 KTA PDHPP Project at Usar, Taluka- Alibag, District- Raigad, in Maharashtra with project cost of Rs 7823 Crores including interest during construction (IDC) of Rs 722 Crores and foreign currency components of Rs 1749 Crores.

Further GAIL's Board in 459th meeting held on 31st July 2023 has approved the revised cost of the project to Rs. 11,256 Crore including interest during construction (IDC) of Rs 784 Crores and foreign currency components of Rs 1400 Crores and the returns are enumerated below:

Base: On previous 5 Yrs. Historical Pricing	Project-IRR	Equity-IRR	Average DSCR
Project cost: 11,256 Crore	12.50	15.54	1.59

Expected returns from the project as per FAR are mentioned below:

Year	2026	2027	2028	2029	2030	2031	2032	2033	2034
PAT (Cr.)	60	254	426	537	450	687	783	867	757
EBIDTA (Cr.)	524	1251	1480	1578	1400	1652	1717	1766	1558
EBIDTA/Ton (Rs.)	10480	25020	29600	31560	28000	33040	34340	35320	31160

Year	2035	2036	2037	2038	2039	2040	2041	2042	2043
PAT (Cr.)	1030	1108	1164	980	1234	1280	1307	1103	1390
EBIDTA (Cr.)	1857	1903	1967	1724	2064	2126	2165	1894	2278
EBIDTA/Ton (Rs.)	37140	38060	39340	34480	41280	42520	43300	37880	45560

Year	2044	2045	2046	2047	2048	2049	2050	2051
PAT (Cr.)	1437	1476	1206	1543	1589	1625	1322	815
EBIDTA (Cr.)	2343	2396	2040	2489	2553	2603	2203	1314
EBIDTA/Ton (Rs.)	46860	47920	40800	49780	51060	52060	44060	26280

Scheduled Completion date: Oct-2025

PP Project, Pata:

GAIL is having gas based petrochemical complex at Pata for utilization of C2 /C3 fractions of natural gas for production of Polyethylene i.e. LLDPE& HDPE. In the Gas cracker unit (GCU) of Pata, Ethylene is the main product and Propylene is one of the by products. Prior to commissioning of GCU - II, propylene produced from GCU - I was of chemical grade (Min. 95% Propylene). GCU-II has been designed for producing Polymer Grade Propylene (Min. 99.5% Propylene) and provision has been made for processing the chemical grade produced from GCU - I for production of polymer grade propylene. Polypropylene is the

fastest growing polymer and Indian demand is expected to increase at a CAGR of around 9% in the next 5 years. Special grades of Polypropylene i.e. Impact and random are considered as premium grades; However, ethylene is required for the production of Impact and random grades. Since ethylene is available at Pata, hence special grades of PP (Impact and Random) can be produced at Pata. The Board of Directors in its 398th meeting held on 05th March 2019 had approved setting up 60 KTA PP unit at GAIL Pata at a project cost of Rs. 910.56 Crore. M/s EIL has been awarded EPCM Consultant for the Project on 12.07.2019 and M/s Grace Technology, Inc has been selected as licensor of PP unit on 04.09.2020. Further Board in its 452nd Board meeting held on 05.12.2022. has accorded approval for revised project cost of Rs 1299.02 Crore.

Financial results for the project:

PP PROJECT PATA	
Project cost (in Crore)	1,299
PIRR	11.22%
EIRR	13.16%

Cost of Production & Margins:

Project Name	PP PATA				
Project Life	25 Years				
Project Capacity	49.74 KTPA				
Main Input	Propylene = 49 KTA				
Main Output	Poly Propylene (PP) = 49.74 KTA				
Completion Date	March'25				
Project Cost	Rs 1299 Cr				
PIRR	14.16%				
Period Ending	31Mar25	31Mar30	31Mar35	31Mar40	31Mar45
Year	3	8	13	18	23
Particulars	INR Per MT	INR Per MT	INR Per MT	INR Per MT	INR Per MT
Sales (A)	1,46,878	1,79,897	2,20,339	2,69,872	3,30,541
Variable Cost	94,202	1,21,332	1,42,674	1,75,061	2,15,332
Raw Material Cost	83,343	1,01,697	1,24,431	1,52,257	1,86,319
Chemicals Cost	4,681	11,761	7,604	9,705	12,387
Utilities Cost	4,776	6,079	7,758	9,902	12,638
Packaging Material & Others	1,402	1,795	2,881	3,197	3,989
Fixed Cost	34,427	28,653	28,401	34,292	41,949
Total Cost (B)	1,28,629	1,49,985	1,71,076	2,09,353	2,57,281
Margin (A-B)	18,249	29,913	49,263	60,519	73,260
Propylene Cost	83,073	1,01,748	1,24,621	1,52,637	1,86,951
Margin (USD)	227	345	527	601	675
Margin %	12%	17%	22%	22%	22%

Schedule Completion date: June-2025

IPA Project, Usar:

Board in its 451st Meeting held on 4th November 2022 approved the proposal for setting up 50 KTA Isopropanol Unit (IPA) at GAIL, Usar, Maharashtra at a project cost of Rs 530.04 Cr. (including IDC and taxes).

Licensor Selection for the same is under PBO stage.

Scheduled Completion: The project mechanical completion is envisaged within 24 months from date of signing of Licensor agreement. Period for pre-commissioning/commissioning has been considered as 2 months.

2. Kindly provide the outlook for investment in GMPL (erstwhile JBF Petrochemicals) and expected return along with competitors, demand of output product (PTA) in India and future outlook. Kindly provide the deficit between PTA imports and indigenous production for last 5 years.

Asia is the largest consumer of Purified Terephthalic Acid (PTA) in the world, largely driven by consumption in China. India is second largest consumer of PTA, primarily an import driven market and imported predominantly from China. PTA demand is further expected to increase in the coming years on account of growing demand from polyester based textile and packaging industry. The growth of demand of polyester in India is directly related to the GDP growth.

Synergy for GAIL

- GAIL has been in the business of petrochemicals (PE and PP) for ~23 years and has developed expertise in all aspects of running integrated petrochemical complexes.
- The opportunity offers GAIL an additional product slate (PTA) which is in sync with its existing petrochemical capabilities.
- PTA demand is increasing at an annual growth rate of over ~6%
- GAIL's marketing function has rich experience in petchem marketing having offices on pan-India basis. PTA can be marketed using the existing set-up.

Domestic capacity of PTA in India in 2024 was 6.3 MMTPA. Further, it is anticipated that 2 million tonnes of capacity shall be added by GAIL and IOCL by FY 2025-26. However, PTA requirement in India is projected to grow upto 14.7 MMTPA in 2035. Hence, 100% of PTA produced from GMPL can be consumed in domestic market.

3. GAIL's strategy on Power Sector (Solar, Wind & Gas based). Does GAIL have any expansion plans for renewable energy? Further, is GAIL in planning to operate few stranded Gas based power plants?

At present, GAIL's RE capacity is ~ 145 MW comprising of ~ 118 MW Wind and ~ 27 MW Solar. Further, RE capacity addition of ~ 40 MW is being done in the short term through Ground-mounted, Rooftop and Floating Solar.

GAIL aims to install 1.7 GW of renewable energy capacity by 2030 (*source: GAIL sustainability report 2023-24*)

GAIL is committed to achieve its Net Zero target of 100% reduction in Scope-1 & Scope-2 emissions by 2035. This requires ~ 3,500 MW (Mega-Watt) of Renewable Energy (RE) to meet

the Net Zero targets towards existing Business requirement. Further, necessary action is being undertaken to set up the additional RE capacity within next 5-7 years. GAIL is exploring both Organic as well as Inorganic routes to expand its RE portfolio through participation in RE tenders, acquisition of operating RE assets.

Gas based Power Plants:

On 08.08.2024, GAIL entered into a Memorandum of Understanding (MoU) with M/s Rajasthan Rajya Vidyut Utpadan Nigam Limited (RRVUNL) to optimise the operations of RRVUNL's gas-based power plants in Rajasthan. Under this MoU, land for 1,050MW of RE projects will also be allocated to this SPV. SPV will sell the bundled power to the discoms. Currently, it owns two operational gas-based power plants in Dholpur & Ramgarh districts of Rajasthan having installed capacities of 330 MW and 270.50 MW respectively. In this regard, GAIL is in process of appointment of consultant for carrying out due diligence studies. Based on the outcome of due diligence, GAIL will take equity in the SPV.

4. Details and update the status of new projects:

A. Propane dehydrogenation and Polypropylene Plant at Usar (PDH PP)

GAIL is setting up 500 KTA Propane Dehydrogenation & Polypropylene (PDH-PP) plant at Usar, Taluka- Alibag, District- Raigad, in Maharashtra. GAIL's PDH-PP project in Usar, Maharashtra, will be **India's first** propane dehydrogenation (PDH) plant, with propylene production seamlessly integrated into a polypropylene (PP) plant. The feedstock for the PDH-PP Project is Propane. 500KTA Propylene along with 25 KTPA of Hydrogen and 19 KTA of C4 Mix will be produced from PDH Unit.

Major units of this upcoming petrochemical complex are PDH unit, PP unit and Utilities & Offsite.

Propane received through pipeline shall be stored in mounded bullets and will be fed to PDH unit where Propylene will be produced through catalytic dehydrogenation process. Hydrogen and C4 Mix will be produced as by products. M/s Lummus Technology LLC, USA is licensor for PDH Unit.

Propylene will be used as raw material in PP unit where it will be converted to Poly propylene pellets as the final product. M/s Grace Technologies Inc., USA is the Licensor of Polypropylene (PP) Unit.

Polymer pellets will be stored in Silos and sent to bagging unit for packaging in 25 Kg bags. The product bags will be dispatched by road transport in trucks.

The plant shall be capable of production of following **six grades of Homopolymer** Propylene:

S. No.	End Use	% Share
1	TQ Film	10
2	Woven sacks	40
3	Fiber & Filaments	05
4	BOPP Films	10
5	Injection Molding	30
6	Extrusion / Thermoforming	05

Physical Progress of the project is 82%.

B. Polypropylene Unit at Pata (PP-Pata)

GAIL (India) Limited has a petrochemical complex at Pata for production of linear low density and high-density polyethylene from ethylene produced from C2/C3 (ethane/Propane) cracking. While in the process, the cracker is also producing around 50 KTPA of Polymer grade Propylene which is at present sold locally as final product. On the basis of the same, M/s GAIL is setting up 60 KTPA Polypropylene unit at Pata utilizing the Propylene produced at site.

Product Profile: The desired product profile for the Project is indicated below:

S. No.	End Use	% Share
1	Homo	75
2	Impact	20
3	Random	05

Physical Progress of the project is 95%.

C. GMPL (erstwhile, JBF Petrochemicals Limited)

GAIL Mangalore Petrochemicals Limited (GMPL) erstwhile JBF Petrochemicals Limited, set up a **1.25 million tonnes per annum capacity** Purified Terephthalic Acid (PTA) plant at Mangalore Special Economic Zone.

Expected Investment and Returns:

Capex: Total Financial implication ~ **Rs. 4,200 crores**

(Rs 2100 Cr.-- upfront & Rs 2100 Cr. -- future capex over next two years)

Funding by GAIL -: Rs. 1,476 Crore TL & Rs. 808 Crore Equity Contribution as on 31.01.2025

TL @ Repo +1.3%, 4 years Moratorium Period,

Repayment to GAIL starts from June 2027 till June 2037.

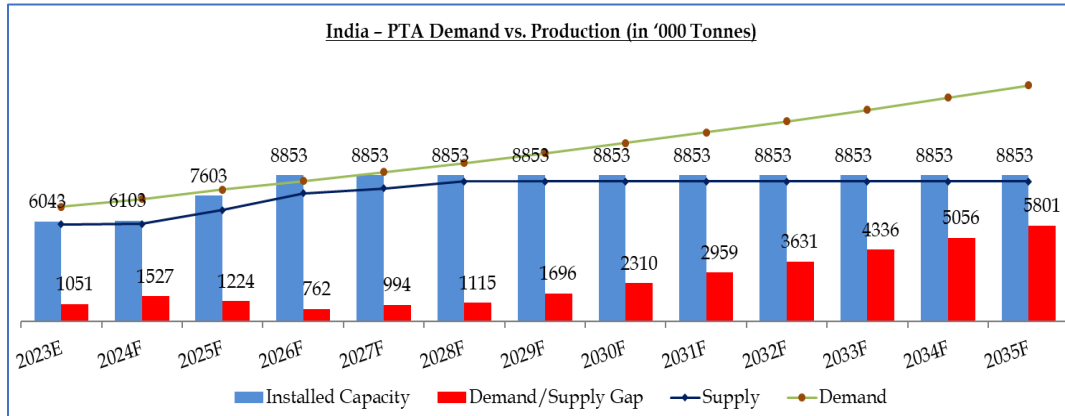
PIRR – 12.83%

EIRR - ~18%

Status: Operational Activities expected to be completed by **February, 2026.**

PTA Market and Consumers:

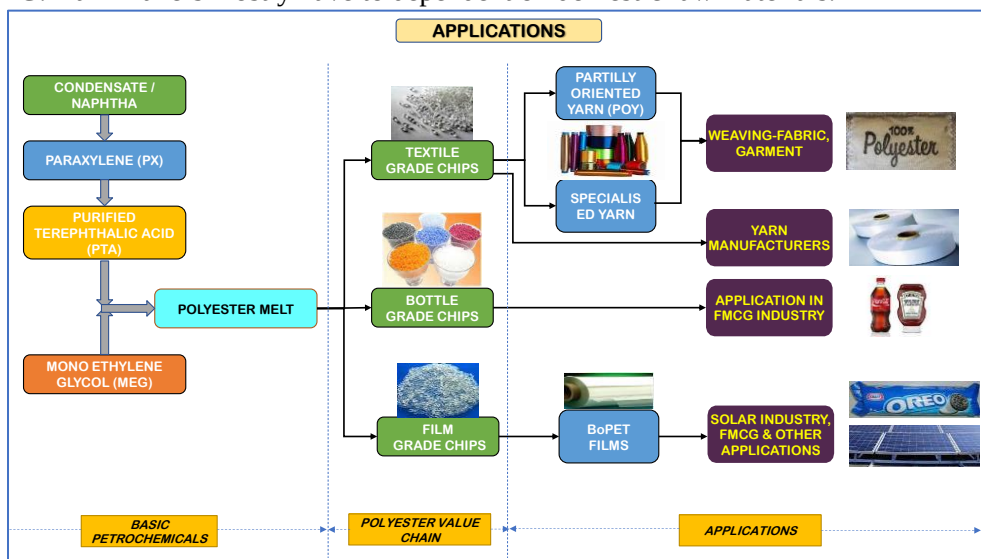
- JBFPL (now GMPL) was set up for backward integration project with technology license by BP Corporation, North America (now Ineos).
- Indian Market currently has a production capacity of 6.04 Million Tonnes Per Annum (MMTPA) of PTA, which is expected to reach 8.85 MMTPA by 2035. The Indian market has been facing supply shortages since mid-September 2020. Currently India is importing high levels of PTA.
- Reliance Industries, IOCL and MCPI are the major PTA producers in India with a total installed capacity of 4.22 MMTPA, 0.55 MMTPA and 1.27 MMTPA respectively. Additionally, IOCL is setting-up a greenfield PTA plant of 1.20 MMTPA capacity at Paradip, Orissa which is expected to be operational by 2024-25.



• **Major Probable Customers:**

Bhilosa Industries Private Ltd.	771.05 TMT
Wellknown Polysters P Ltd	136.68 TMT
Filatex India	105.31 TMT
Indo Rama Synthetic	89.42 TMT
Sanathan Textiles Ltd.	86.42 TMT

- The main feedstock of the plant is paraxylene (PX) which was to be sourced from across-the-fence facilities of Mangalore Refinery and Petrochemicals Limited (MRPL) and secured by a supply contract. However, MRPL has unilaterally terminated the contract during the moratorium period of CIRP and the matter is sub-judice. GAIL is in discussions with various leading PX suppliers in Asia and Middle East for sourcing PX on delivered basis including developing import infrastructure (at Mangalore Port)
- PTA and Mono-Ethylene Glycol (MEG) are two key raw materials used to make synthetic yarn viz., Partially Oriented Yarn, Polyester Staple Fibre, Polyester Staple Fibre Fill, Poly Ethylene Terephthalate, etc. There are very few manufacturers and suppliers of polyester raw materials PTA and MEG. Yarn makers mostly have to dependent on domestic raw materials.



D. 50 KTA Isopropanol Unit at Maharashtra

Board in its 451st Meeting held on 4th November 2022 approved the proposal for setting up 50 KTA Isopropanol Unit (IPA) at GAIL, Usar, Maharashtra at a project cost of Rs 530.04 Cr. (including IDC and taxes). EIL is the EPCM for the Project.

The project will be producing 50 KTPA of IPA product from hydrogenation of Acetone which will work as raw material for the Project.

Licensor Selection for the same is under PBO stage.

The project mechanical completion is envisaged within 24 months from date of signing of Licensor agreement. Period for pre-commissioning/commissioning has been considered as 2 months.

Capacity	50 KTPA
Project Cost	Rs. 530 Cr
Completion date	24 months after licensor selection. As on 30.06.2024, Licensor selection is under progress and Cryogenic Nitrogen Plant is at awarding stage.

Specialty chemicals emerged as one of the initiatives for consideration in GAIL Strategy 2030. As per FICCI report, Specialty chemicals industry stood at around 32 Billion USD in 2020 and expected to grow at 12% CAGR over the next 5 years. As per IHS report, prepared for GAIL, demand of IPA in the India stood at 190 KTA in the year 2020. The demand is expected to increase with a CAGR of 7.1% from 2021-2026. India has the total production capacity of 100 KTA at present. Investment for 50 KTA IPA Project, Usar was granted for production of Iso propyl alcohol. The feed for iso propyl alcohol plant is acetone and hydrogen. Acetone shall be sourced from outside through tankers and unloaded at Usar site and then stored in feed tanks. Hydrogen will be sourced from PSA of PDH Unit through pipeline and stored in Hydrogen Bullet.

	7- years Historical Average
Project IRR (PIRR)	15.35%
Equity IRR (EIRR)	23.27%
Avg. DSCR	2.05

E. Acquiring equity in LNG Liquefaction Project/Plant in USA

To expand presence along the LNG value chain and to secure firm LNG supply, GAIL is exploring LNG offtake and Equity option in Natural Gas (NG) liquefaction project in USA. In this regard, GAIL floated an Expression of Interest (EOI) on 11th April 2025 seeking equity option (ranging from 10% to 26% equity) in the existing / upcoming NG liquefaction project in USA along with supply of 1 MMTPA LNG on FOB basis. Due date of submission of EOI is 28th April 2025.

F. EOI for Ethane Vessels Carrier (Water Borne)

GAIL is exploring the opportunities for Time Chartering of Very Large Ethane Carriers (VLECs) to cater petrochemical production requirements in India.

To pursue diversification of feedstock for existing petrochemical plants, GAIL is touching all aspects of ethane value chain including sourcing from ethane-surplus countries, water borne transportation, development of import terminal and last mile pipeline connectivity.

G. GAIL's 26% Stake acquisition in LNG Japonica

In January 2023, GAIL acquired 26% equity in M/s LNG Japonica Shipping Corporation Limited (Japonica) from M/s Mitsui O.S.K. Lines, Ltd, Tokyo (MOL) at par for US\$ 260. It also offered GAIL an opportunity to partner with one of the leading shipping players in the world and to gain experience in LNG shipping industry.

Japonica operates LNG vessel - GAIL Bhuwan and GAIL has signed Time Charter Party Agreement (TCP) with Japonica for charter hiring of GAIL Bhuwan for a period of 5 years from 20.03.2021.

With the above acquisition, GAIL received Right Of First Offer (ROFO) rights on GAIL Bhuwan i.e. upon expiry of existing TCP or any future TCP, Japonica shall offer GAIL Bhuwan first to GAIL for TCP. Accordingly, discussion on extension of TCP is underway. In December 2023, GAIL received a dividend of US\$ 117,000 from Japonica for FY2023. Also, dividend of around US\$ 1 million for FY2024 is expected in Q2/Q3 -2025.

5. Kindly provide a brief write-up on GAIL's investment in IGX.

Indian Gas Exchange Ltd. (IGX) is the India's first Gas Exchange that features multiple buyers and sellers, on an online platform, to trade natural gas at designated physical hubs in India. IGX is promoted by its parent company i.e. IEX, which is the largest power trading company of India. IGX started its operations on 15.06.2020 as Gas Trading Platform. After the notification of Gas Exchange Regulations (GER) by PNGRB, IGX is operating as the Gas Exchange since 10.12.2020.

GAIL acquired equity stake of 5% in IGX with an equivalent equity investment of Rs.3.75 crores. The agreement to give this effect was signed on 09.02.2021. As per the GER, a Member of a Gas Exchange cannot take more than 5% equity stake in that particular Gas Exchange in which it is a member. Hence, GAIL being a member in IGX was allowed to take only 5% equity shares in IGX.

Upon issue of equity shares to the IGX ESOP Trust (ESOP Shares) on 19.10.2021, the shareholding % of GAIL became 4.93% from 5%

6. What is the status of TAPI pipeline?

GAIL, under the aegis of Government of India, is pursuing the TAPI Pipeline project to receive Natural Gas supply from the Galkynysh fields in Turkmenistan with State Concern Turkmenogas endorsed as the Consortium Leader. TAPI Pipeline Company Limited (TPCL) was incorporated in 2014 in Isle of Man. The proposed 56" x 1849 kms (Turkmenistan – 214 km, Afghanistan – 816 km and Pakistan – 819 km) of pipeline is expected to carry 90 MMSCMD of natural gas, of which India and Pakistan are expected to offtake 38 MMSCMD each. Afghanistan has contracted

for 1.5 – 4 MMSCMD. The landfall point of the pipeline in India has been proposed to be at Fazilka (Punjab). TPCL has its Project office in Dubai.

The following agreements have been signed:

- Government level: Gas Pipeline Framework Agreement (GPFA), Inter Governmental Agreement (IGA)
- Company Level: Gas Sales & Purchase Agreement (GSPA), Operations Agreement (OA), Shareholders Agreement (SHA) and Investment Agreement (IA)
- The major outstanding issues are (i) Economic Viability, (ii) Security of Supply and (iii) Tie-up of Debt and Equity for the Project. Since the signing of GSPA in May 2012, the global energy markets have undergone a sea change. Accordingly, efforts are underway between GAIL and Turkmen gas for arriving at mutually acceptable GSPA terms and conditions. Further, in order to fulfil the contractual assurances regarding security of natural gas supplies considering the geo-political situation in the regions, three Business Principles were proposed by GAIL for inclusion in SHA. In order to move forward with the project, an early resolution of the outstanding issues i.e. three Business Principles and GSPA re-negotiations are considered essential.

7. Kindly provide the status of GAIL's investment in startups. Which are GAIL's major focus areas for investment?

Start-up Initiative of GAIL “Pankh” launched in July 2017 to invest in promising Start-Ups under DPIIT Guideline. As per DPIIT guideline definition an entity is considered as Start-Up up to 10 years from the Date of its Incorporation / Registration, whose Turnover in any Financial year has not exceeded Rs. 100 crore and the entity should work towards Innovation, Development or Improvement of Products or Processes or Services, or if it is a scalable business model with a high potential of employment generation or wealth creation and it should not have been formed by splitting up or reconstruction of an existing Business.

GAIL Start-up initiative continues to foster innovation, entrepreneurship, and sustainable development by investing in promising and high-growth potential Start-Ups across diverse sectors and aims to derive long-term strategic benefits such as access to new technologies, technical know-how, and societal impact.

GAIL solicits investment proposals from the Start-Up operating in the area mainly includes Natural gas, Pipeline O & M, Petrochemical, energy, renewable & alternate energy source, IOT, emerging mobility, health, Social, environment, Safety & security, other emerging section like e-commerce, Fintech, IOT etc.

In recognition of its expanding vision and increasing opportunities, during FY 2024–25 GAIL has increased the Start-Up investment fund, increasing corpus to be invested in Start-Ups from Rs 100 crore to Rs 500 crore.

GAIL has so far supported 38 Start-Ups with a total investment commitment of approx. Rs. 77.38 crore and a total disbursement of approx. Rs. 68.09 crore. The impact of these investments is significant, with Start-Ups contributing to improved operational efficiency, green energy transition, and inclusive access to essential services. GAIL supported Start-Ups are operating in various areas viz. Bio-Polymer, Bio-fuel, AI, AI-Logistics, Electric Mobility, Nutrition, Rural Development, Health, Pipeline Inspection, Energy, Renewable Energy,

Environment, CBG, IOT, Industry 4.0 (process improvement), Fintech sector with credit line product of Social cause, SaaS software, Space industry with green Propellant etc.

In FY 2024-25, GAIL has signed investment agreements with 3 Start-Ups with a commitment of approx. Rs 6.69 crore. Moreover, during the year, 04 Start-Ups supported by GAIL raised follow-on funding at higher valuations and so far 13 nos. GAIL supported Start-Up raised follow up funding at higher valuation, reflecting the growing notional value of the initial investments.

Till date, GAIL has opened 9 nos. Rounds for solicitation of Investment proposal from promising Start-Ups. Recently, GAIL has opened its 10th Round of online solicitation inviting investment proposals from the Start-Ups operating in the Focus Areas (Core and Non-core Area) on 1st April' 2025, which will remain open till 31st May' 2025.

In acknowledgment of these efforts, GAIL was honoured with the prestigious “**Start-Up Excellence Award**” in the “Investment in Start-Ups” category at the Governance Now 11th PSU Awards. These achievements reaffirm GAIL’s position as a leading PSU investor in Start-Ups and its commitment to innovation-led, inclusive, and sustainable growth.

8. Writeup on the following:

• Small Scale LNG

GAIL has set up two no’s of Small-Scale NG Liquefaction plant (SSLNG) of design capacity of 2 x 18 Ton LNG per Day with associated facilities at Vijaipur plant in Guna district of M.P.

Project Status:

Both the skid has been commissioned.

Major Facilities in the Project:

Natural Gas Liquefaction skid of 2 x 18 Ton/Day capacity.

Natural Gas conditioning/ Pre-treatment

LNG Storage tank of 2 x 56 KL

LNG Tanker loading facility.

Feature of SSLNG plant:

- Small, Modular, Scalable, and can be easily relocated as per requirement.
- Easy and fast installation with High Efficiency.
- Easy to maintain & operate these units from remote areas.
- LNG could be produced in close proximity to the sources and also for the users due to its compact and modular design with lesser land requirement for setup.

Benefits of the Project:

Opportunities to develop the Geographical Areas (GAs), presently, without access to main NG pipeline and/or which are far from coastal LNG terminals.

- SSLNG has the potential to revolutionize India’s gas usage landscape and serve as a significant step towards sustainability, economic growth and efficient monetization of resources. SSLNG will help in distribution of natural gas through liquefaction in new CGD areas, liquefaction of gas at isolated fields and will support setting-up of LNG fueling stations and in bunkering.
- Government of India has identified LNG as one of the new age fuels to cater to the transportation sector especially Heavy Diesel Vehicles and mining sector. GAIL has also been

exploring the possibility of using LNG in the transport sector and in other technically feasible sectors such as mining, marine and railways etc.

- As a step in this direction, GAIL has procured 02 (two) Small scale LNG skids which were first of its kind in India and successfully commissioned both the skids.
- GAIL is also exploring the possibility of small-scale manufacturing facilities in India under Atmanirbhar Bharat initiative.

• Ethane Cracker:

GAIL is exploring the possibility of setting up **1,500 KTPA Greenfield Ethane Cracker Project** for production of various type of petrochemicals.

Project Configuration:

Cracker Unit shall be designed to produce 1500 KTPA ethylene, based upon which following downstream units will be set up.

S. No.	Unit Description	Capacity
1	Cracker Unit	1500 KTPA Ethylene
2	LLDPE/HDPE Swing	650 KTA
3	Dedicated HDPE	500 KTA
4	Poly Vinyl Chloride	600 KTA
5	Chlor – Alkali (Cl ₂ basis)	360 KTA
<i>Intermediate Units like B- 1, EDC, VCM shall be set up as a part of Petrochemical complex</i>		

Location & Land required ~ 800 hectares land at Sehore, Madhya Pradesh

Feed stock requirement: Around 1800 KTA Ethane feedstock will be required, which will be imported at Dabhol terminal from US. Further Ethane shall be transferred to the petrochemical complex through a dedicated feedstock pipeline

Construction Period: 72 Months from zero date

Project Cost: ~ Rs 73,236 Cr. with Chlor Alkali unit and about Rs 66444 Cr. without Chlor Alkali unit (Inclusive of all taxes, duties, Interest during Construction (IDC) and margin money)

Incentives Required: Incentives required in the mix of Capital and Operational expenditure. Incentive amount of ~ Rs 36,500 Cr. (NPV @ ~ Rs 10, 700 Cr.) has been considered which includes Capital Grant, annual interest free loan, Annual Grant for interest payment, SGST refund and exemption on power charges & duty.

• Biofuels

GAIL is exploring opportunities in both Compressed Bio Gas (CBG) and 1G Ethanol domains.

- (i) **CBG:** GAIL has commissioned its first CBG plant of 5 Tons per Day (TPD) CBG production capacity at Ranchi.

GAIL Board has approved acquisition of equity in M/s Leafiniti Bioenergy Private Limited (LBPL), subsidiary of M/s TruAlt Bioenergy Limited (TBL), and setting up of 10 CBG plants through LBPL, subject to DIPAM approval. A proposal has been submitted for DIPAM approval. GAIL has signed MOU with VERBIO India Private Limited on 25.10.2024 for jointly exploring setting up Agri Residue based CBG projects in India, while also exploring option of equity acquisition in VERBIO's existing 33 TPD CBG plant in Punjab.

Further, GAIL has also selected a party to explore formation of a Joint Venture for setting up of Municipal Solid Waste (MSW) based CBG projects across the country.

- (ii) **Ethanol:** GAIL along with its Joint Venture (JV) partners envisage to set up First Generation (1G) 500 Kilo Liters Per Day (KLPD) capacity of grain-based ethanol plant in Rajasthan. GAIL Board has accorded investment approval for equity participation in the JV company for the project subject to approval of DIPAM. A proposal has been submitted for DIPAM approval.

City Gas Distribution (CGD)

1. How many PNG connections, CNG conversions, Industrial and Commercial connections and volumes do we see in the GAs directly or through JVs/Subsidiaries under GAIL in next five years?

GAIL along with its City Gas Distribution (CGD) entities is authorized to develop CGD networks in 72 Geographical Areas (GAs) across the country. Of these, 25 GAs were authorized in the 9th and 10th CGD Bidding Rounds, 5 GAs were authorized in the 11th and 11A CGD Bidding Rounds, and 2 GAs have been recently authorized in the 12th Bidding Round. As per the Minimum Work Program (MWP) committed to PNGRB for aforementioned 32 GAs, around 1,600 CNG stations and around 70 lakhs domestic PNG connections are to be provided by the end of FY 29-30.

2. At what % vehicles are being converted in to CNG vehicle and its market shares in new vehicles?

As per Vaahan Dashboard, around 83 lakh CNG based vehicles have been registered in the country, representing ~2.2% of the total vehicles in India.

As per PNGRB's report titled "India's Natural Gas Demand Projection for 2030-2040" published in April 2025, **CNG Four-wheeler segment is growing at a decadal rate of 12%**, while the CNG Threewheeler segment, CNG segment is expanding at 43% over the same period. Even, Twowheeler segment -CNG segment has started gaining traction following the launch of India's first CNG- powered bike.

3. Brief of GAIL's presence in CGDs across India? How many GAs are being served directly by GAIL, and how many through its JVs and subsidiaries?

In the early 1990s, GAIL launched pilot projects in Delhi and Mumbai leading to start of commercial operations of City Gas projects. Later, CGD projects in these cities were rolled out through JVCs viz. Indraprastha Gas Limited (IGL) and Mahanagar Gas Limited (MGL). The impact of these ventures was remarkable from the onset and has transformed the lives of the people of these two cities.

Based on the success of IGL and MGL, GAIL has further set up six more CGD JVCs/Subsidiaries viz. Aavantika Gas Limited (AGL), Bhagyanagar Gas Limited (BGL), Central UP Gas Limited (CUGL), Green Gas Limited (GGL), Maharashtra Natural Gas Limited (MNGL), and Tripura Natural Gas Company Limited (TNGCL) to serve GAs across various States.

GAIL Gas Limited, a wholly owned subsidiary of GAIL (India) Limited was incorporated in 2008 to develop and operate CGD networks in Uttar Pradesh, Uttarakhand, Haryana, Madhya Pradesh, Rajasthan, Jharkhand, Odisha, Karnataka, and Chhattisgarh.

Further, JVCs of GAIL Gas Limited are implementing CGD projects in various GAs across the States of Assam, Andhra Pradesh, Goa, Gujarat, Rajasthan and Uttarakhand.

Bengal Gas Company Limited (BGCL), a subsidiary of GAIL (India) Limited was incorporated in 2019 to cater to the GA of Kolkata and adjoining areas in West Bengal.

GAIL is directly serving 6 GAs viz. Varanasi, Patna, Jamshedpur, Ranchi, Bhubaneswar and Cuttack, whereas, 66 GAs are being served through its JVCs and Subsidiaries.

4. As GAIL is having direct as well as presence through its JV in CGD business, what is GAIL's strategy for CGD business in India?

While allocating domestic Natural Gas, CGD sector has been accorded highest priority by the Government of India. Considering long term growth prospects, GAIL's CGD entities have been participating in various CGD Bidding Rounds conducted by PNGRB from time to time.

PNGRB last year concluded the 12th CGD Bidding Round for the development of new CGD networks. Considering the strategic benefits, GAIL's CGD entities participated in the bidding round and won the authorizations for Manipur State and Mizoram State.

5. GAIL's Strategy for listing of GAIL Gas Ltd. and/or other JVs under CGD basket? Possibilities on integration of other City Gas JVs with GAIL Gas like MNGL, CUGL etc. Kindly brief on GAIL's Strategy for disinvestment / consolidation of entities in its CGD ventures?

To unlock the value and bring synergy in its CGD business, proposals regarding integration of GAIL's CGD entities are at various levels of discussions. Depending on the prevailing market conditions and considering shareholders' interests, decision will be taken by the Board.

GAIL at present is not interested in disinvestment in its CGD ventures. On the other hand, GAIL is focused on expanding its CGD business.

6. Kindly brief on CGD authorizations to GAIL/JVs in various bidding rounds. Further kindly brief on MWP targets for these CGDs along with timeline.

Presently, GAIL's CGD entities are authorized to develop CGD networks in 72 GAs across the country. As per the MWP commitments to PNGRB, GAIL's CGD entities have committed to achieve around 87 lakh DPNG connections and around 2,000 CNG stations by FY32.

7. What is the Capex plan & expected profitability for next 3 years of CGDs under GAIL, its subsidiaries and CGD JVs?

During FY 24-25, the Capex of GAIL's CGD entities was around Rs.4,000 crores in developing the CGD infrastructure. The Capex plan of GAIL's CGD entities for the next three years is estimated to be more than 10,000 crores.

8. Total CGD volume (PNG, CNG, Industrial, commercial) in India for previous two years & Projected volumes for FY 24-25 along with GAIL's share? Share of Domestic Gas in total CGD sales?

As per PPAC data, the total CGD volume (PNG, CNG, Industrial, Commercial) in India for FY 22-23, FY 23-24, FY 24-25 (till February) is tabulated below:

S. No.	FY	Total CGD volume (MMSCMD)	Share of Domestic Gas (MMSCMD)	Share of Domestic Gas (%)
1	FY 22-23	33.0	24.3	73.6%
2	FY 23-24	36.9	27.4	74.4%

S. No.	FY	Total volume CGD (MMSCMD)	Share Domestic Gas (MMSCMD)	Share Domestic Gas (%)
3	FY 24-25 (till Feb.)	41.0	27.4	66.7%

The average CGD volume (PNG, CNG, Industrial, commercial) for the GAs authorized to GAIL and its CGD JVCs / Subsidiaries during FY22-23, FY23-24 and FY24-25 (tentative) is tabulated below:

S. No.	FY	Average CGD volume (MMSCMD)
1	FY 22-23	17.2
2	FY 23-24	18.0
3	FY 24-25 (tentative)	19.5

9. Does electric vehicles growth and next-generation of emission norms upcoming BS-VII fuel quality norms (on gasoline and diesel) would pose threat to CNG market in long-term?

The adoption of Electric vehicles (EVs) could cut demand for CNG in the long term. However, in the near term, adoption of EVs faces multiple barriers such as price, battery life, and charging infrastructure.

10. What is the process of award of CGD License in new GAs, what are the factors considered by PNGRB while issuing the license?

The License for GAs are awarded as per the PNGRB (Authorizing Entities to Lay, Build, Operate or Expand City or Local Natural Gas Distribution Networks) Regulations 2008 and its amendments thereof. As per the recent amendments, while awarding new licenses, PNGRB follows a two-stage e-bidding process wherein at stage 1 entities are required to qualify technically before their respective Financial bids are opened in stage 2. The Financial bids are called on the basis of the following bidding parameters:

S. No.	Bidding Criteria	Weightage (%)	Explanation
1	Lowness of transportation rate for CGD – in rupees per million British Thermal Unit (Rs. / MMBTU)	10	Bidder is required to quote transportation rate for CGD only for the first contract year which shall not be less than Rs. 30 /MMBTU. Rates for the subsequent contract years shall be derived considering the quoted rate and allowed escalation.
2	Lowness of transportation rate for CNG -in rupees per kilo gram (Rs. / kg)	10	Bidder is required to quote transportation rate for CNG only for the first contract year which shall not be less than Rs. 2 /kg. Rates for the subsequent contract years shall be derived considering the quoted rate and allowed escalation.
3	Highness of number of CNG Stations (online and daughter booster stations) to be installed within 8 contract years from the date of authorization	20	-

S. No.	Bidding Criteria	Weightage (%)	Explanation
4	Highness of number of domestic piped natural gas connections to be achieved within 8 contract years from the date of authorization	50	-
5	Highness of inch-kilometer of steel pipeline (including sub-transmission steel pipelines) to be laid within 8 contract years from the date of authorization	10	-

11. What is the period for which CGD license is granted and what are the covenants for the same?

- 25 years exclusivity for laying, building, operating or expanding CGD infrastructure.
- 5/8 years of exclusivity from the purview of common carrier or contract carrier.

12. Threat on growth plans in CNG due to high Gas price scenario?

India witnessed an increase in gas prices under the APM regime to \$8.57 per MMBTU during the second half of FY 22-23. The revised prices were the highest ever since the new gas formula was announced, though much lower than the prices of imported LNG delivered to India.

Higher gas price increases input costs to CGD entities. With the rise in gas prices, CGD entities had raised the Retail Selling Price (RSP) of CNG to partially offset the impact of increased gas costs. The rise in RSP of CNG had a marginal impact on demand in the short term. In light of the revision in the formula for pricing of natural gas, as approved by the Union Cabinet in April, 2023, price of domestic gas allocated to CGD entities has softened and CNG RSP has also come down.

Government has put the CGD sector (domestic and transport) under highest category when allocating domestic Natural Gas. The sector enjoys an economic advantage when compared to alternative fuels such as Petrol and Diesel. The high prices of LNG in the international market are expected to lower in future. As such significant impact on the demand for CNG is not envisaged.

13. Does re-shifting of allocation of domestic gas from CGD sector poses threat to the CGD sector? How will it impact?

Yes, re-shifting of allocation of domestic Natural Gas from CGD sector poses a threat to the CGD sector. A reduction in allocation of domestic Natural Gas for the CGD sector would force CGD entities to increase the proportion of higher-priced imported gas, resulting in a significant jump in the retail prices. The direct correlation between CGD volume growth and the extent of domestic gas allocation is self-evident.

PNGRB has allocated new GAs in the 11th and 12th CGD Bidding Rounds, and reduction in gas allocation would mean that the authorized CGD entities would have to buy imported LNG to supply as CNG to automobiles and PNG to households.

The CGD sector is already facing some competition from Electric Vehicles (EVs). Therefore, reducing the domestic gas allocation will deter conversion of Diesel/Petrol vehicles to CNG. Though CNG is an environmentally-friendly fuel, cost economics plays a significant role in converting to CNG.

Therefore, the economic viability of the CGD sector might be impacted since high gas prices may bring the CNG at par Diesel and Petrol, eroding the incentive for users to convert vehicles to CNG.

14. Pricing outlook for CNG/PNG/in the short, medium and long term?

In recent past, India witnessed an increase in APM gas prices to \$8.57 per MMBTU, which were the highest ever since the new gas formula was announced. Since higher gas price increases input costs to CGD entities, CGD entities had raised the Retail Selling Price (RSP) of CNG and domestic PNG (DPNG) to partially offset the impact of increased gas costs. The rise in RSP of CNG and domestic PNG had a marginal impact on demand in the short term given the inherent benefits associated with these fuels.

As per the revision in the formula for pricing of natural gas, as approved by the Union Cabinet in April, 2023, price of domestic gas allocated to CGD entities has softened and the RSPs of CNG and DPNG have also come down. As per the new pricing formula, APM gas shall be capped to \$6.5 per MMBTU for a period of 2 years with an annual increment of \$0.25 per MMBTU thereafter. Industry and concerned stakeholders (including consumers) have taken Government's pricing reforms in good stead and hence, pricing outlook seems optimistic in the medium term. APM and UBP prices from 1st April, 2025 are \$6.75/MMBTU and \$6.87/MMBTU respectively in the country.

Government of India has put the CGD sector (domestic and transport sector) under highest category when allocating domestic Natural Gas. The sector enjoys an economic advantage when compared to alternative fuels such as Petrol, Diesel, LPG etc. The prevailing high prices of LNG in the international market are also expected to lower in future. As such significant impact on the demand for CNG and DPNG due to pricing is not envisaged. Nonetheless, given decreasing domestic Natural Gas production and consequent cut in gas allocation for the CNG (T) sector, prices of CNG to retail customers have increased.

15. Details in Brief about blending of Hydrogen with Natural Gas? In which sectors/CGDs, GAIL is mixing the hydrogen and what %? Further GAIL's and Government future strategies in this regard?

The Hon'ble Prime Minister of India launched the National Hydrogen Mission (NHM) in 2021. The Mission aims to aid the government in meeting its climate targets and making India a *Green Hydrogen* hub. The overarching objective of the Mission is to make India the Global Hub for production, usage and export of *Green Hydrogen* and its derivatives. The Mission will lead to significant de-carbonization of the economy, reduced dependence on fossil fuel imports, and enable India to assume technology and market leadership in the field of *Green Hydrogen*. To achieve the above objectives, the Mission aims to build capabilities to produce at least 5 Million Metric Tonne (MMT) of *Green Hydrogen* per annum by 2030, with potential to reach 10 MMT per annum with the growth of the export markets.

In line with the NHM, GAIL commissioned India's first *Green Hydrogen* production plant at Vijaipur in May, 2024. The 10 MW Proton Exchange Membrane (PEM) Electrolyzer can produce 4.3 TPD '*Green*' Hydrogen. Moreover, GAIL has also conducted a pilot project for mixing Hydrogen (5% by volume) in the PE network of Aavantika Gas Limited, Indore. The hydrogen blended Natural Gas had been supplied to various CGD consumers.

16. Physical Performance & Financial Performance of GAIL's CGDs for FY2024-25

FY2024-25	Cuttack	Jamshedpur	Ranchi	Patna	Bhubaneswar	Varanasi
FINANCIAL PERFORMANCE						
Turnover (Rs. Crores)	27.13	35.04	71.71	281.18	71.74	253.57
PBT (Rs. Crores)	(0.75)	(3.41)	(5.16)	37.87	(4.79)	23.46
PHYSICAL PERFORMANCE						
DPNG Volume (MMSCM)	0.84	0.55	0.88	2.86	1.51	4.27
Commercial PNG (MMSCM)	0.02	0.30	0.14	0.55	0.27	1.08
Industrial PNG (MMSCM)	0.34	-	0.35	0.87	1.51	3.02
CNG (MMSCM)	3.92	5.60	12.06	50.72	10.51	41.90
Total Volume (MMSCM)	5.12	6.45	13.42	55.00	13.80	50.28
Total Volume (MMSCMD)	0.01	0.02	0.04	0.15	0.04	0.14

GAIL Gas

1. List of GAs directly under GAIL Gas. Physical details of PNG and CNG Actual vs targets in each GA for last 2 years.

S.No.	GA Name (GAIL Gas)	Overall Physical Progress as on 31.03.2025
1	Sonepat	89.42
2	Meerut	84.47
3	Dewas	87.04
4	TTZ	100
5	Bengaluru Rural and Urban Districts	100
6	Dakshin Kannada District	25.29
7	Dehradun District	33.08
8	Dhanbad & Giridih Districts	45.85
9	Puri, Ganjam & Nayagarh Districts	60.14
10	Sundargarh & Jharsuguda Districts	38.29
11	Seraikela-Kharaswan District	32.06
12	West Singhbhum District	14.02
13	Raisen, Sehore & Sajahpur Districts	24.67
14	Mirzapur, Chandauli & Sonbhadra Districts	59.25
15	Gajapati, Kandhamal, Boudh and Sonepur Districts (11th Bidding Round)	3.4
16	Kondagaon, Bastar, Sukma, Narayanpur, Bijapur and Dantewada Districts (11 A Bidding Round)	1

Note: Overall physical progress is calculated for achievement against MWP targets for DPNG Connections, CNG Stations and P/I (Inch-km).

2. GA wise profitability of GAIL Gas for last 3 years and future outlook. Which GAs you see as most emerging GA in terms of growth?

Period	9001	9003	9004	9018	Bulk	TTZ	9th Round	10th Round	11th Round	Other GA's	HO	Total
FY 24-25	Dewas	Meerut	Sonepat	Bengaluru								
Physical Performance (Quantity Data (MMSCM))												
Industrial excluding TTZ	10.95	4.33	39.25	105.10	-	0.10	2.51	13.78	-	-	-	176.02
Industrial TTZ	-	-	-	-	-	463.93	-	-	-	-	-	463.93
Commercial	0.22	0.87	1.36	7.47	-	1.18	0.11	0.08	-	-	-	11.30
Domestic	3.15	5.84	3.27	13.04	-	1.79	0.21	0.20	-	-	-	27.49
PNG Sales (MMSCM)	14.32	11.04	43.89	125.60	-	467.00	2.83	14.07	-	-	-	678.74
CNG Sales (MMSCM)	8.40	59.44	56.02	155.95	-	61.86	28.09	25.57	0.03	-	-	395.35
Bulk Sales (MMSCM)	-	-	-	-	1,595.47	-	-	-	-	-	-	1,595.47
Total Quantity (MMSCM)	22.73	70.48	99.90	281.55	1,595.47	528.86	30.91	39.64	0.03	-	-	2,669.56

Period	9001	9003	9004	9018	Bulk	TTZ	9th Round	10th Round	11th Round	Other GA's	HO	Total
FY 24-25	Dewas	Meerut	Sonepat	Bengaluru								
Total Quantity (MMSCMD)	0.06	0.19	0.27	0.77	4.37	1.45	0.08	0.11	0.00			7.31
Turnover (Cr.)												
- Industrial excluding TTZ	61.05	26.08	221.82	580.24	-	0.57	13.17	73.76	-	-	-	976.69
- Industrial TTZ	-	-	-	-	-	1,498.88	-	-	-	-	-	1,498.88
- Commercial	1.15	5.21	7.65	45.52	-	6.07	0.57	0.45	-	-	-	66.62
- Domestic	12.47	25.00	14.79	49.64	-	7.43	0.98	0.92	-	-	-	111.24
PNG Sales	74.68	56.30	244.26	675.41	-	1,512.95	14.72	75.12	-	-	-	2,653.43
CNG Sales	43.65	277.14	283.78	801.98	-	330.04	151.80	128.62	0.12	-	-	2,017.14
Bulk Sales	-	-	-	-	7,263.92	-	-	-	-	-	-	7,263.92
Total Turnover (Rs. Crs)	118.33	333.44	528.04	1,477.39	7,263.92	1,842.99	166.51	203.74	0.12	-	-	11,934.49
Average price (Rs./SCM)												
- Industrial excluding TTZ	55.76	60.24	56.51	55.21	-	56.82	52.51	53.51	-	-	-	55.49
- Industrial TTZ	-	-	-	-	-	32.31	-	-	-	-	-	32.31
- Commercial	52.85	59.88	56.17	60.93	-	51.26	51.79	53.97	-	-	-	58.97
- Domestic	39.55	42.84	45.22	38.08	-	41.64	47.17	45.64	-	-	-	40.46
Average Price - PNG (Rs/SCM)	52.14	51.01	55.66	53.77	-	32.40	52.09	53.40	-	-	-	39.09
Average Price - CNG (Rs/SCM)	51.95	46.62	50.66	51.43	-	53.35	54.04	50.30	49.87	-	-	51.02
Average Price - BULK (Rs/SCM)	-	-	-	-	45.53	-	-	-	-	-	-	45.53
Financial Information												
Revenue from Operations - Sales	124.44	372.07	567.68	1,590.57	7,263.92	1,886.94	185.45	220.23	0.14	-	-	12,211.44
Revenue from Operations - Others	2.25	0.85	0.49	4.99	0.01	3.36	6.96	0.07	-	0.36	0.00	19.33
Interest Income	-	-	-	0.03	-	-	-	-	-	-	34.10	34.13
Other Income	0.28	0.54	0.51	0.57	-	0.25	12.01	0.05	-	-	1.60	15.81
Total Income	126.97	373.46	568.69	1,596.16	7,263.92	1,890.55	204.41	220.35	0.14	0.36	35.70	12,280.71
Gas Consumed except gas loss	74.88	231.99	350.83	1,001.27	7,181.01	1,531.44	124.90	160.70	0.22	0.00	-	10,657.23
Gas Loss	0.94	5.11	6.68	(3.80)	-	(3.80)	2.22	2.84	0.00	-	-	10.20
Gas Cost	75.83	237.10	357.50	997.46	7,181.01	1,527.64	127.12	163.54	0.22	0.00	-	10,667.42
Excise Duty	8.27	38.67	39.65	113.46	-	44.25	20.74	15.40	-	-	(0.00)	280.44
Employee Benefit Expenses	2.04	3.81	2.51	17.78	-	14.12	5.70	4.69	0.33	-	21.75	72.73
Consultancy & Legal Charges	0.05	0.03	0.16	0.19	0.03	0.09	0.06	0.02	0.00	-	2.07	2.68
Electricity Charges	0.27	0.68	0.47	7.63	-	2.48	2.49	2.04	0.02	0.00	0.52	16.61
Rates & Taxes	0.02	0.09	0.01	0.23	0.00	0.06	0.18	0.01	0.00	-	0.01	0.61
Rent-Office & Others	0.20	1.97	0.66	0.38	0.00	0.46	1.14	0.05	0.05	-	5.79	10.69
Repairs and Maintenance	7.09	24.21	22.80	41.30	-	23.06	23.92	18.90	0.22	0.61	0.04	162.16
Security Expenses	0.07	1.07	0.15	2.13	-	1.50	1.55	0.88	0.03	0.17	(0.00)	7.55
Stores & Spares Consumed	0.12	1.13	1.27	1.59	-	1.40	0.13	0.01	-	-	0.15	5.79
Vehicle Hire & Running Expenses	0.66	0.99	0.94	6.36	-	6.36	4.05	7.19	0.09	-	0.42	27.06
Fuel Expenses	1.95	8.32	8.42	14.65	-	7.30	5.14	2.18	-	-	-	47.94
Insurance Charges	-	-	-	0.01	-	0.00	-	-	-	-	14.01	14.03
Other Expenses	2.71	7.75	10.20	77.29	0.03	8.99	5.69	3.50	0.81	0.00	33.51	150.49
EBIDTA	27.69	47.63	123.96	315.69	82.86	252.84	6.52	1.95	(1.63)	(0.43)	(42.58)	814.49
Finance Cost	0.18	0.02	0.24	16.97	0.00	3.46	19.72	0.98	0.09	-	7.28	48.94

Period	9001	9003	9004	9018	Bulk	TTZ	9th	10th	11th	Other	HO	Total
FY 24-25	Dewas	Meerut	Sonepat	Bengaluru			Round	Round	Round	GA's		
Depreciation and Amortization Expenses	5.99	11.10	12.01	56.02	0.02	19.02	26.88	18.76	0.12	0.29	0.47	150.68
Total Expenses	105.45	336.95	456.99	1,353.46	7,181.09	1,660.19	244.49	238.14	1.98	1.08	86.03	11,665.84
PBT before Corporate Allocation	21.52	36.51	111.71	242.70	82.84	230.36	(40.08)	(17.79)	(1.84)	(0.72)	(50.33)	614.87
Corporate Expenses Allocation	2.16	6.47	9.87	27.66	-	32.81	3.22	3.83	0.00	-	(86.03)	0.00
PBT after Corporate Allocation	19.35	30.04	101.83	215.04	82.84	197.55	(43.31)	(21.62)	(1.84)	(0.72)	35.70	614.87

For Period 2023-24:

Particulars	Dewas	Meerut	Sonepat	Bengaluru	Bulk Trading	TTZ	9th Round GAs	10th Round GAs	CO	Total
Physical Performance										
Quantity Data (MMSCM)										
Industrial excl. TTZ	10	6	37	92	-	-	1	9	-	157
Industrial TTZ	-	-	-	-	-	466	-	-	-	466
Commercial	0	1	1	7	-	1	0	0	-	10
Domestic	3	5	3	8	-	1	0	0	-	20
PNG Sales (MMSCM)	13	12	41	107	-	468	2	9	-	652
CNG Sales (MMSCM)	7	48	47	83	-	53	15	13	-	267
Bulk Sales (MMSCM)	-	-	-	-	1,447	-	-	-	-	1,447
Total Quantity (MMSCM)	20	60	88	190	1,447	521	17	22		2366
Total Quantity (MMSCMD)	0.05	0.16	0.24	0.52	3.95	1.42	0.05	0.06	-	6.46
Financial Information										
Revenue from Operations - Sales	115	328	530	1,088	6,786	1,841	100	135	-	10,925
Revenue from Operations - Others	3	1	1	5	-	2	5	0	0	19
Interest Income	-	-	-	-	-	-	-	-	14	14
Other Income	0	0	0	1	-	1	0	0	1	4
Total Income	118	330	531	1,094	6,786	1,844	106	136	15	10,962
Gas Cost	71	206	317	724	6,717	1,480	74	112	-	9,701
EBIDTA	23	47	140	174	69	249	(6)	(7)	(97)	593
Finance Cost	0	0	0	16	0	2	10	1	4	33
Depreciation and Amortization Expenses	6	11	12	52	0	18	15	12	1	126
Total Expenses	101	294	403	988	6,718	1,615	136	156	117	10,529
PBT before Corporate Allocation	17	35	128	106	69	229	(30)	(20)	(102)	434
Corporate Expenses Allocation	3	9	15	31	-	52	3	4	(117)	-
PBT after Corporate Allocation	14	26	113	75	69	177	(33)	(24)	15	434

Period 2022-23:

Particulars	Dewas	Meerut	Sonepat	Bengaluru	Bulk Trading	TTZ	9th Round GAs	10th Round GAs	CO	Total
Physical Performance										
Quantity Data (MMSCM)										
Industrial excl. TTZ	11	6	35	103	-	-	0	7	-	162
Industrial TTZ	-	-	-	-	-	472	-	-	-	472
Commercial	0	0	1	5	-	1	0	0	-	8
Domestic	3	3	3	6	-	1	0	0	-	17
PNG Sales (MMSCM)	13	10	39	115	-	474	0	7	-	658
CNG Sales (MMSCM)	6	45	46	39	-	49	10	8	-	203
Bulk Sales (MMSCM)	-	-	-	-	1,214	-	-	-	-	1,214
Total Quantity (MMSCM)	19	55	85	154	1214	523	10	15	0	2075
Total Quantity (MMSCMD)	0.05	0.15	0.23	0.42	3.33	1.43	0.03	0.04	-	5.68
Financial Information										
Revenue from Operations - Sales	107	323	526	952	6,511	1,946	61	87	-	10,513
Revenue from Operations - Others	3	1	5	4	-	1	2	0	0	18
Interest Income	-	-	-	0	-	-	-	-	10	10
Other Income	0	0	1	2	-	0	0	0	1	5
Total Income	110	324	531	957	6,511	1,947	64	87	11	10,545
Gas Cost	76	216	354	702	6,456	1,604	54	88	0	9,552
EBIDTA	17	41	110	144	54	256	(26)	(19)	(58)	519
Finance Cost	0	0	0	15	-	1	0	0	7	24
Depreciation and Amortization Expenses	4	9	10	45	0	13	7	6	1	95
Total Expenses	98	292	431	874	6,457	1,705	97	112	78	10,145
PBT before Corporate Allocation	12	32	100	83	54	242	(33)	(25)	(67)	400
Corporate Expenses Allocation	2	6	10	18	-	38	1	2	(78)	-
PBT after Corporate Allocation	10	26	90	65	54	204	(34)	(27)	11	400

3. Listing of GAIL Gas and expected timeline?

Tentative timeline is one year (subject to approval)

4. List of GAs through subsidiary/JV wise of GAIL Gas. Physical & financial details of each CGD entity for last 5 years.

List of GAs through Subsidiary/JVs

S.No.	Name of Joint Venture Company	GA Name
1	VGL	Vadodara District
2	APGDC	East Godavari District
3		West Godavari District

S.No.	Name of Joint Venture Company	GA Name
4	RSGL	Kota
5		Sheopur and Gwalior (Except Area Already Authorized) Districts
6	HNGPL	Haridwar District
7	GNGPL	North Goa District
8	PBGPL	Cachar, Hailakandi & Karimganj Districts
9		Kamrup & Kamrup Metropolitan Districts

Physical Details of GAs through Subsidiary/JV:

Name of JV	VGL - VADODARA	HNGPL - Haridwar	RSGL - KOTA	GNGPL - Goa	GGPL - APGDC	PBGPL - Assam
DPNG (No.)						
Achieved in FY2020-21	30,334	15,411	6,382	2,106	47,711	285
As on 31.03.2021	1,78,119	20,946	16,758	5,949	1,00,232	285
Achieved in FY2021-22	19273	3721	2325	3915	675	55
As on 31.03.2022	1,97,392	24,667	19,083	9,864	1,00,907	340
Target for FY 2022-23	25,000	4,000	12,000	2,000	15,000	4,000
Achievement in FY 2022-23	24,261	642	3,006	500	0	373
As on 31.03.2023	2,21,653	25,309	21,789	10,364	1,00,907	713
Target for FY 2023-24	20,000	2,500	12,000	1,000	15,000	8,750
Achievement in FY 2023-24	24,072	8	12,544	600	0	4,264
As on 31.03.2024	2,45,725	25,317	34,333	10,964	1,00,907	4,977
Achievement in FY 2024-25	25,932	130	20,518	688	0	4,829
As on 31.03.2025	2,71,657	25,447	54,851	11,652	1,00,907	9,806
CNG STATIONS (No.)						
Achieved in FY2020-21	8	3	3	3	8	0
As on 31.03.2021	27	6	11	6	31	0
Achieved in FY2021-22	8	0	0	1	1	0
As on 31.03.2022	34	6	11	6	32	0
Target for FY 2022-23	6	3	2	4	3	10
Achievement in FY 2022-23	6	1	1	1	1	5
As on 31.03.2023	40	7	12	7	33	5
Target for FY 2023-24	6	2	4	3	2	17
Achievement in FY 2023-24	6	1	3	2	0	8
As on 31.03.2024	46	8	15	9	33	13
Achievement in FY 2024-25	6	2	3	1	0	4
As on 31.03.2025	52	10	18	10	33	17
Pipeline Network Infrastructure (in Kms.)						
Steel Pipe length in km as on 31.03.2021	54.55	41.92	40.12	37.33	93.05	3.55
MDPE/ Thermoplastic pipes length in km as on 31.03.2021	2,790.50	480.85	135.92	111.49	619.66	23.14
Total network length in km as on 31.03.2021	2,845.05	522.77	176.04	148.82	712.70	26.69
Steel Pipe length in km as on 31.03.2022	57.62	35.62	40.39	43.77	97.44	3.55

Name of JV	VGL - VADODARA	HNGPL - Haridwar	RSGL - KOTA	GNGPL - Goa	GGPL - APGDC	PBGPL - Assam
MDPE/ Thermoplastic pipes length in km as on 31.03.2022	3,181.50	689.30	185.54	117.16	622.67	30.16
Total network length in km as on 31.03.2022	3,239.12	724.92	225.93	160.93	720.11	33.71
Steel Pipe length in km as on 31.03.2023	59.12	35.62	40.39	44.30	110.27	19.15
MDPE/ Thermoplastic pipes length in km as on 31.03.2023	3,516.26	695.82	251.62	124.76	639.79	48.19
Total network length in km as on 31.03.2023	3,575.39	731.44	292.01	169.06	750.05	67.34
Steel Pipe length in km as on 31.03.2024	62.34	35.65	40.46	49.83	111.80	58.84
MDPE/ Thermoplastic pipes length in km as on 31.03.2024	3,800.58	716.61	271.16	173.32	642.22	586.38
Total network length in km as on 31.03.2024	3,862.92	752.26	311.62	223.15	754.01	645.22
Steel Pipe length in km as on 31.03.2025	65.50	35.65	40.56	55.23	112.28	87.26
MDPE/ Thermoplastic pipes length in km as on 31.03.2025	4,213.02	721.00	314.68	182.27	648.12	1,533.37
Total network length in km as on 31.03.2025	4,278.52	756.65	355.24	237.50	760.40	1,620.63

Financial Details of GAs through Subsidiary/JVs (in Rs. Crores):

As on 31st March 2025: (₹ crores)

Particulars	APGDC	VGL	RSGL	HNGPL	GNGPL	PBG
	31st Mar'2025	31st Mar'2025	31st Mar'2025	31st Mar'2025	31st Mar'2025	31st Mar'2025
Total Profit/Loss of JV Company	-13.00	36.66	6.56	7.93	3.84	-13.15
Percentage Holding	50.00%	17.07%	50.00%	50.00%	50.00%	26.00%
Share of profit/ (loss) for the year:	-6.50	6.26	3.28	3.97	1.92	-3.42

(₹ crores)

Description	APGDC	VGL	RSGL	HNGPL	GNGPL	PBGPL
Summary of Balance Sheet						
Assets						
Non-Current	655.15	523.05	183.60	201.85	215.22	423.81
Current	26.65	123.54	43.83	18.42	11.30	34.56
Total	681.81	646.59	227.44	220.27	226.52	458.37
Equity	182.04	341.31	179.37	102.07	82.35	140.49
Liabilities & Provisions						
Non-Current	321.38	111.87	12.23	73.87	72.39	281.46
Current	178.38	193.41	35.84	44.32	71.78	36.42
Total	681.81	646.59	227.44	220.27	226.52	458.37

As on 31st March 2024: (₹ crores)

Particulars	APGDC	VGL	RSGL	HNGPL	GNGPL	PBG
	31st Mar'2024	31st Mar'2024	31st Mar'2024	31st Mar'2024	31st Mar'2024	31st Mar'2024
Total Profit/Loss of JV Company	-32.97	34.86	4.53	5.96	1.20	-11.68
Percentage Holding	50.00%	17.07%	50.00%	50.00%	50.00%	26.00%
Share of profit/ (loss) for the year:	-16.48	5.95	2.27	2.98	0.60	-3.04

(₹ crores)

Description	APGDC	VGL	RSGL	HNGPL	GNGPL	PBGPL
Summary of Balance Sheet						
Assets						
Non-Current	641.16	445.76	177.14	197.04	201.66	285.50
Current	29.14	107.31	42.44	20.43	8.18	40.90
Total	670.30	553.07	219.59	217.47	209.83	326.40
Equity	165.96	306.72	177.33	94.14	78.51	122.39
Liabilities & Provisions						
Non-Current	323.78	76.10	12.63	86.29	73.62	160.12
Current	180.56	170.24	29.63	37.04	57.71	43.89
Total	670.30	553.07	219.59	217.47	209.84	326.40

As on 31st March 2023: (₹ crores)

Particulars	APGDC	VGL	RSGL	HNGPL	GNGPL	PBG
	31st Mar'2023	31st Mar'2023	31st Mar'2023	31st Mar'2023	31st Mar'2023	31st Mar'2023
Total Profit/Loss of JV Company	-35.37	-10.51	8.83	2.2724	1.78	-3.24
Percentage Holding	50.00%	17.07%	50.00%	50.00%	50.00%	26.00%
Share of profit/ (loss) for the year:	-17.68	-1.79	4.42	1.14	0.89	-0.84

(₹ crores)

Description	APGDC	VGL	RSGL	HNGPL	GNGPL	PBGPL
Summary of Balance Sheet						
Assets						
Non-Current	616.42	423.29	160.99	192.29	188.82	149.84
Current	39.26	93.95	39.23	37.35	14.35	33.26
Total	655.68	517.24	200.21	229.64	203.17	183.10
Equity	195.39	271.87	165.93	88.18	77.31	119.07
Liabilities & Provisions						
Non-Current	322.96	69.73	9.87	97.88	78.60	4.97
Current	137.33	175.64	24.41	43.58	47.26	59.06

Description	APGDC	VGL	RSGL	HNGPL	GNGPL	PBGPL
Total	655.68	517.24	200.21	229.64	203.17	183.10

As on 31st March 2022: (₹ crores)

Particulars	APGDC	VGL	RSGL	HNGPL	GNGPL	PBG
	31st Mar'2022	31st Mar'2022	31st Mar'2022	31st Mar'2022	31st Mar'2022	31st Mar'2022
Total Profit/Loss of JV Company	-0.02	16.00	8.13	3.3428	0.02	1.16
Percentage Holding	50.00%	17.07%	50.00%	50.00%	50.00%	26.00%
Share of profit/ (loss) for the year:	-0.01	2.73	4.07	1.67	0.01	0.30

(₹ crores)

Description	APGDC	VGL	RSGL	HNGPL	GNGPL	PBGPL
Summary of Balance Sheet						
Assets						
Non-Current	573.15	381.46	138.24	188.63	161.44	69.19
Current	45.11	83.29	48.17	10.04	3.75	36.94
Total	618.27	464.75	186.42	198.66	165.19	106.13
Equity	296.08	282.38	157.75	43.14	55.53	100.40
Liabilities & Provisions						
Non-Current	177.42	67.96	10.53	113.50	84.48	0.54
Current	144.77	114.41	18.14	42.02	25.17	5.19
Total	618.27	464.75	186.42	198.66	165.19	106.13

As on 31st March 2021:

(₹ crores)

Particulars	APGDC	VGL	RSGL	HNGPL	GNGPL	PBG
	31st Mar'2021	31st Mar'2021	31st Mar'2021	31st Mar'2021	31st Mar'2021	31st Mar'2021
Total Profit/Loss of JV Company	(5.47)	2.8214	11.85	0.6737	-0.09	1.36
Percentage Holding	50.00%	17.07%	50.00%	50.00%	50.00%	26.00%
Share of profit/ (loss) for the year:	-2.73	0.48	5.93	0.34	-0.04	0.35

(₹ crores)

Description	APGDC	VGL	RSGL	HNGPL	GNGPL	PBGPL
Summary of Balance Sheet						
Assets						
Non-Current	539.34	340.97	127.01	161.37	135.72	14.44
Current	46.02	66.78	47.24	28.57	4.51	87.22
Total	585.36	407.74	174.25	189.94	140.23	101.66
Equity	190.62	266.38	150.26	40.04	48.09	99.24

Description	APGDC	VGL	RSGL	HNGPL	GNGPL	PBGPL
Liabilities & Provisions						
Non-Current	237.63	76.73	9.05	129.33	81.59	
Current	157.11	64.63	14.94	20.56	10.54	2.42
Total	585.36	407.74	174.25	189.94	140.23	101.66

5. Detailed CAPEX plans for next 5 years along with source of funding. Actuals for previous two years (in Rs/Crores).

CAPEX	Actuals		BE 25-26	Projections			
Particulars	FY 23-24	FY 24-25	FY 25-26	FY 26-27	FY 27-28	FY 28-29	FY 29-30
SMDT	99	86	137	96	67	47	33
Bangaluru	103	80	126	95	71	53	40
9th Round	263	237	256	269	282	296	311
10th Round	123	72	94	103	114	125	138
11th Round	0	14	41	45	50	55	60
Others / Operational CAPEX	10	5	145	75	75	75	75
Total Capex Requirement	598	494	799	683	659	651	657
Sources of Funds							
Internal accruals (including Equity contribution)	248	54	359	683	659	651	657
Long Term Loan	302	440	440	-	-	-	-

6. Key financial ratios of GAIL Gas & its Subsidiaries/JVs for previous three years and expected CAGR for next 3 years against each of them.

- Revenue
- Operating income
- PBT
- PAT
- Net-worth
- Long term debt
- Short term debt
- Working Capital

Particulars	FY 22-23	FY 23-24	FY 24-25	FY 25-26	FY 26-27	FY 27-28
Revenue	10394	10762	11934	14597	15758	17089
CAGR%	53%	4%	11%	22%	8%	8%
EBITDA	519	593	814	948	1121	1337
CAGR%	8%	14%	37%	16%	18%	19%
PBT	400	434	615	724	862	1043
CAGR%	4%	8%	42%	18%	19%	21%
PAT	297	323	451	538	640	775

Particulars	FY 22-23	FY 23-24	FY 24-25	FY 25-26	FY 26-27	FY 27-28
CAGR%	5%	9%	39%	19%	19%	21%
Net-worth	2953	3143	3604	3969	4441	5024
CAGR%	22%	6%	15%	10%	12%	13%
Long term debt	933	1189	1505	1745	1545	1345
Short term debt	-	-	-	-	-	-
Working Capital	-	-	-	-	-	-

**7. Share of retails customers (CNG/PNG) and share of bulk customers in total revenue.
Details of big industrial customers and expected new tied up in near future.**

Particulars	2023-24	2024-25
Total Sales	10,944.11	12,230.78
CNG/PNG	4,417.25	4,966.86
Bulk	6,526.86	7,263.92
% share of CNG/PNG Customers	40%	41%
% share of Bulk Customers	60%	59%

8. Product wise (CNG/PNG/Industrial) profitability for previous three years and expected CAGR for next 3 years

PARTICULARS	FY 22-23	FY 23-24	FY 24-25	FY 25-26	FY 26-27	FY 27-28
Revenue (Net):	₹ Crore	₹ Crore	₹ Crore	₹ Crore	₹ Crore	₹ Crore
CGD (PNG & CNG)	2240	2651	3468	4527	5264	6149
TTZ_UPM	1643	1507	1499	1583	1583	1583
Natural Gas Trading	6511	6786	7264	8487	8911	9357
Total	10,394	10,944	12,231	14,597	15,758	17,089
EBITDA:						
CGD (PNG & CNG)	332	393	599	724	893	-233
TTZ_UPM	133	131	133	137	137	137
Natural Gas Trading	54	69	83	87	91	96
Total	519	593	815	948	1121	0
PBT:						
CGD (PNG & CNG)	215	236	399	502	636	812
TTZ_UPM	131	129	133	135	135	135
Natural Gas Trading	54	69	83	87	91	96
Total	400	434	615	724	862	1043
PAT:						
Total	297	323	451	538	640	775

9. GA wise total no. of PNG connections, CNG stations, CNG vehicle conversions added during previous three years and year wise projection for addition in next 3 years.

S.No.	GA Name	DPNG Connections						
		Target for FY 2022-23	Achieved in FY 2022-23	Target for FY 2023-24	Achieved in FY 2023-24	Target for FY 2024-25	Achieved in FY 2024-25	Cumulative till Mar'25
1	Sonepat	8,000	416	15,000	3,500	11,400	5,669	38,830

S.No.	GA Name	DPNG Connections						
		Target for FY 2022-23	Achieved in FY 2022-23	Target for FY 2023-24	Achieved in FY 2023-24	Target for FY 2024-25	Achieved in FY 2024-25	Cumulative till Mar'25
2	Meerut	20,000	585	20,000	3,385	16,200	6,669	63,155
3	Dewas	5,000	2,022	10,000	3,912	8,300	788	22,738
4	TTZ	6,000	0	10,000	3,522	5,200	1,863	24,563
5	Bengaluru	20,000	34,818	50,000	37,008	12,500	35,984	3,33,574
6	Dakshin Kannada	25,000	7,376	20,000	7,682	14,200	10,000	45,387
7	Dehradun	21,000	4,192	20,000	3,069	12,200	4,465	35,272
8	Dhanbad-Giridih	4,000	2,548	10,000	56	5,200	162	12,716
9	Puri-Ganjam-Nayagarh	4,000	668	10,000	1,634	3,100	741	8,303
10	Sundargarh-Jharsuguda (Rourkela)	7,000	505	10,000	2,603	5,700	6,524	16,729
11	Seraikela-Kharaswan	15,000	1,392	10,000	0	5,200	331	13,839
12	West Singbhum	600	125	2,000	0	2,000	0	125
13	Raisen-Sehore-Sajahpur	40,000	1,876	20,000	1,579	3,900	2,915	7,502
14	Mirzapur-Chandauli-Sonbhadra	15,000	1,769	20,000	745	15,300	1,933	9,589
15	Gajapati,Kandhamal, Boudh and Sonapur districts	-	-	3000	0	3,790	57	57
16	Kondagaon, Bastar, Sukma, Narayanpur, Bijapur and Dantewada	-	-	1000	0	6060	0	0
	Total	1,90,600	58,292	2,31,000	68,695	1,30,250	78,101	6,32,379

S.No.	GA Name	CNG Stations						
		Target FY 2022-23	Achieved in FY 2022-23	Target for FY 2023-24	Achieved in FY 2023-24	Target for FY 2024-25	Achieved in FY 2024-25	Cumulative till Mar'25
1	Sonepat	8	2	10	5	5	7	38
2	Meerut	5	0	5	3	5	4	26
3	Dewas	2	3	5	0	3	2	8
4	TTZ	15	6	15	5	5	6	53
5	Bengaluru	30	40	45	44	38	38	191
6	Dakshin Kannada	10	8	12	6	10	11	52
7	Dehradun	8	7	10	1	6	4	25

8	Dhanbad-Giridih	5	4	7	3	6	3	18
9	Puri-Ganjam-Nayagarh	4	1	5	3	5	4	13
10	Sundargarh-Jharsuguda (Rourkela)	8	4	6	4	4	1	13
11	Seraikela-Kharaswan	10	7	10	2	4	6	24
12	West Singbhum	1	2	2	0	1	0	2
13	Raisen-Sehore-Sajahpur	15	9	15	7	5	5	39
14	Mirzapur-Chandauli-Sonbhadra	12	9	10	8	5	6	38
15	Gajapati,Kandhamal, Boudh and Sonapur districts	3	0	5	3	5	4	7
16	Kondagaon,Bastar, Sukma, Narayanpur, Bijapur and Dantewada districts	3	0	5	0	4	3	3
	Total	140	102	167	94	111	104	**521

* * The total 521 CNG Stations of GAIL Gas exclude 29 numbers of upgraded CNG Stations.

* Apart from that 120 CNG stations are mechanically completed.

S.No.	GA Name	Targets for FY 2025-26 to 2027-28					
		DPNG			CNG Stations		
		Target for FY 2025-26 (P)	Target for FY 2026-27 (P)	Target for FY 2027-28 (P)	Target for FY 2025-26 (P)	Target for FY 2026-27 (P)	Target for FY 2027-28 (P)
1	Sonepat	14000	15000	15000	6	6	7
2	Meerut	16000	18000	18000	6	6	7
3	Dewas	5000	6000	6000	3	4	4
4	TTZ	5000	5500	6000	6	6	7
5	Bengaluru	30000	35000	40000	40	45	52
6	Dakshin Kannada	12000	13000	14000	10	13	14
7	Dehradun	15000	16500	18000	7	8	8
8	Dhanbad-Giridih	5000	5500	6000	7	7	8
9	Puri-Ganjam-Nayagarh	3000	3500	4000	6	6	7
10	Sundargarh-Jharsuguda (Rourkela)	5500	6000	6500	5	5	5
11	Seraikela-Kharaswan	7000	7500	8000	4	5	5
12	West Singbhum	600	700	700	1	1	1

S.No.	GA Name	Targets for FY 2025-26 to 2027-28					
		DPNG			CNG Stations		
		Target for FY 2025-26 (P)	Target for FY 2026-27 (P)	Target for FY 2027-28 (P)	Target for FY 2025-26 (P)	Target for FY 2026-27 (P)	Target for FY 2027-28 (P)
13	Raisen-Sehore-Sajahpur	4000	4500	5000	6	6	7
14	Mirzapur-Chandauli-Sonbhadra	15000	16500	18000	6	6	7
15	Gajapati,Kandhamal, Boudh and Sonepur districts	4000	4500	5000	6	6	7
16	Kondagaon,Bastar,Sukma, Narayanpur, Bijapur and Dantewada districts	5000	5500	6000	4	5	5
	Total	146100	163200	176200	120	135	150

10. Does GAIL Gas have plans to acquire other GAs? Further, kindly provide brief if GAIL Gas has any plans for consolidation of its GAs.

GAIL Gas is planning to acquire Gwalior & Sheopur GA (GA 10.23) from M/s Rajasthan State Gas Limited after approval of the PNGRB board. No plan for consolidation of its GA.

11. Kindly brief about adverse impact on profitability due to deallocation of APM Gas. How does GAIL Gas plan to regain the lost profit?

Deallocation of APM Gas to priority sector of GAIL Gas has increased cost of goods sold and accordingly same has been reflecting through increase in RSP of CNG in H2 of FY 2024-25. However, entire increase in cost of goods sold due to deallocation of APM gas can't be passed thru to customers and part of the same is being done in phased manner. Further, GAIL Gas is trying to address the impact on profitability through following measures:

- Entire New Well Gas (NWG) offered to GAIL Gas in lieu of reduction in APM allocation has been purchased. NWG is slightly higher priced than APM gas but it is cheaper than RLNG.
- For priority sector, GAIL Gas regularly participates in HPHT gas bidding at IGX platform and arrange for cheaper domestic HPHT gas for CNG segment. HPHT gas is competitively priced as per MoPNG guidelines and currently it is cheaper than the imported RLNG.
- GAIL Gas has also entered into competitively price mid-term RLNG purchase agreements with GAIL which will help in rebalancing sourcing portfolio in cost effective manner
- GAIL Gas is also trying to implement more volume centric business approach and hence efforts are being made to increase volume so that our total margins remains intact although per unit margin is reduced.

12. Kindly provide the unit economics for D-PNG and CNG i.e. the breakup of various costs from purchase of Gas till last mile delivery. Further, kindly brief about the average cost of installation per D-PNG connection.

Following are the main elements of Cost Break Up from Purchase of Gas till last mile delivery of Product.

Sl. No.	Particular	Particular
	DPNG	CNG
01	Gas Cost	Gas Cost
02	Marketing Margin Charged by Supplier	Marketing Margin Charged by Supplier
03	Tax on Purchase	Tax on Purchase
04	Transmission Charges on Purchased Gas	Transmission Charges on Purchased Gas
05	GST on Transmission Charges	GST on Transmission Charges
06	Supply and Distribution Cost of CGD entity	Supply and Distribution Cost of CGD entity
07	Tax on sales	Excise duty
08	Nil	Tax on sales

CNG and DPNG Cost Break Up for GAIL Gas GAs in April 2025:

CNG Floor Price [COST PLUS]	UOM	Dewas	Meerut	Sonepat	TTZ UP Part	TTZ Raj. Part	Bengaluru	Dehradun	DK	Mandi deep	MCS	Dhanbad	Rourkela	Puri	Adityapur
APM	Rs/MMBTU	214	196	204	188	166	165	202	196	176	204	235	496	218	204
NAPM	Rs/MMBTU	194	224	234	224	198	145	183	177	159	182	212	447	197	184
HPHT	Rs/MMBTU	-	389	401	-	-	357	-	-	-	245	-	-	-	-
NWG	Rs/MMBTU	-	105	44	94	83	49	-	-	-	-	-	-	-	-
PF RLNG	Rs/MMBTU	511	-	-	200	184	205	211	-	-	354	653	-	-	484
LT-RLNG	Rs/MMBTU	-	-	-	-	-	-	-	-	-	-	-	-	-	313
GORGON	Rs/MMBTU	-	-	-	-	-	-	-	601	-	-	-	-	-	-
HH	Rs/MMBTU	-	-	-	274	278	24	390	-	663	-	30	-	730	-
Gas Cost at CGS	Rs/MMBTU	918	914	883	981	909	946	987	975	998	985	1,130	942	1,145	1,185
Non Gas Cost	Rs/MMBTU	371	371	371	371	371	371	371	371	371	371	371	371	371	371
Total Gas Cost	Rs/MMBTU	1,289	1,284	1,254	1,352	1,280	1,317	1,358	1,345	1,369	1,356	1,501	1,313	1,516	1,556
GGL Marketing Margin	Rs/MMBTU	1.29	1.28	1.25	1.35	1.28	1.32	1.36	1.35	1.37	1.36	1.50	1.31	1.52	1.56
CNG Floor Price [Cost Plus]	Rs/MMBTU	1,291	1,286	1,255	1,353	1,282	1,318	1,359	1,347	1,370	1,357	1,503	1,315	1,517	1,558
CNG Floor Price [Cost Plus]	Rs/SCM	51	50	49	53	50	52	53	53	54	53	59	52	59	61
CNG Floor Price [Cost Plus]	Rs/Kg	68.05	67.79	66.19	71.34	67.57	69.51	71.66	71.01	72.24	71.54	79.22	69.32	80.00	82.14
	UOM	Dewas	Meerut	Sonepat	TTZ UP Part	TTZ Raj. Part	Bengaluru	Dehradun	DK	Mandi deep	MCS	Dhanbad	Rourkela	Puri	Adityapur
Gas Cost at CGS	Rs/MMBTU	738	787	815	787	787	752	885	901	738	815	924	937	937	929
Non Gas Cost	Rs/MMBTU	225	225	225	225	225	225	225	225	199	225	225	225	225	225
Total Gas Cost	Rs/MMBTU	962	1,012	1,040	1,012	1,012	977	1,110	1,126	936	1,040	1,148	1,162	1,162	1,154
GGL Marketing Margin	Rs/MMBTU	0.96	1.01	1.04	1.01	1.01	0.98	1.11	1.13	0.94	1.04	1.15	1.16	1.16	1.15
DPNG Floor Price [Cost Plus]	Rs/MMBTU	963	1,013	1,041	1,013	1,013	978	1,111	1,127	937	1,041	1,150	1,163	1,163	1,155
DPNG Floor Price [Cost Plus]	Rs/SCM	37.77	39.69	40.80	39.69	39.69	38.34	43.55	44.18	36.75	40.80	45.06	45.60	45.60	45.28
DPNG Floor Price [Cost Plus]	Rs/SCM	37.77	39.69	40.80	39.69	39.69	38.34	43.55	44.18	36.75	40.80	45.06	45.60	45.60	45.28

DPNG – Per connection cost estimation:

SN	Item Details	Unit Price (Rs per connection)
1	Material Cost (including Regulator, Meter, isolation valves, GI Pipes, MDPE service line Pipes (32 mm & 20 mm), Cu pipe, warning plate marker)	6,862.85
2	Service cost per DPNG connection (including 8 meters of MDPE Laying, 14 meters of GI Laying with accessories, Installation of Meter, Regulator, Flexible Hose, anaconda pipe, Conversion of Domestic appliance, Laying of 5 meters of 32 mm Pipe)	12,450.00
3	Total	19,312.85
4	GST @ 18%	3,476.3
5	Total including GST	22,789.2

13. Kindly brief about the benefits of PNG over LPG for household use.

The major benefits of PNG over LPG w.r.t. Household usage are as per following

- Continuous supply
- No Hassles of Booking
- No advance payment
- No space occupancy like LPG cylinder
- More safe

14. Broad details of cash flow envisaged during next three years

- Operating activities
- Investing activities
- Financing activities

GAIL Gas - Statement of Projected Cash-flow			
Year	31-Mar-26	31-Mar-27	31-Mar-28
Operating Activities	417	554	645
Investing activities	-799	-683	-659
Financing activities- Loan (Long Term)	440	-	-

Pipeline Projects

1. Updated Status of ongoing Pipeline Projects and Expected commissioning timelines.

a. Jadishpur Haldia Bokaro Dhamra Pipeline Project (3289 Km)

With an aim to ensure cleaner and quality energy in Eastern India at an affordable Price, GAIL is executing Jagdishpur – Haldia & Bokaro – Dhamra Natural Gas Pipeline (JHBDPL) Project (2563 Km), popularly known as the Pradhan Mantri Urja Ganga Project. Further, the pipeline is also extended from Barauni to Guwahati (via Siliguri & Bongaigaon) (726 Km), as an integral part of JHBDPL, to connect North East Region (NER) with the National Gas Grid.

The JHBDPL including BGPL is passing through Uttar Pradesh (335 km), Bihar (719 km), Jharkhand (500 km), West Bengal (696 km), Odisha (785 km) & Assam (254 Km). JHBDPL has revived fertilizers plant of HURL Gorakhpur, HURL Sindri, HURL Barauni and meeting demand of M/s Matix Fertilizers at Durgapur. Further, the network will enable supply of cleaner fuel for domestic & industrial customers in eastern part of India enroute the pipeline including Haldia Refinery. The salient details of Integrated JHBDPL is as below:

State Covered: Uttar Pradesh, Bihar, Jharkhand, Odisha, West Bengal & Assam

Total Length: 3289 Km

Commissioned: 3118 Km

Approved Cost: Rs. 17,405 crores

Expenditure: Rs. 16,281 crores

Physical progress: 98.6 %

3,118 Km of JHBDPL has been commissioned, including connectivity to HURL Gorakhpur, HURL Sindri, HURL Barauni, Matix Fertilisers, & IOCL Barauni.

The section-wise details of JHBDPL are as below:

- JHBDPL Phase-1 (754 Km): Operational
- Dhamra-Angul Pipeline (Section-2A of JHBDPL- **413 Km**): Operational
- Dobhi-Durgapur Pipeline (Section-2B of JHBDPL – **482 Km**): Operational
- Bokaro-Angul Pipeline (Section-3A of JHBDPL – **620 Km**): Operational
- Barauni-Guwahati Pipeline (**717 Km**): Operational
- Duragpur-Haldia Pipeline (Section-3B of JHBDPL):
 - State Covered-West Bengal
 - Total Length-294 Km
 - Anchor Load Customer-IOCL Haldia
 - Project approved cost-Rs. 2,433 crores
 - Expenditure-Rs. 2,167 crores
 - Anticipated completion- 31-Dec-25
 - Physical progress-89.1%

Present Status:

- Durgapur - IP8 -Kolkata section (132 Km): Operational
- **IP8 - Haldia section (162 Km): Work under progress.**

Present Issues:

- Slow RoU opening in West Bengal. 289 Km RoU (out of 294 Km) released till date.
- 3 Km RoU affected in Uluberia-I/Amta-II block of Howrah district due to possession issue.

- Presence of numerous fisheries in the ROU and slow compensation disbursement to fisheries owners/operators in West Bengal, presently affecting 8 Km of pipeline section.

b. KKB MPL – II (30”/24” X 901 km)

Kochi-Koottanad-Bangalore-Mangalore Pipeline (KKB MPL) Project was conceptualized in the year 2007 by Govt. of India as part of the ambitious National Gas Grid (NGG) and GAIL was authorized by MoPNG to lay this gas pipeline project in the states of Kerala, Karnataka and Tamil Nadu. This 30”/24” Pipeline with total capacity of 16 MMSCMD evacuates the re-gasified LNG (RLNG) from Kochi LNG terminal to transport to various consumers in the state of Kerala, Tamil Nadu & Karnataka, thereby connecting these three southern states with National Gas Grid. This pipeline caters to the needs of house-holds as PNG (Piped Natural Gas) for cooking purpose, hotel industries, CNG (Compressed Natural Gas) to vehicles in the transport sector, Power plants, Fertilizer plants and many other industries. The salient details of project are as below:

State Covered:	Tamil Nadu (322 Km), Kerala (514 Km), Karnataka (58 Km)
Anchor Load Customer:	MCFL, OMPL & MRPL in Mangalore (Commissioned)
Project approved cost:	Rs. 5,909 crores
Expenditure:	Rs. 4,176 crores
Anticipated completion:	31-Dec-25
Physical progress (%):	93.7%

Status:

- Kochi-Koottanad-Mangaluru (450 Km) Operational. Commissioned on 23.11.2020.
- Koottanad to Bengaluru (441 Km): 129 km Operational.
- Work in progress for balance Krishnagiri to Coimbatore Section (~ 322 Km). Target completion by Dec’25.

Present Issues:

- Krishnagiri to Coimbatore Section: Hindrances in execution of laying works along NH-844, NH-44 and NH-544 due to space constraint. Pipeline is being laid beneath the service road after receipt of working permission for BT cutting from NHAI. In-principal approval received, working permission along NH 844, NH 44, NH 544 and methodology for execution awaited from NHAI

c. Srikakulam-Angul pipeline project (24”/18”/12” X 743 km)

GAIL (India) Limited is laying the natural gas pipeline from Angul Odisha to Srikakulam Andhra Pradesh known as Srikakulam Angul Natural Gas Pipeline Project, with an aim to supply of clean and quality energy in Odisha & Andhra Pradesh. This pipeline has a strategic importance as a part of national gas grid and it would connect with GAIL’s Jagdishpur Haldia Bokaro Dahmara Pipeline (JHBDPL) network at Angul and with Kakinada Srikakulam Pipeline at Srikakulam. The pipeline will act as vital infrastructure for offtake of gas from existing/upcoming LNG terminals like Ennore, Dhamra, Gangavaram (A.P.), Gopalpura FSRU etc.

The salient details of project are as below:

State Covered: Andhra Pradesh (125 Km) & Odisha (618 Km)

Anchor Load Customer: Utkal Alumina, NALCO, Vedanta

Project approved cost: Rs. 2810 crores

Expenditure: Rs. 2133 crores

Physical progress: 94%

Status:

- Mainline (421 Km): Completed in Dec'24. PESO approval in progress. Gas-in done in 86 Km pipeline section.
- Spur line (322 Km): Work is under progress. Anticipated to be completed by Jun'25

Present Issues: In spur line, Stage -1 permission received for 54 Km forest section, however, working permission is still awaited.

d. Dhamra-Haldia pipeline project (18" X 253 km)

Dhamra Haldia Pipeline Project is being laid as an expansion of Pradhan Mantri Urja Ganga (PMUG) pipeline project named as Jagdishpur- Haldia & Bokaro Dhamra Pipeline Project (JHBDPL) to meet the additional gas demand of IOCL Haldia and demand of CGD GA enroute in Odisha and West Bengal. The Pipeline is passing through 03 districts in the states of Odisha (154 Km) & 02 districts in the states West Bengal (99 Km). The salient details of project are as below:

State Covered: Odisha (154 Km), West Bengal (99 Km)

Anchor Load Customer: IOCL Haldia

Project approved cost: Rs.1031 crores

Expenditure: Rs. 823 crores

Physical progress (%): 86

Anticipated Completion: 31-Dec-2025

Status:

- Odisha (154 Km): **Completed.**
- West Bengal (99 Km): Work is under progress.

Present Issues:

- Slow RoU opening in West Bengal.
- Presence of numerous fisheries in the ROU and slow compensation disbursement to fisheries owners/operators in West Bengal, presently affecting 8 Km of pipeline section.

e. Mumbai-Nagpur-Jharsuguda P/L Project (24"/18"/12" X 1755 km)

Mumbai-Nagpur-Jharsuguda Pipeline is a 1755 KM pipeline originating from Mumbai and ending at Jharsuguda. Also, its spur line will extend from Nagpur (Maharashtra) to Jabalpur (Madhya Pradesh). It is also a part of National Gas Grid (NGG) and will pass through 25 districts in the states of Maharashtra (12 Districts), Madhya Pradesh (3 Districts), Chhattisgarh (9 Districts) & Odisha (1 District). It will originate from Thane (Maharashtra) & will connect to JHBDPL in Jharsuguda (Odisha).

The pipeline shall ensure availability of clean fuel and cater the demand of CGD, industries and anchor load customers. Availability of domestic gas may help in development of major industries like Power, Fertilizer and other industries enroute pipeline in future.

Pipeline is passing through several Industrial Zones viz. Mhaskal, Aurangabad, Wardha, Amravati, Jalgaon, Nagpur, Gondia in Maharashtra; Jabalpur & Chhindwara in Madhya Pradesh; Raipur, Durg, Bilaspur, Korba & Raigarh in Chhattisgarh; Jharsuguda in Odisha.

State Covered	Maharashtra (963 Km), Madhya Pradesh (255 Km), Chhattisgarh (448 Km), Odisha 89 (Km).
Anchor Load Customer:	CGD enrooted the Pipeline
Project approved cost:	Rs.7844.25 crores
Expenditure:	Rs. 6224.3 crores
Physical progress	93.2%
Anticipated completion date:	June'25

Status: Work is under progress.

Present Issues:

- Hindrance exists in MSRDC RoW having length 0.2 Km affection 0.4 Km pipeline section.
- Around 6.1 Km pipeline laying in 5 patches is affected due to non-availability of space in NH Corridor in Nagpur-Jharsuguda Section (NH-44 Parallel in Nagpur District). BT cutting permission sought from NHAI.

f. Gurdaspur Jammu Pipeline

GJPL Pipeline project is passing through 05 districts in the state of Punjab and J&K, same will cater the demands of domestic & industrial consumers enroute the pipeline.

State Covered: Punjab (75 Km), J&K (77 Km)

Project approved cost: Rs. 522 crores

Expenditure: Rs. 120.7 crores

Physical progress (%): 32

Anticipated Completion: Jul'26

• Status:

- Contract for laying works awarded on 09.09.2024 for Jammu Section
- Work is under progress.

Present Issues:

- Forest permission for 62.8 Km length in Punjab is yet to be received

g. C2-C3 Pipeline – Vijaipur to Pata

C2-C3 is dedicated liquid hydrocarbon pipeline which will transport C2-C3 extracts from GAIL's Vijaipur plant to its Pata petrochemical plant as a feedstock for polymer production.

Govt./Board Sanction Date: 16.05.24

State Covered: MP (193.5 Km), UP (163.2 Km)

Project approved cost: Rs. 1792 crores

Expenditure: Rs. 3.47 crores

Physical progress (%): 13.7

Anticipated completion date: Jan'27

Status:

- PMC (Pipeline): Work awarded to M/s EIL on 21.10.2024. Kick-off meeting conducted on 24.10.2024.
- Line Pipe Package: Awarded on 03.04.2025.
- Laying Work Package: Under evaluation.

Present Issues: NIL

h. Last Mile Connectivity (LMCs) / Hookups:

Work is in progress for following LMCs:

S No.	Connectivity Name	Capacity (MMSCMD)	Target Completion Date
1	LMC to M/s Gold Plus Float Glass Private Ltd (GPFGPL), Belagavi, Karnataka (8" X 36 Km)	0.2	Apr'25
2	LMC to JSW Steel Coated Products Ltd (JSWSCPL) Kamleshwar, at Nagpur in Maharashtra from MNJPL (4" X 3.8 Km)	0.15	Apr'25
3	LMC to M/s Brahmani River Pellets Ltd from IP-02 on Dhamra Angul Pipeline section of JHBDPL	0.25	Oct'25
4	LMC to M/s JSL, Kalinganagar, Jajpur, Odisha from JHBDPL network	0.15	Oct'25
5	LMC to M/s GMPL Mangalore, Karnataka	0.50	June'25
7	LMC to JSW-BPSL	1.05	June'25
8	LMC to JSW Steel Coated Products, Nagpur	0.05	June'25
9	LMC to Jindal Saw Ltd, Nashik	0.05	July'25
10	LMC to Godawari Power and Ispat Ltd (GPIL)	0.13	Dec'25

Further, 44 Nos. of CGD Hookups are in progress which shall be commissioned in next 12 months, subject to handover of plot by CGD entity.

2. Kindly provide a brief on major reasons for delay in execution of Pipeline Projects? What are the possible solutions?

Major reasons for delay in pipeline projects are outlined below:

➤ RoU Acquisition Issues:

- Stiff Resistance / Demand of Higher Compensation/Re-routing by Land Owners: In some of the cases farmers / landowners put up resistances, either individually, through public representatives or through Unions, in laying of pipelines through their land. They demand re-routing of pipeline or very high Land & Crop compensation.
- Non-availability of Land Possession Certificate (LPC) or updated Land records: Land records/owner ship details have not been updated in some cases, which leads to difficulty in RoU acquisition process including disbursement of Land & Crop compensation.
- For re-opening of ROU for repairs/ replacement/ capacity augmentation of existing Pipelines: In re-opening of RoU, crop compensation is being paid. However, Landowners/farmers demand

land compensation afresh although the pipeline work is done in the acquired ROU for which land compensation is already paid.

- Non-availability of Govt. land for Sectionalizing Valve / Intermediate Pigging stations & dispatch / receiving terminals

➤ **Crossing/Laying Permissions from various authorities**

- Long drawn process leads to delay in getting Forest permission. Additionally, amendment in Forest Conservation [Van (Sanrakshan Evam Samvardhan)] rules, Nov'23, requires user agency to provide Compensatory Afforestation (CA) land leading to further delay.
- Other Crossing permission has to be taken from various authorities such as NHAI, Railways, Water bodies, CRZ, Wildlife Corridor, Mangroves etc.
- In case of change of mode of crossing, afresh application is required which takes Additional time for permission.

➤ **Difficulties in laying the pipeline along NH/SH Roads:**

- The NH/SH ROW is often restricted due to the space already occupied by the highway infrastructure. This limits the area available for laying pipelines.
- Many highways have utilities like electricity poles, drainage systems, and communication cables within the ROW, leads to limited space for pipeline laying which required special permission from authorities to BT cutting of service road.
- In densely populated or urbanized areas, difficulties are faced in laying pipeline due to constrained working space, restricted areas, encroachment issues, existence of utilities, restricted time for working etc.
- Working along busy highways requires strict measures to ensure the safety of both workers & commuters, this leads to slow work progress and limited working hours.
- In under construction highway concurrent laying of pipeline requires additional efforts due to left over debris/boulders and restrictions in simultaneous working

➤ **Delay in appointment/Non-availability of dedicated Competent Authority (CA):**

- As per Petroleum & Mineral Pipeline (P&MP) Act 1962, acquisition of RoU is to be done through Competent Authority (CA) appointed for the project. Competent Authority and revenue staff are deputed or authorized from revenue deptt. of the state Govt. It has been observed that in many cases there is either delay in deputation from the State Govt. or in some cases CA is not given on dedicated basis for the project.

➤ **Possible solutions to reduce delays in Pipeline projects are given below:**

- Single window system for all permission required in laying the pipelines.
- Many projects are currently experiencing delays due to resistance from farmers and their demands for enhanced compensation. To address this issue effectively, a compensation framework should be developed in collaboration with stakeholders and the state government, ensuring that the needs and concerns of landowners are adequately addressed.

- Timely deployment of competent authorities for acquisition of ROU will be ensured by early engagement with state governments to avoid delay in ROU acquisition.
- PM Gati Shakti National Master Plan will play critical role in implementation of pipeline projects by streamlining coordination with various stakeholders across India.
- Timely flagging of issues on PMG portal results in quick resolution of issues being faced at various levels in state/central governments.
- Local communities, indigenous groups, and landowners to be engaged, to build trust and address their concerns proactively.

3. Brief Details on Last Mile Connectivity (LMCs) executed by GAIL in last 5 years (In terms of Length, number of customers connected, incremental capacity and Investment?)

S No	Period	Last Mile Connectivity (LMC) Commissioned (Nos)	Length of LMCs (Km)	Cost of LMCs (Rs Crores)	CGD Hookup Commissioned (Nos.)
1	2020-21	8	51	97	16
2	2021-22	4	50	164	27
3	2022-23	5	22	69	27
4	2023-24	8	27	85	26
5	2024-25	2	29	46	27
		27	179	461	123

4. Detailed CAPEX of GAIL in last two years along with projected Capex figures for FY25-26 & FY26-27.

Details of Capex incurred by GAIL from FY2019-20 till FY2024-25 and estimated capex from 2025-26 till 2027-28 is as per the table below:

Particulars	FY 2019-20	FY 2020-21	FY 2021-22	FY 2022-23	FY 2023-24	FY 2024-25	FY 2025-26 [E]	FY 2026-27 [E]	FY 2027-28 [E]
Pipeline Projects & CGD	4,647	4,744	4,458	5,000	3,682	2,481	2,967	5,287	2,849
Petrochemical Projects	6	119	383	1,366	3,300	2,653	4,429	1,345	324
Exploration & Production	141	93	349	307	37	113	275	546	66
Net Zero/Renewables	-	-	-	6	122	77	1,228	4,034	3,653
Operational Capex & Other Projects	670	1,261	1,608	1,356	3,285	4,570	1,966	1,938	1,972
Total	5,463	6,217	6,798	8,034	10,426	9,894	10,865	13,150	8,774
Equity Contributions to JVs/ Subsidiaries	651	764	940	1,065	1,000	618	1,920	369	843
Total Capital Outlay	6,114	6,982	7,738	9,099	11,426	10,512	12,784	13,518	9,617
(-)Govt. Grant	1,552	728	590	-	-	-	-	-	-
Total Capital Outlay	4,562	6,253	7,148	9,099	11,426	10,512	12,784	13,518	9,617

5. Does GAIL have plan to bid for new LPG pipelines recently notified by PNGRB? If yes, how many pipelines will be bid by GAIL (kindly provide name, length and estimated capex).

PNGRB has issued public notice Ref: PNGRB/Auth/3-PPPL (20) /LPG dt 10.12.2024 for public consultation for 9 LPG pipelines as below:

S no	Proposed Pipeline (start point-End point)	Bottling plant to be connected (Nos)	Length (km)	Capacity (in TMTPA)
1	Cherlapally - Nagpur	6	453	510
2	Shikarpur-Hubli-Goa	7	508	630
3	Mumbai-Aurangabad-Jalgaon	6	521	510
4	Paradip-Raipur	6	555	540
5	Gwalior-Haldwani	7	580	510
6	Jalandhar-Jammu	4	230	300
7	Ennore-Puducherry	4	160	540
8	Ennore-Gummidipoondi	2	32	240
9	Chennai Petroleum Corporation Limited-Cauvery Basin Refinery-Trichy-Madurai	8	431	510

Almost, five of these nine pipelines are emanating from existing LPG pipelines.

Accordingly, GAIL vide letter dt 08.01.2025 has sought information from PNGRB on the capacity available of the already existing pipelines and the capacity utilization of the bottling plants which would be served by new proposed pipelines.

Based on information provided by PNGRB, further decision on these pipelines would be taken.

Konkan LNG Limited (KLL)

1. Brief on Physical and Financial Performance during previous three years and projection for next two years.

Financial Year	2022-23 (Audited)	2023-24 (Audited)	2024-25 (Unaudited)	2025-26 (Projected)	2026-27 (Projected)
Number of Cargoes	19	17	21	34	39
Revenue from Operation (Rs. Cr)	400.86	406.87	478.24*	800.96	963.23
PBT (Rs. Cr)	-180.61	-327.17	-172.70	-145.14	-76.33
PAT (Rs. Cr)	-450.84	-246.57	-125.11	-303.87	-94.18

*It does not include sale of LNG of Rs 300.00 Crores

2. What is the status for the break water facility at KLL Dabhol? What will be available capacity thereafter?

Construction of breakwater is completed in April-2025. After completion of breakwater project KLL port shall receive All-Weather Port status from statutory authorities. With all weather port status, KLL will be able to unload LNG cargos throughout the year including monsoon season.

3. Is there any time and cost overrun in the ongoing project?

Yes. Time overrun is 32 Months & Cost overrun is Rs. 125 Cr. (approx.)

4. How many Cargoes were received in previous and target for next two years? How many cargoes will be handled post breakwater facility completion?

Financial Year	2022-23	2023-24	2024-25	2025-26 (Projected)	2026-27 (Projected)
Number of Cargoes	19	17	21	34	39

5. Future outlook of Konkan LNG?

KLL has recently completed one of the major project of construction of breakwater which will facilitate KLL being all weather port. Other projects such as ambient air heating system, Captive Power Plant and expansion of Truck Loading facility are under execution which will enhance self-sufficiency, reliability, and profitability for KLL in future. In addition, KLL is in process to expand its existing capacity of 5MMTPA to 15 MMTPA in two stages. In the first stages capacity will be enhanced from 5 MMTPA to 6.3 MMTPA by debottlenecking of existing plant. In the second phase capacity will be enhanced from 6.3 MMTPA to 15 MMTPA by including other products (ethane & propane) in the portfolio

6. Investment and Impairment history of GAIL's capital in RGPPL & KLL.

Chronology of Equity Changes

KLLs equity structure has changed significantly since its One-Time Settlement (OTS) with its lenders. A summary of the major events is provided below:

S No.	Particulars	Change in Equity (Including CCCPS)
1	One Time Settlement: GAIL (India) Ltd. provided Inter Corporate Loan i.e. (ICL-1) to Konkan LNG Ltd (KLL) and debt settlement was executed on 27th March 2020 with lenders along with novation of residual debt from lenders to GAIL as ICL-2. Consequent upon the debt settlement, the stake owned by the lenders was transferred to GAIL, which result in increase in its shareholding.	56.70% to 77.32%.
2	Share Purchase Agreement between GAIL & NTPC: A Share Purchase Agreement was executed between GAIL (India) Limited and NTPC Limited on 23rd February, 2021 wherein GAIL & NTPC agreed to fully exit from RGPPL & KLL respectively by swapping entire equity of NTPC in KLL with GAILs entire equity in RGPPL.	77.32 % to 92.15%
3	Partly Paid up into Fully Paid up: Balance Call of INR 4.5 per share was called in the month of September 2021. Accordingly, shares were allotted to GAIL and its shareholding increased.	92.15% to 93.01%
4	Right Issue of Equity Shares: 23,10,00,000 Equity Shares of INR 10/- each were offered on right basis to the shareholders of the company in proportion to their shareholding during December 2022. GAIL (India) Limited paid application money for entitled shares and for unsubscribed shares. Accordingly, shares were allotted to GAIL on 27th December 2022 and its shareholding increased.	93.01% to 94.26%
5	Right Issue of Equity Shares: 10,00,00,000 Equity Shares of INR 10/- each were offered on right basis to the shareholders of the company in proportion to their shareholding during March 2023. GAIL (India) Limited paid application money for entitled shares and for unsubscribed shares. Accordingly, shares were allotted to GAIL on 27th March 2023 and its shareholding increased.	94.26% to 94.68%

- GAIL is the majority shareholder of KLL and currently holds 94.68% of the Company's share capital, and the remaining 5.32% is held by MSEB Holding Company Limited.
- GAIL (India) Limited as the promoter of KLL made additional capital contribution of INR 331.04 Cr in FY23 in line with approved previous business plan.

Shareholding Details of KLL:

Shareholder	% Ownership (31-Mar-25)
GAIL	94.68%
MSEB	5.32%
TOTAL	100%

**The ownership has been calculated assuming conversion of CCPS into Equity.*

7. What is the plan for power intake for KLL given the situation that RGPPL plant is under shutdown? What is the stop gap arrangement? Is there any capacity constraint / additional financial implication w.r.t. Stop Gap arrangement and what is the present status?

KLL is in process of installing natural gas based captive power plant to cater power requirement of the terminal. KLL's present power requirement is being fed through 33KV grid power.

8. Kindly provide brief on construction Central Heating system. What is the scheduled completion date, capex?

KLL is in process to install the ambient air-based heating system wherein ambient air will be utilized to gain heat required for regasification of LNG. It is expected to complete the project by June-2026. Approximate capex in Rs. 599 crores.

9. Kindly provide a brief about capacity expansion plans of KLL.

KLL is in process to expand its existing capacity of 5MMTPA to 15 MMTPA in two stages. In the first stages capacity will be enhanced from 5 MMTPA to 6.3 MMTPA by debottlenecking of existing plant. In the second phase capacity will be enhanced from 6.3 MMTPA to 15 MMTPA by including other products (ethane & propane) in the portfolio.

Exploration & Production

1. Details of blocks that GAIL got under different licensing policies. Details of relinquished blocks with capital costs incurred over them.

GAIL has, till date, participated in 39 E&P blocks, 3 CBM Blocks in India and 4 blocks overseas (46 blocks). Out of these, 29 E&P blocks, 3 CBM Blocks in India and 2 blocks overseas (34 blocks) have been relinquished. At present, GAIL has participating interest in 10 E&P blocks in India and 2 blocks overseas. The details are as follows:

Bid Round	Year	Awarded	Relinquished	Currently participating
NELP-I	2000	2	2	-
NELP-II	2001	6	5	1
NELP-IV	2004	2	1	1
NELP-V	2005	3	2	1
NELP-VI	2007	15	15	-
NELP-VII	2008	1	1	-
NELP-VIII	2010	2	2	-
NELP-IX	2012	5	1	4
OALP-I/HELP	2018	1	-	1
OALP-VII/HELP	2022	1	-	1
Sub-Total		38	29	9
Farm-in (Pre-NELP)	2002	1	-	1
Overseas Blocks	2002 - 2007	4	2	2
CBM-III	2007	3	3	-
Total E&P Blocks		46	34	12

Details of Relinquished Blocks

20 blocks have been relinquished by GAIL over the past 8 - 10 years. Year wise relinquishment details are as follows: FY 2013-14 (10 blocks), 2014-15 (5 blocks), 2015-16 (1 block), 2016-17 (2 blocks), 2017-18 (2 blocks), however, no blocks were relinquished after FY 2017-18.

Blocks relinquished during FY 2013-14 (10 blocks of NELP-VI) are as follows:

S. No.	Block Name	Operator	GAIL's PI
1.	KG-DWN-2004/1	ONGC	10%
2.	KG-DWN-2004/2	ONGC	10%
3.	KG-DWN-2004/3	ONGC	10%
4.	KG-DWN-2004/5	ONGC	10%
5.	KG-DWN-2004/6	ONGC	10%
6.	CY-DWN-2004/1	ONGC	10%
7.	CY-DWN-2004/2	ONGC	10%
8.	CY-DWN-2004/4	ONGC	10%

9.	CY-PR-DWN-2004/2	ONGC	10%
10.	KG-ONN-2004/2	GSPC	40%

Blocks relinquished during FY 2014-15 (5 blocks) are as follows:

S. No.	Block Name	Operator	GAIL's PI	NELP
1.	AN-DWN-2009/13	ONGC	10%	NELP-VIII
2.	AN-DWN-2009/18	ONGC	10%	NELP-VIII
3.	AA-ONN-2003/1	JOGPL	35%	NELP-V
4.	CY-DWN-2004/3	ONGC	10%	NELP-VI
5.	CY-PR-DWN-2004/1	ONGC	10%	NELP-VI

Blocks relinquished during FY 2015-16 (1 block) are as follows:

	Block Name	Operator	GAIL's PI	NELP
1.	RJ-ONN-2004/1	GAIL	22.225%	NELP-VI

Blocks relinquished during FY 2016-17 (2 blocks) are as follows:

	Block Name	Operator	GAIL's PI	NELP
1.	CY-ONN-2005/1	GAIL	40%	NELP-VII
2.	GK-OSN-2010/2	ONGC	10%	NELP-IX

Blocks relinquished during FY 2017-18 (2 blocks) are as follows:

	Block Name	Operator	GAIL's PI	NELP
1.	MN-OSN-2000/2	ONGC	20%	NELP-II
2.	AN-DWN-2003/2	Eni	15%	NELP-V

Expenditure on blocks which have been relinquished till March 2025 is **Rs. 1,801 Crores.**

2. Current number of blocks with GAIL with the stages they are currently in and What is the outlook of GAIL's E&P business in India or overseas?

The details of the activities in the blocks by stages are as below:

Production Stage (5 blocks):

- I. Crude Oil production is in progress from 2 Cambay onland JV blocks (i.e. CB-ONN-2000/1 & CB-ONN-2003/2) @ 215 Bopd (GAIL Share) where GSPC is the Operator.
- II. Natural Gas production is in progress from 2 Myanmar offshore blocks (i.e. A1 & A3) @ 1.17 MMCMD (GAIL Share) where POSCO-International Limited is the Operator. Further, Field Development Activities for Phase-IV is in progress.
- III. CB-ONN-2010/11 (Cambay Onland, Gujarat State), Operator: GAIL. Production of Crude Oil has commenced since 18th March'23. Currently, production is in progress @ 20 Bopd.

Development Stage (2 blocks): Development activities are in progress in 2 blocks:

- I. AA-ONN-2002/1 (Tripura Onland, Tripura State), Operator: JOGPL. Follow-up with MOEF for approval of appraisal and development locations is in progress.

- II. CB-ONN-2010/8 (Cambay Onland, Gujarat State), Operator: Bharat Petro Resources Limited. Operator and JV Partners proposed for relinquishment of the block due to project being economically unviable. MC meeting held on 04.04.2025 for relinquishment of the block, MOM is awaited.

Appraisal Stage (2 Blocks): Appraisal activities are in progress in 2 blocks:

- I. CY-OS/2 (Cauvery Offshore): Hardy (75%, Operator) and GAIL (25%) are the partners in the block CY-OS/2. Due to the dispute regarding the nature of discovery, arbitration was initiated between MoP&NG/DGH and Consortium. The Arbitration was awarded in favour of the consortium on 2nd February, 2013. The GOI filed an appeal petition against the arbitration award at Delhi High Court on 13 Aug'13 for which the last hearing was held on 13.04.2024 and the proposal is relisted on 13.08.2025.
- II. GK-OSN-2010/1 (Gujarat Kutch Offshore). Operator: ONGC. Further studies are in progress to decide the way forward in the block. Operator has sought time extension from DGH to carry out the activities in the block.

Exploration (3 Blocks): Exploration activities are in progress in 3 blocks:

- I. AA-ONN-2010/2 (Assam Arakan, Assam State): Operator: OIL. Drilling of 1 Exploratory well completed on 11.01.2024. Preparation for drilling of 2nd well is in progress.
- II. CB-ONHP-2017/12 (Cambay Onland, Gujarat State), Operator: GAIL: Drilling of 1 well, Narmada#1 completed in Sept' 24. The well was plugged and abandoned due to non-discovery of Hydrocarbon. Relinquishment process in progress.
- III. RJ-ONHP-2021/1 (Rajasthan onland), Operator-GAIL: This Block was awarded during FY 2022-23 for which Revenue Sharing Contract (RSC) was signed on 28.06.2022 and PEL was granted by Govt. of Rajasthan on 30.11.2022. Acquisition, Processing & Interpretation of 50 LKM 2D and 80 Sq.Km. of 3D Seismic Data completed. Currently, tendering process for various drilling related activities is in progress to commence the drilling of 2 exploratory wells from Q2 of 2025-26.

3. Revenue, profitability and capex for FY24 & FY25; Projection for next three years.

Revenue, profitability and capex for FY24 & FY25 is as per the table below:

Figures in Rs. Crores

S.No.	Financial Year	Revenue	Profitability	Capex
1	FY 2023-24	1,087.13	313.05	160.36
2	FY 2024-25	1,102.28	462.28	86.26

Estimated projection for next three years is as below:

Figures in Rs. Crores

Financial Year	2025-26	2026-27	2027-28
E&P Revenue	1,092.00	1,073.00	1,115.92
Royalty	91.00	93.49	97.06
Govt Share of Profit Petroleum	311.00	328.11	353.55
Oil & Gas Producing expenses	92.00	114.56	108.46
Survey & LD	14.00	12.00	-
Drywell Expenses	0.00	10.00	10.00

Financial Year	2025-26	2026-27	2027-28
Depreciation	80.00	90.25	95.14
Profit (-)/Loss (+) on Exchange	-	-	-
PROFIT BEFORE TAX (PBT)	504.00	424.59	451.71

The projected Capex (BE) for next three years is as below: *Figures in Rs. Crores*

Blocks	FY 2025-26	FY 2026-27	FY 2027-28
Domestic	38	27	50
Overseas	237	519	65
Total E&P Capex	275	546	115

The investment in Domestic Blocks is mainly towards 1 GAIL operated Block RJ-ONHP-2021/1, 1 GSPC operated Block CB-ONN-2000/1 and 1 Onland Block in Tripura. The investment in Overseas Blocks is toward A-1 & A-3 Blocks, Myanmar.

4. Overseas investments in E&P, details of blocks GAIL is holding in international territories.

GAIL has participating interest in 3 overseas acreages. Out of these, 2 are in Myanmar and 1 is in USA. The details of Oil and Gas fields acquired in foreign countries are as below:

(i) Block A-1 & A-3, Myanmar

GAIL is having 8.5% participating Interest (PI) in Blocks A-1 and A-3, Myanmar. Operator for these blocks is POSCO International Corporation (51%). Other partners are ONGC Videsh Limited (17%), Myanma Oil and Gas Enterprise (15%) and Korea Gas Corporation (8.5%). Production of gas has started from Block A-3 since 22nd June 2013 and from Block A-1 since 10th January 2014. GAIL's investment in Myanmar blocks is 402.50 MMUS\$.

(ii) Eagle Ford Shale Asset, USA

During September 2011, GAIL Global (USA) Inc. ("GGUI"), a wholly owned subsidiary of GAIL in USA, executed a Purchase and Participation Agreement with Carrizo Oil & Gas Inc. ("Carrizo") and acquired 20% working interest (4,040 net acres) in Eagle Ford Shale acreage in Texas, USA. During 2018, GGUI divested around 2,502 acres to M/s EP Energy. On December 20, 2019, M/s Callon (Eagle Ford) LLC ("Callon") acquired Carrizo. After this acquisition, Callon with remaining working interest i.e. 80% was functioning as Operator for joint venture activities. Effective October 1, 2021, there was a change in operatorship to Texas American Resources II, LLC, ("TAR") in respect of approx. 2,059 acres. Callon continue to act as the Operator in the remaining contract areas of 811 acres.

Effective October 1, 2023, Ridgemar Energy Operating, LLC ("Ridgemar") became Operator of GGUI for 811 acres. Effective January 31, 2025, M/s. Crescent Energy acquired Ridgemar Energy's acreage of 811 acres. Current acreage position of GGUI is as tabulated below:

Operator	Acreage	No. of Wells
TAR	2,059	36
Crescent Energy	811	56
Total	2,870	92

Current net production of GGUI is around 600 Boepd. Most of the GGUI acreage is located in the volatile oil window and have high liquid percentage. Total investment by GGUI in these assets is US\$ 297 million till March, 2025.

5. How revenue gets generated in overseas blocks, what is the price/revenue formula for GAIL for these blocks?

(i) Block A-1 & A-3, Myanmar

Gas production from 2 blocks in Myanmar (A-1 and A-3) has already reached its peak production of 14 MMSCMD (at 100% share). Domestic gas is sold to Myanma Oil and Gas Enterprise (MOGE) and Export gas to China is sold to China National United Oil Corporation (CNUOC) as per Domestic / Export Gas Sales and Purchase Agreements signed with them respectively. Year on year production is planned considering Annual Contracted Quantities (ACQ) by the buyer as per provisions of Export GSPA and Domestic GSPA. From A-1 and A-3 blocks and Offshore Pipeline Project, revenue is generated from the gas sales produced from A-1, and A-3 blocks and from Offshore Pipeline Tariff.

The price formula is as follows:

Gas Price Formula: $7.73923 [0.2(CPI_y / CPI) + 0.2(OM_y / OM) + 0.5(F_y / F) + 0.1]$

Where, CPI= 209.47633, OM= 187.05833 and F= 87.89692 and CPI_y, OM_y and F_y are variable as per the below descriptions.

CPI_y (Consumer Price Index) = 12 months period average of 6 month prior to Recalculation Date for the Consumer Price Index in United State of America for all urban consumers (CPI-U), as published by US Department of Commerce, Economic and Statistics Administration and Bureau of Economic Analysis

OM_y (Producer Price Index) = 12 months period average of 6 month prior to Recalculation Date for Producer Price Index for Field and Gas Field Machinery, published by the US Department of labor, Bureau of Labor statistics

(ii) Eagle Ford Shale Asset, USA

In case of GGUI, Operator is selling GGUI's share of hydrocarbon production. GGUI crude oil sales are linked to WTI with a MEH (Magellan East Houston Terminal) differential. Similarly, natural gas sales are linked to Houston Ship Channel (HSC) and having a basis of Henry Hub (HH).

**6. What is the contribution of Myanmar in terms of revenues and physical performance?
What is the peak volume and revenue from Myanmar gas block?**

Contribution of Gas Sales (at 100% Share) in Myanmar is as follows. GAIL's share is 8.5%.

FY	MMSCF (100%)	MMm3 (100%)	MMSCMD (100%)	MMSCMD (GAIL Share)
2013-14	37,571	1,064	2.92	0.25
2014-15	134,758	3,816	10.5	0.89
2015-16	181,956	5,153	14.1	1.20
2016-17	171,969	4,870	13.3	1.13
2017-18	168,498	4,772	13.1	1.11
2018-19	142,641	4,040	11.1	0.94
2019-20	216,720	6,138	16.8	1.43
2020-21	199,317	5,645	15.5	1.32
2021-22	185,481	5,253	14.4	1.22
2022-23	195,309	5,531	15.1	1.28
2023-24	173,022	4,899	13.4	1.14
2024-25	176,965	5,012	13.7	1.16
Total	1,984,207	56,193		

Peak volume from these blocks is 500 MMSCFD (14.16 MMSCMD) which is expected to remain there till the year 2033.

GAIL Share of Year-wise revenue from Myanmar blocks is as follows

Year	Amount (Rs. Crore)
2013-14	170.21
2014-15	600.06
2015-16	716.63
2016-17	581.06
2017-18	598.32
2018-19	603.70
2019-20	943.35
2020-21	823.44
2021-22	814.11
2022-23	1,108.75
2023-24	986.78
2024-25	1,059.00
Total	9,005.41

Revenue from Myanmar contribute ~92% of total revenues of E&P business.

7. Can India attain self-sufficiency in hydrocarbon energy market? On what resources India can raise its hope for energy security? How GAIL view itself in India's self-sufficiency program?

A comparison of domestic production and import for the past few years is as follows:

Year	Crude Oil (MMT)		Natural Gas (MMSCM)	
	Production	Import	Production	Import
2013-14	37.79	189.24	35,410	
2014-15	37.46	189.43	33,660	
2015-16	36.94	202.85	32,250	
2016-17	36.01	213.93	31,897	24,849
2017-18	35.68	220.43	32,649	27,439
2018-19	34.20	226.50	32,875	28,740
2019-20	32.20	227.00	31,180	33,680
2020-21	30.23	193.76	28,342	32,326
2021-22	29.70	212.20	34,024	31,906
2022-23	29.15	230.87	33,573	27,021
2023-24	24.86	212.58	32,646	28,396
2024-25	22.48	202.96	36,135	36,500

India imported around ~90% of its crude oil requirement and ~50% of its natural gas requirement during FY 2024-25.

The present target is to reduce this dependency by 10%. The process of exploration in India has been eased to a great extent through policy liberalization & ease of doing business. The major challenge to domestic oil and gas production are the prices, both in the domestic market and international bench mark prices of these hydrocarbons.

As the activities in blocks awarded during the past 4 – 5 years moving forward and considering dynamic pricing of gas and crude oil to cover the costs, domestic production is expected to increase further.

Furthermore, certain demand will be met through Compressed Bio Gas (CBG), renewable energy and other Bio-Fuels like ethanol and will reduce import dependency.

GAIL's E&P activities are very limited as compared to the national scale which is predominantly carried out by ONGC and OIL, two National Oil Companies. The efforts to participate in domestic bidding round in proven Category – I and Category – II basins will continue to add to the efforts of these NOC and contribute towards securing self-sufficiency of the nation.

8. How many type of policies GoI has adopted in past to distribute exploration licenses to entities?

- Government of India awarded blocks for exploration and production of oil and gas as per New Exploration Licensing Policy (NELP) during years 1999 to 2011/12. Nine bid rounds were held under this Policy and 254 blocks were awarded to various companies / consortiums. ONGC, OIL and other Public Sector Companies also participated in these bid rounds on non-preferential basis. The awarded blocks under NELP regime (254 blocks in 9 bid rounds) are located in onland

(114), offshore shallow water (59) and offshore deepwater (81) areas. Currently, only 29 blocks are operational.

- Hydrocarbon Exploration Licensing Policy (HELP) was launched w.e.f. 1st July 2017. Open Acreage Licensing Policy (OALP) is one of the key program of HELP. (a) Single license that will cover exploration & production of all types of hydrocarbon viz. conventional oil & gas, coal-bed methane, shale oil (b) Revenue Sharing Model: No cost recovery; Submission of Expression of Interest basis which Government would call for competitive bids (c) Full marketing & pricing freedom of gas produced on arm's length basis.

Under **HELP**, 9 **OALP**, 3 Discovered Small Fields (**DSF**) and 1 Special CBM bid round have been held. Under **OALP** bid rounds, 172 blocks (with total area of 378,056 KM²) have been awarded and under **DSF**, which was held during the year 2016, 2019, 2022 and 2024 wherein total of 87 contract areas have been awarded.

9. Any updates on discovery at Mahanadi (0.72tcf; offshore), Tripura (71bcf; on-land) and any other discovery?

- **What shall be the breakeven gas price for the above discoveries?**
- **When is the production expected to begin?**

- i) In Tripura Onland Block (AA-ONN-2002/1, NELP-IV, Operator JOGPL Private Limited, GAIL Share 80%), development activities have been initiated in the block. The project is waiting on some statutory approvals and production from the field may take about 2 - 3 years after all pending approvals are received. The break-even gas price for this discovery would be to the tune of US\$ 6.05/MMBTU. Considering the prevailing domestic gas prices, execution strategy for development activities is being re-worked out and the development activities would be synchronized with movement in domestic gas prices and also in-sync with gas pipeline infrastructure development in the region.
- ii) Mahanadi Offshore Block (MN-OSN-2000/2, NELP-II, ONGC): The block has been relinquished by consortium during July 2017 (ONGC, GAIL, OIL & IOCL) as the prevailing gas prices were not conducive to support development of deep-water offshore block, since the breakeven price for development was around US\$ 10/MMBTU and retention of block for long period was not allowed.

10. Any Latest Development in E&P segment with details thereon. Further, details about significant activities/events undertaken or to be undertaken.

The details about significant activities / events undertaken are as below:

- i) Production of Crude Oil has commenced from GAIL operated block CB-ONN-2010/11 since 18th March'23 @ 20 Bopd.
- ii) Acquisition, Processing & Interpretation of 50LKM 2D and 80 Sq.Km. of 3D Seismic Data completed in the block RJ-ONHP-2021.
- iii) Tendering process for various drilling related activities is in progress to commence the drilling of 2 exploratory wells from Q2 of 2025-26.

- iv) A-1 & A-3 Myanmar Offshore Blocks: Phase-IV Development activities for Mahar gas discovery are in progress.
- v) Participation in ongoing domestic bids rounds.

Treasury

1. Breakup of Loan outstanding as on 31.03.2025 along with interest rates and repayment schedule thereof.

<i>Long Term Loan Outstanding as on 31.03.2025</i>					
Loans	INR (Rs. Cr.)	Interest Rate	Interest Basis	Rate	Repayment terms
OIDB Loan -I	531.25	7.45%	Fixed USD taken)	(INR Swap	Yearly in 8 installments started from Mar-23 to Mar-30
OIDB Loan -II	75.00	7.39%	Fixed USD taken)	(INR Swap	Yearly in 8 installments starting from Apr-23 to Apr-30
OIDB Loan -III	37.50	6.42%	Fixed		Yearly in 8 installments started from June-23 to June-30
OIDB Loan -IV	700.00	7.00%	Floating		Yearly in 8 installments starting from March-27 to March-34
HDFC TERM LOAN – I	1,333.33	7.20%	RBI Repo+0.95		Quarterly in 48 installments started from June-21 to Mar-33
HDFC TERM LOAN – II	750.00	6.93%	RBI Repo+0.68		Quarterly in 16 installment started from June-22 to Mar-26
Bank of India Term Loan	1,586.67	6.85%	RBI Repo+0.60		Quarterly in 15 installment started from June-23 to Dec-26
Central Bank of India Term Loan	1,476.00	7.25%	RBI Repo+1.00		Quarterly in 32 installment starting from Aug-25 to May-33.
7.34% BONDS 2027 SERIES – I	1,575.00	7.34%	Fixed USD taken for Coupon)	(INR Swap	Bullet Repayment in Dec-2027
Punjab National Bank	3,500.00	7.25%	RBI Repo+1.00		Quarterly in 32 installment starting from Dec-25 to Sep-33.
Total Loans	11,564.75				

2. What is the average cost of debt, quantum of current debt, how much of it is foreign denominated and how much is hedged against forex fluctuations?

Average cost of debt for the FY 2024-25 was 7.31%, quantum of current debt is Rs. 2,305 (long term loan) crores. The FX Debt is NIL as on 31st March, 2025.

3. Details of Borrowings availed by GAIL in last 3 financial years. Details of loan repayment during FY2024-25 and scheduled repayments for next 5 years.

Year	Type of Borrowing	Rs. Crores
2024-25	Long Term Borrowing	-

Year	Type of Borrowing	Rs. Crores
	Short Term Borrowing	3,500
	Total	3,500
2023-24	Long Term Borrowing	5,676
	Short Term Borrowing	10,000
	Total	15,676
2022-23	Long Term Borrowing	4,975
	Short Term Borrowing	-
	Total	4,975

TOTAL LOAN REPAYMENTS FY 2024-25		
DATE OF PAYMENT	LOAN	Total (Rs In Crore)
27-04-2024	OIDB-100 CR	13
13-05-2024	UBOI STL 400	400
13-05-2024	SBI STL 2500	1,000
20-06-2024	SBI STL 2500	400
21-06-2024	SBI STL 2500	505
24-06-2024	SBI STL 2500	595
30-06-2024	HDFC RTL-I	42
30-06-2024	HDFC RTL-I	188
30-06-2024	BOI RTL-I	227
Total Q1 FY24-25 PAYMENT		3,368
30-09-2024	HDFC RTL-I	42
30-09-2024	HDFC RTL-I	188
30-09-2024	BOI RTL-I	227
01-07-2024	OIDB-50 CR	6
Total Q2 FY24-25 PAYMENT		462
31-12-2024	HDFC RTL-I	42
31-12-2024	HDFC RTL-I	188
31-12-2024	BOI RTL-I	227
Total Q3 FY24-25 PAYMENT		456
31-03-2025	HDFC RTL-I	42
31-03-2025	HDFC RTL-I	188
31-03-2025	BOI RTL-I	227
15-01-2025	OIDB-850	106
Total Q4 FY24-25 PAYMENT		562
GRAND TOTAL		4,848

Repayment Schedule FY2025-26 till FY2033-24 (₹ crores)											
Loan Facility	long term loan o/s As on Mar-24	long term loan o/s As on Mar-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32	2032-33	2033-34
OIDB Loan	1,469	1,344	125	213	213	213	213	106	88	88	88
HDFC TERM LOAN-I	1,500	1,333	167	167	167	167	167	167	167	167	-

Repayment Schedule FY2025-26 till FY2033-24 (₹ crores)											
Loan Facility	long term loan o/s As on Mar-24	long term loan o/s As on Mar-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32	2032-33	2033-34
HDFC TERM LOAN-II	1,500	750	750	-	-	-	-	-	-	-	-
Bank of India Rupee Term Loan	2,493	1,587	907	680	-	-	-	-	-	-	-
BONDS 2027	1,575	1,575	-	-	1,575	-	-	-	-	-	-
Central Bank of India Rupee Term Loan	1,476	1,476	138	185	185	185	185	185	185	185	46
Punjab National Bank Rupee Term Loan	3,500	3,500	219	437	437	437	437	437	437	437	219
Total	13,513	11,565	2,305	1,681	2,576	1,001	1,001	895	876	876	352

4. Kindly provide Credit Rating status of GAIL with downside risk factors by well-known rating agencies.

GAIL is rated as “AAA with stable outlook” by all the three domestic rating agencies who have rated GAIL i.e. by M/s ICRA, CARE and India ratings.

In the International market GAIL’s issuer rating by both Moody’s and Fitch Ratings has been capped at par with the Sovereign Rating.

Rating by M/s Moody’s is “Baa3 with Stable outlook”

Rating by Fitch Ratings is “BBB- with Stable outlook”

The major downside risks identified by the rating agencies are as follows:

- Continued growth in GAIL’s cyclical and unregulated business segments which have a much higher business risk than its core gas transmission business.
- Vulnerability of GAIL’s trading segment to energy prices.
- Margin risk to the Petrochemical segment.
- Commodity price risks associated with petrochemicals, LPG and liquid hydrocarbons (LHC) businesses.
- Significant off-balance sheet liabilities.

5. Details regarding issuance of Bond(s) with maturity date. What %age of total Loan requirement is expected to be fulfilled through issuance of Bonds in future & likely yield (next 5 years)?

There are No bonds issuance done in current FY however GAIL has raised INR Bonds of Rs. 1,575 crores in FY23 @ 7.34% coupon rate for 5 years with bullet repayment on maturity. Further the share of the INR denominated bonds will depend on the market scenario and cash flow of GAIL at the time of availing the loan.

6. What are the steps the company has taken in last three years to bring down the Finance cost? Please quantify.

In the past years, GAIL has undertaken various activities including swapping INR liability to US liability, negotiation of interest rates on the existing loans, tying up loans at very competitive rates linked to RBI Repo Rate etc. to bring down its cost of borrowing.

7. Any new/innovative idea/plans for raising forex borrowings which GAIL has (including from GIFT city), also indicate associated benefits thereof (separately for each such plan / ideas).

In the FY 25, GAIL has revised its Forex policy which enables the borrowings from FX market using natural hedge available with GAIL.

Further GAIL has also incorporated a Wholly Owned Subsidiary (WoS) in GIFT City, Gandhinagar on 07th April, 2025 for conducting global treasury Centre activities and Ship leasing activities. In view of the above, GAIL after assessment of various scenarios such as market timing, cash flow etc. may avail the FX loans in the near future.

8. Any new strategy adopted for investment of surplus funds and also indicate associated benefits thereof.

Last year, GAIL has started investment in new short-term instrument .i.e. TREPS (Treasury bills Repurchase) and CROMS (Clearcorp Repo Order Matching System) to improve the interest income and to avoid the blockage of funds in CLTD (Corporate Link Term Deposit) where the interest is eligible only after 7 days of deposit.

Benefit of TREPS In Comparison with Mutual Fund (MF) :-

- 1) Limit on Investment:** - The Limit for investment of surplus funds in Mutual Fund is restricted to maximum 30%, However, there is no limit for investment in TREPS/CROMS
- 2) Better Return:** - Due to levy of Stamp Duty on Liquid/Overnight Mutual Fund and Exit Load on Liquid Fund, the return on Mutual Fund reduces significantly and considering the same, investment in TREPS/CROMS is expected to yield higher returns especially for shorter duration (less than a week)

In Comparison with CLTD (Corporate Link Term Deposit) Flexibility: - Unlike in CLTD, where the interest is eligible only after 7 days of deposit. The return is available in TREPS/CROMS even for 1 day, which facilitates the better management of funds

9. Detail of Cash & Bank Balances as on 31.03.2025 (separately for each head) & status of threshold limit applicable for buy-back of shares as per DIPAM guidelines.

Following are the details w.r.t. Cash/Bank balances being managed by Corporate Treasury (as on 31.03.2025):

S.No.	Particulars	₹ crores
1	Cash at Bank and Corporate Liquid Term Deposits [CLTD] *	887
2	Working Capital Loan	-2012
	Net Amount [1-2]	-1125

* Including Funds of EEFC Account (\$52.15 Million & EURO 0.24 Million)

Corporate Social Responsibility (CSR)

1. Kindly provide a brief write up of GAIL's policy on CSR.

The salient features of Corporate Social Responsibility Policy of GAIL include:

- a) **Composition of CSR Committee** – The CSR Committee of GAIL presently comprises of C&MD - Chairperson, Director (HR) and two Independent Directors.
- b) **Mandatory spends on CSR** – 2% of avg. net profit of the preceding three financial years.
- c) CSR projects to be undertaken in areas and domains identified under Schedule VII with special focus on:
 - Nutrition, Health and Sanitation and Drinking Water projects (*GAIL Arogya*)
 - Education initiatives (*GAIL Ujjawal*)
 - Livelihood Generation and Skill development initiatives (*GAIL Kaushal*)
 - Rural Development (*GAIL Unnati*)
 - Women Empowerment initiatives (*GAIL Sashakt*)
 - Care of the elderly and differently abled (*GAIL Saksham*)
 - Environment centric initiatives (*GAIL Harit*)
- a) Majority of CSR programmes shall be implemented in 'local areas', which are within a radius of 100 Kms adjoining GAIL installations, which are largely located in remote areas/along the GAIL pipeline.
- b) Regular Monitoring of Projects in form of Progress reports, site visits, video conferencing etc.
- c) Reporting on CSR Policy in the form of an Annual Report as per prescribed format.
- d) General Provisions – No contribution to political parties, surplus from CSR activities not to be a part of business profit etc.

2. Kindly provide a brief on various CSR initiatives of GAIL and the positive outcomes of each of the initiatives.

GAIL undertakes its CSR initiatives in line with the activities listed under Schedule VII of Companies Act with special focus on 7 focus areas collectively named *GAIL Hriday*. Major initiatives under each of the focus areas are as below:

GAIL Ujjwal (Education initiatives)

- GAIL Utkarsh is a flagship CSR project of GAIL providing free residential coaching for Engineering and Medical entrance exams. The project is implemented across 3 Centres- Kanpur (100 students for JEE), Varanasi (80 girl students- 30 for JEE and 50 for NEET) and Haldwani (50 students for JEE). More than 2200 students have been impacted under this project since its inception. 168 out of 180 students undergoing coaching for engineering entrance exams have cleared JEE Mains 2025.
- Education support is provided to underprivileged students in rural and urban slums. School bags cum convertible study table have also been distributed to children to motivate them to attend school and improve their posture.
- Support has been provided to government schools to improve their infrastructure through drinking water provision, sanitary napkin vending machines, incinerators etc. to provide better facilities and improve the attendance of students.

- Smart Classes and STEM Resource Centres have also been set up in Government schools to enhance the learning experience of students by incorporating interactive digital tools and support learning in Science, Technology, Engineering, and Mathematics.

GAIL Arogya (Nutrition, Health and Sanitation and Drinking Water projects)

- Through a flagship project of GAIL, free door step delivery of primary healthcare services is provided in remote locations adjoining GAIL Work centres and pipelines through operation of Mobile Medical Units (MMUs). Approx. 60 MMUs have catered to health needs of around 10 lakh beneficiaries in FY 2024-25.
- Health care has also been provided through operation of STI clinics, conducting health camps in villages, and providing requisite medical equipment for government hospitals.
- Nutrition kits have been distributed to malnourished women & children to tackle anaemia and improve nutritional status.
- RO water plants and hand pumps have been installed to provide safe and clean drinking water to the public.

GAIL Kaushal (Livelihood Generation and Skill development initiatives)

- GAIL has been operating 02 Skill Schools at Guna in Madhya Pradesh and Nagaram, East Godavari in Andhra Pradesh. Skill training is provided to youth beneficiaries in hydrocarbon in trades like CNC Operator, Industrial welder, Industrial electrician, Draughtsman Mechanical, etc.
- Under the aegis of MoPNG, Skill Development Institutes have been setup by Oil & Gas PSUs to provide training and equipping individuals with vocational skills and knowledge to enhance their employability and career prospects. GAIL is the Nodal PSU for the SDI at Raebareli.
- GAIL Unnati (Rural Development)
- To improve the rural infrastructure, GAIL has taken up projects to install street lights improving lighting in rural locations thereby improving the safety of the area as well.
- Project on repairing dilapidated houses in Almora, Uttarakhand was also taken up in FY 2024-25.

GAIL Saksham (Care of the elderly and differently abled)

- Aids and assistive devices were provided to approx. 785 Divyangjan beneficiaries in FY 2024-25 to significantly enhance their ability to participate in daily life and achieve greater independence.
- Non-formal Education, Speech Therapy for EWS Children with Hearing Impairment was also provided in FY 2024-25.

GAIL Sashakt (Women Empowerment initiatives)

- Support for weaving blankets, vocational skills, and sustainable livelihood opportunities were provided to women to ensure social progress, economic growth of them and their families, and sustainable holistic development. The projects addressed systemic inequalities and promoted gender equality.

GAIL Harit (Environment centric initiatives)

- CSR programmes on environment sustainability include production of bio-degradable Coir Fibre Based Root Trainer (CRT) Planting Stock, installation of Solar street lights and Solar RO water plants.

3. How much amount has been spent by GAIL on CSR during previous years (i.e., from FY 2020-21 to FY 2024-25) with breakup in various schemes (separately for each FY)?

The expenditure made by GAIL towards CSR activities during the last 05 years is as below:

The focus area wise expenditure of GAIL in the last 5 years is as below:

S No	Financial Year	CSR Expenditure (Rs. in Cr.)
1	2020-21	147.67
2	2021-22	204.97
3	2022-23	99.10
4	2023-24	175.71
5	2024-25	183.77

Expenditure (Rs. in Cr)					
Focus Area under GAIL CSR Policy	FY 2020-21	FY 2021-22	FY 2022-23	FY 2023-24	FY 2024-25*
Arogya (Health)	40.26	36.19	27.61	90.03	108.25
Kaushal (Skill Development)	13.76	22.56	5.98	7.79	12.98
Sashakt (Women Empowerment)	1.86	0.95	31.79	8.13	4.51
Ujjawal (Education)	22.82	18.12	12.47	24.49	35.48
Unnati (Rural Development)	4.19	4.64	1.83	13.28	3.39
Saksham (Care for Elderly and Differently Abled)	3.08	2.36	4.03	1.80	3.22
Harit (Environmental Projects)	1.11	20.99	7.57	19.34	5.25
Others (Expenses incurred on Impact Assessment, Admin Overheads and other areas as included under Schedule VII)	60.60	99.17	7.82	10.85	10.69
TOTAL	147.67	204.97	99.10	175.71	183.77

*Tentative expenditure, subject to change upon finalization of annual accounts for FY 2024-25.

Taxation

1. View on GST on Natural Gas (which includes expected benefits, key hurdles, impact on GAIL, etc.)

GST on five petroleum products viz petroleum crude, High speed Diesel (HSD), motor spirit (commonly known as petrol), natural gas and aviation turbine fuel supply of petroleum crude, high speed diesel, motor spirit (commonly known as petrol), natural gas and aviation turbine fuel, shall be levied with effect from such date as may be notified by the Government on the recommendations of the Council and therefore legacy taxes continue to be applicable on these products.

Expected benefit to gas sector

Natural Gas is primarily an industrial input used as fuel/feedstock by industries like power, fertilizers, petrochemical, LPG and City Gas Distribution (CGD). Natural Gas is an environment – friendly fuel unlike other fuels viz. Crude, Coal and other alternate fuels which pollute the environment due to higher carbon emission. With the increasing level of pollution, thrust of the government is to increase the use of greener fuel. All other polluting fuels are being kept under the ambit of the GST and the environment friendly fuel is kept outside the GST. Bringing Gas under GST will have a significant role in limiting CO2 emissions, global warming and climate change.

Currently, VAT rates on Natural Gas vary from Nil to 20% across different States, with some States levying an additional cess on taxable turnover. Such differential tax treatment and variation in rates act as a barrier in free and smooth trade of Natural Gas across the country and further aggravate the competitiveness of industries using Natural Gas as an input. Such distortion in tax treatment having effect on price for consumers situated in different states can be removed only by bringing Natural Gas under GST.

Inclusion of Natural Gas under GST will also promote the Natural Gas based industries as seamless credit of GST will be available. Natural gas is one of the major industrial inputs used as fuels in industries and non-inclusion of Natural Gas under GST lead to absorption of VAT paid on procurement of Natural Gas in their cost of production which is rendering large, medium and small industries using Natural Gas economically unviable and uncompetitive.

Key hurdles in inclusion of Natural Gas under GST

- Petroleum products contribute a significant amount of revenue to the states.
- Levying VAT on these products, the state governments have control over their revenues. Bringing petroleum under GST will take away that control as the Centre controls fixing the rate of GST.
- Central Govt. is open to the possibility, but a majority of the states are against the inclusion of petroleum products under the GST.
- Many states are still against the inclusion of petroleum products under GST.
- Under GST, even under the 28% tax slab, states will earn a lot less compared to what they currently earn from petrol and diesel sales as value-added tax (VAT).

Impact on GAIL

- I. Inclusion of GST will have positive impact on overall gas industry including GAIL.
- II. There will be significant reduction in the landed cost of Natural Gas to the customer which will increase the overall demand of Natural Gas. This will have favorable impact on the GAIL's top line.
- III. Substantial reduction in cost and time of tax compliances and litigations.

2. Brief on Advance Pricing Agreement (APA)

In terms of Income Tax Act, 1961, an Advance Pricing Agreement (APA) is an agreement between a tax payer and tax authority determining the transfer pricing methodology for pricing the tax payer's international transactions with its related party for future years. An APA provides the benefit of tax certainty with respect to tax litigations in respect to international transactions covered by the APA.

When an APA is finalized between Assessee & Tax authorities of country where the assessee is located, it is called **Unilateral Advance Pricing Agreement**. Whereas, when it involves the tax payer, associated enterprise (AE) of the tax payer in the foreign country, tax authority of the country where the tax payer is located, and the foreign tax authority, the APA is called **Bilateral Advance Pricing Agreement**.

GAIL signed 1st Unilateral Advance Pricing Agreement (UAPA) with **Central Board of Direct Taxes (CBDT)** in February 2023 for the intercompany transactions relating to the sourcing of liquefied natural gas (LNG) from US through GAIL Global USA LNG LLC ('GGULL'). The covered period of above agreement was from **FY 2017-18 to FY 2021-22**.

Further, a fresh cum renewal application for UAPA was filed in March 2024 for the covered period from FY 2024-25 to FY 2028-29 with roll back period of two years from FY 2022-23 to FY 2023-24. GAIL has successfully signed its second **Unilateral Advance Pricing Agreement (APA)** with the **Central Board of Direct Taxes (CBDT)** on 27.03.2025.

This agreement reaffirms our commitment to **tax transparency, certainty, and compliance**. Securing another APA strengthens our strategic approach to **transfer pricing** and ensures alignment with regulatory frameworks, reinforcing our dedication to long-term growth and governance excellence.

Salient Features of the APA signed: -

- Final TP margin payable to GGULL fixed at 8.5% on Value Added Expenses (VAE) as against proposed TP margin of 10% on VAE
- NIL guarantee fees on all the fund based & non- fund based corporate guarantees.
- Covered period for the 2nd UAPA is FY 2022-23 to FY 2028-29.

3. Brief on any changes during FY 2024-25 in taxation (VAT / CST etc.) having impact on Natural Gas prices.

In order to promote the use of Natural Gas, some of the states have reduced the VAT rate on Natural Gas, PNG and CNG during the FY 2024-25. Details of such reductions are as under:

State	Product	Old Rate	New Rate
Andhra Pradesh	NG	24.50%	5.00%
	CNG	24.50%	5.00%
	PNG	5.00%	5.00%

State	Product	Old Rate	New Rate
Assam	NG	14.50%	14.50%
	CNG	14.50%	5.00%
	PNG	14.50%	14.50%
Bihar	NG	20.00%	20.00%
	CNG	20.00%	12.50%
	PNG	20.00%	12.50% (Domestic & Commercial PNG) & 5% (Industrial PNG)
Chhattisgarh	NG	25.00%	14.50%
	CNG	25.00%	14.50%
	PNG	25.00%	14.50%
Rajasthan	NG	10.00%	10.00%
	CNG	14.50%	7.50%
	PNG	10.00%	7.50%
Uttarakhand	NG	20.00%	20.00%
	CNG	20.00%	10.00%
	PNG	20.00%	5.00%

State / UT wise VAT rate on Natural Gas, CNG and PNG					
State / UT	NG	CNG	PNG	NG sold for Fertilizer Manufacturing	NG sold for Power Generation
Andhra Pradesh	5.00%	5.00%	5.00%	5.00%	5.00%
Arunachal Pradesh	20.00%	20.00%	20.00%	20.00%	20.00%
Assam	14.5% (First Point Sale)	5.00%	14.50%	14.50%	14.50%
Bihar	20%	12.50%	12.50% (Domestic & Commercial PNG) & 5% (Industrial PNG)	20%	20%
Chandigarh	5%	12.50%	5%	5.00%	5%
Chattisgarh	14.50%	14.50%	14.50%	14.50%	14.50%
Dadra & Nagal Haveli and Daman & Diu	6.00%	6.00%	6.00%	6.00%	6.00%
Delhi	Exempt for supply to Delhi Govt. - power companies & for CNG for sale in Delhi otherwise 5.00 %	0%	5%	5%	0%
Goa	12.50%	4.00%	4.00%	12.50%	12.50%
Gujarat	15% / 6% for prescribed purposes	5% for retail consumers	5% for domestic use	15%	15%

State / UT wise VAT rate on Natural Gas, CNG and PNG					
State / UT	NG	CNG	PNG	NG sold for Fertilizer Manufacturing	NG sold for Power Generation
Haryana	6.00% + Surcharge @ 5.00% of Tax rate	5.00 % + Surcharge @ 5.00% of Tax Rate	(5.00 % for domestic use otherwise 6.00%) + Surcharge 5% of Tax rate	6.00% + Surcharge @ 5% of 6%	6.00% + Surcharge @ 5% of 6%
Himachal Pradesh	4.00% if used as industrial input otherwise 13.75%	4.00% if used as industrial input otherwise 13.75%	4.00% if used as industrial input otherwise 13.75%	4% (Considering Industrial Input)	4% (Considering Industrial Input)
Jammu Kashmir	14.50%	14.50%	14.50%	14.50%	14.50%
Jharkhand	14%	14%	14%	14%	14%
Karnataka	5.50%	5.00%	5.50%	5.50%	5.50%
Kerala	5.00%	5.00%	5.00%		
Madhya Pradesh	14%	14%	14%	14%	14%
Maharashtra	3%	3.00%	3.00%	3.00%	3.00%
Manipur	13.50%	13.50%	13.50%	13.50%	13.50%
Meghalaya	14.50%	14.50%	14.50%	14.50%	14.50%
Mizoram	13.50%	13.50%	13.50%	13.50%	13.50%
Nagaland	14.50%	14.50%	14.50%	14.50%	14.50%
Orissa	5% for prescribed purposes otherwise 15%	15%	5% for domestic use otherwise 15%	5.00%	15.00%
Punjab	3.00% + Surc. @ 10.00% of Tax Rate	3.00% + Surc. @ 10.00% of Tax Rate	3.00% + Surc. @ 10.00% of Tax Rate	3.00% + Surc. @ 10.00% of Tax Rate	3.00% + Surc. @ 10.00% of Tax Rate
Puducherry	14.50%	14.50%	5% - Domestic use 7.5% - Industrial & Commercial use	14%	10%
Rajasthan	10.00%	7.50%	7.50%	0.00%	10.00%
Sikkim	13.50%	13.50%	13.50%	13.50%	13.50%
Tamil Nadu	5.00%	5.00%	5.00%	5.00%	5.00%
Telangana	14.50%	14.50%	5% for domestic use otherwise 14.50%	14.50%	14.50%
Tripura	4% VAT + 19.5% Road Development Cess (Total 24.28%)	4% VAT + 19.5% Road Development Cess (Total 24.28%)	4% VAT + 19.5% Road Development Cess (Total 24.28%)	4% VAT + 19.5% Road Development Cess (Total 24.28%)	4% VAT + 19.5% Road Development Cess (Total 24.28%)
Uttarakhand	20%	10.00%	5.00%	5.00%	0% (0% for sale of RLNG to gas based power plants)
Uttar Pradesh	10.00%	12.50%	10.00%	14.50%	10.00%
West Bengal	5.00% if used as fuel otherwise 14.50 %	5.00% if used as fuel otherwise 14.50 %	5.00% if used as fuel otherwise 14.50 %	14.5% for feed and 5% for fuel component	5.00%

Sustainability Development

1. Kindly provide a table on GAIL's Renewable energy portfolio as on 31.03.2025. Further, kindly include ongoing/planned renewable projects with scheduled completion dates.

GAIL is planning to ramp up its Renewable Energy capacity to ~3500 MW by 2035 from the current 135 MW. Presently, the following RE Projects are under different stages of execution:

- 170 MW Wind Power Project in Maharashtra
- 100 MW Solar Power Project in Uttar Pradesh
- 17.5 MW Floating Solar Plant at Pata, Uttar Pradesh
- 14 MW Roof Top Solar across Installations
- 12 MW Solar Plant at Vijaipur, Madhya Pradesh
- 7.75 MW Floating Solar at Vijaipur, Madhya Pradesh

2. Kindly provide a brief on projects undertaken by GAIL for achievement of Net Zero?

GAIL has taken following decarbonization projects/initiatives for achievement of Net Zero:

- **Electrification of NG-based equipment:** GAIL has converted 01 Natural Gas turbine driven Compressor to Electrical drive (3.6 MW).
- **Renewable Energy (RE):** GAIL has a current RE portfolio of 135 MW. Presently, the following RE Projects are under different stages of execution:
 - 170 MW Wind Power Project in Maharashtra
 - 100 MW Solar Power Project in Uttar Pradesh
 - 17.5 MW Floating Solar Plant at Pata, Uttar Pradesh
 - 14 MW Roof Top Solar across Installations
 - 12 MW Solar Plant at Vijaipur, Madhya Pradesh
 - 7.75 MW Floating Solar at Vijaipur, Madhya Pradesh
- **CBG:** GAIL has commissioned 05 TPD Municipal Solid Waste (MSW) based CBG plant at Ranchi. The CBG plant was dedicated to the Nation by the Honorable Prime Minister of India on 2nd October 2024.
- **Green Hydrogen:** GAIL is operating a 10 MW PEM Electrolyser-based Green Hydrogen plant of capacity 4.3 TPD at Vijaipur, Madhya Pradesh since May'2024. GAIL also commissioned India's maiden pilot project of blending Hydrogen in City Gas Distribution (CGD) network of M/s Aavantika Gas Limited (AGL), Indore, M.P.
- **Energy Efficiency:** GAIL has implemented an Energy Management System at all its O&M locations and undertakes several energy efficiency improvement initiatives on a regular basis. GAIL's major installations are ISO 50001 Energy Management System certified.
- **Offsetting through Afforestation:** GAIL has planted around ~8.5 lakhs trees across its various locations and aims to enhance afforestation by undertaking extensive tree-plantation across the country. Further, GAIL has entered into an agreement with MP Forest Department for plantation of 75000 trees.

3. What does "net zero" mean in the context of climate change and emissions reduction and GAIL's Target in context with global target?

Global warming is the gradual increase in Earth's temperature observed since the pre-industrial period. Increase in greenhouse gas emission is one of the main reasons of global warming and climate change. The Paris Agreement aims to limit global warming to below 2°C, preferably 1.5°C by 2100, above pre-industrial levels. Reaching net zero means that any GHG emissions made by industrial activities are balanced by an equivalent amount of emissions removed from the atmosphere, resulting in no net increase in atmospheric greenhouse gases.

4. How does GAIL plan to achieve net zero emissions?

GAIL has set an ambitious Net Zero target to achieve 100% reduction in Scope-1 & Scope-2 emissions by the year 2035.

GAIL has already taken actions on decarbonization Levers as follows:

- **Electrification of NG-based equipment:** GAIL has converted 01 Natural Gas turbine driven Compressor to Electrical drive (3.6 MW). The electrification of other major Prime movers (36 machines) is planned in phased manner.
- **Renewable Energy (RE):** GAIL is planning to ramp up its Renewable Energy capacity to ~3500 MW by 2035 from the current 135 MW. Presently, the following RE Projects are under different stages of execution:
 - 170 MW Wind Power Project in Maharashtra
 - 100 MW Solar Power Project in Uttar Pradesh
 - 17.5 MW Floating Solar Plant at Pata, Uttar Pradesh
 - 14 MW Roof Top Solar across Installations
 - 12 MW Solar Plant at Vijaipur, Madhya Pradesh
 - 7.75 MW Floating Solar at Vijaipur, Madhya Pradesh
- **CBG:** GAIL has planned to set up 26 CBG plants by 2030. A 05 TPD Municipal Solid Waste (MSW) based CBG plant at Ranchi was commissioned and dedicated to the Nation by the Honorable Prime Minister of India on 2nd October 2024.
- **Green Hydrogen:** GAIL is operating a 10 MW PEM Electrolyser based Green Hydrogen plant of capacity 4.3 TPD at Vijaipur, Madhya Pradesh since May'2024. GAIL also commissioned India's maiden pilot project of blending Hydrogen in City Gas Distribution (CGD) network of M/s Aavantika Gas Limited (AGL), Indore, M.P.
- **CO2 Valorization:** GAIL is operating microalgae-based CO₂ capture plant of 01 TPD capacity at its Petrochemical Plant at Pata (U.P). GAIL is further exploring various technologies for CO₂ Valorization to Bio-Ethanol and Sustainable Aviation Fuel (SAF).
- **Energy Efficiency:** GAIL has implemented an Energy Management System at all its O&M locations and undertakes several energy efficiency improvement initiatives on a regular basis. GAIL's major installations are ISO 50001 Energy Management System certified.
- **Offsetting through Afforestation:** GAIL has planted around ~8.5 lakhs trees across its location and aims to enhance afforestation by undertaking extensive tree-plantation across the country. Further, GAIL has entered into an agreement with MP Forest Department for plantation of 75000 trees.

- **First Generation (1G) Ethanol:** GAIL is setting up a 500 Kilo Litres Per Day (KLPD) grain based First Generation (1G) Ethanol plant in Rajasthan in a JV mode. GAIL has submitted application to MoP&NG for obtaining approval of DIPAM. The Ethanol produced would be sold to Oil Marketing Companies (OMCs) for blending in Petrol. The project will also produce Distiller's Dried Grains Solubles (DDGS) as a protein rich by-product, which can be used as cattle feed.

5. What are some examples of industries including GAIL for whom it is challenging to achieve net zero target?

Industries such as Oil & Gas, Petrochemicals, Steel, Cement and Fertilizers are classified as "hard-to-abate" sectors due to their high energy intensity and reliance on fossil fuels. Decarbonizing these sectors is challenging because of the inherent emissions in their processes and the current immaturity of low-carbon technologies.

6. What are GAIL's Capex Plans to achieve Net Zero Target? Details in Brief about that and current status thereon.

GAIL has developed a comprehensive roadmap to achieve its Net Zero target, with a detailed assessment of the financial impact of various identified abatement levers. The total Capex is estimated at ~Rs. 38,000 crores by 2035.

7. How does the concept of "net zero" differ from simply reducing emissions?

Net Zero Emissions refer to the balance between the amount of greenhouse gases (GHGs) emitted into the atmosphere and the amount removed from it. An organization achieves Net Zero Emissions when it reduces its emissions as much as possible and then removes or offsets the remaining emissions, so that the net emissions become zero.

8. Writeup on Biofuels, CBG, 1G Ethanol etc. in which GAIL is exploring opportunities.

Biofuels are renewable fuels made from biomass — organic materials such as crops, agricultural residues, and waste. Unlike fossil fuels, biofuels are low-carbon or carbon-neutral, making them an important part of the transition toward clean energy.

1G Ethanol is a type of Bio-ethanol produced from food-based feedstocks such as sugarcane, corn, wheat, or sugar beet through the process of fermentation. GAIL is setting up a 500 Kilo Litres Per Day (KLPD) grain based First Generation (1G) Ethanol plant in Rajasthan in a JV mode. GAIL has submitted application to MoP&NG for obtaining approval of DIPAM. The Ethanol produced would be sold to Oil Marketing Companies (OMCs) for blending in Petrol. The project will also produce Distiller's Dried Grains Solubles (DDGS) as a protein rich by-product, which can be used as cattle feed.

Compressed Biogas (CBG) is a clean, renewable fuel produced by the anaerobic digestion of organic waste such as agricultural residue, cow dung, food waste, etc. It mainly consists of methane (CH₄) and is similar in composition and energy potential to Compressed Natural Gas (CNG).

GAIL has planned to set up 26 CBG plants by 2030. A 05 TPD Municipal Solid Waste (MSW) based CBG plant at Ranchi was commissioned and dedicated to the Nation by the Honorable Prime Minister of India on 2nd October 2024.

9. What role do renewable energy sources play in achieving net zero?

Power generation is one of the largest sources of carbon emissions worldwide. Renewable energy sources like solar, wind etc. generate electricity without burning fossil fuels, reducing GHG emissions.

10. Can you explain the difference between gross emissions and net emissions?

Gross emissions refer to the total amount of greenhouse gases an organization emits annually into the atmosphere, before accounting for any offsets or removals.

Net emissions are what remain after subtracting the amount of GHGs removed from the atmosphere (through carbon sinks, offsets, or removal technologies) from the gross emissions.

11. How do carbon offsetting and carbon credits contribute to achieving net zero?

Carbon Offsetting involves compensating for emissions by supporting projects that reduce, avoid, or remove emissions elsewhere.

Carbon Credits are tradable certificates that represent a reduction, avoidance, or removal of one metric tonne of carbon dioxide (CO₂) or its equivalent.

By purchasing and retiring carbon credits, entities can offset their residual emissions and work towards achieving a balance between emissions and removals, ultimately leading to a net-zero state.

12. What are some potential benefits of reaching a net zero economy?

The benefits of achieving Net Zero economy are as follows:

- Prevents Global warming
- Reduced operational costs through enhanced energy efficiency
- Lower dependence on fossil fuels, thus saving forex.
- Strengthens organization's brand reputation
- Improves ESG rating

13. Are there any countries which have already achieved net zero emissions?

Bhutan and Suriname claim to achieve carbon negative status.

14. What are some potential technological innovations that could help us reach net zero faster?

Technologies for Green Hydrogen, Energy Storage System, Carbon Capture, Utilization & Storage (CCUS) are still evolving and yet to mature for commercial scale deployment. Innovations and government support towards this can help achieve country's Net Zero faster.

Corporate Strategy, Planning & Advocacy

1. What are potential sectors in Indian economy which can increase Gas Share in primary energy mix of country in the decade going forward?

- Govt. of India aims to increase the share of natural gas in the primary energy basket from existing 5.7% to 15% share, which is an increase from current level of ~ 185 MMSCD to ~ 500 MMSCMD which is almost 2.7 times increase from current level. It will need significant efforts from all the stakeholders, policy reforms from the government and huge investment in the entire gas value chain to achieve this target.
- The historical gas usage pattern across different sector has been mentioned below:

Total (MMSCMD)	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
Fertilizer	44.15	48.72	49.53	53.15	57.66	56.11
City Gas Distribution	29.82	25.29	33.36	32.95	39.96	41.40
Power	30.36	29.69	27.83	22.34	24.88	24.32
Refinery	21.33	21.67	14.56	10.71	15.99	15.89
Petrochemicals	9.78	8.42	7.85	5.37	7.31	9.19
Others	19.34	19.96	35.35	36.31	45.59	48.01
Total	154.78	153.74	168.47	160.83	188.39	194.92

Sector(% Share)	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
Fertilizer	28.53%	31.69%	29.40%	33.05%	30.61%	28.79%
City Gas Distribution	19.26%	16.45%	19.80%	20.49%	19.62%	21.24%
Power	19.61%	19.31%	16.52%	13.89%	13.21%	12.48%
Refinery	13.78%	14.10%	8.64%	6.66%	8.49%	8.15%
Petrochemicals	6.32 %	5.47%	4.66%	3.34%	3.88%	4.71%
Others	12.50 %	12.98%	20.98%	22.58%	24.20%	24.63%

Source:PPAC

- It can be clearly seen from the table that there is a shift in the consumption pattern among the top three anchor consumers, while fertilizer industry continues to be the major consumer of Natural Gas, CGD Sector has now surpassed the Power Sector to become the second largest consumer of Natural Gas.
- India's energy sector is undergoing a significant transformation, with a growing focus on cleaner sources of energy. However, natural gas is expected to play a critical role in the country's energy mix for the foreseeable future.
- Next wave of gas demand can come from refineries, chemical process industries, and Iron and Steel plants, both as feedstock, fuel, and process requirements.

- With the completion of the National Gas Grid (NGG), all the refineries currently isolated from the gas grid can move to gas, increasing gas consumption. Gas consumption in the oil refining sector is projected to grow at nearly 9% annually, reaching more than 9 bcm/yr by 2030. This increase will be driven by higher refinery runs at existing units that already use natural gas, and the addition of several new projects and capacity expansions in coming years.
- Further, gas-based blast furnace and gas-based DRI (Sponge Iron) can open up demand for gas. Gas consumption in the sponge iron and steel segment is projected to grow at an annual average rate of 8%, reaching close to 2 bcm/yr by 2030.
- The power sector is considered to be one of the key drivers of any gas-based economy. If we take the example of the USA, over 43% of the electricity comes from gas, 21.4% from renewables, 18.6% from nuclear and 16.2% from coal¹. However, in India over two-thirds of the electricity comes from coal-based plants and the share of gas in electricity generation is merely 5.2%. The Central Electricity Authority (CEA) projects a significant increase in India's peak power demand, from 273 GW in 2025 to 297 GW in 2026-27 and 366 GW by 2031-32. To substantially increase consumption of natural gas in power sector, it has to be used as baseload fuel. Moreover, as India's renewable energy capacity continues to grow, maintaining grid stability has become a pressing concern. Gas-based power plants are suitable to provide excellent grid support capability to facilitate seamless integration of variable renewable energy (VRE) to our grid. Indian power sector needs reforms such as peaking power policy, Gas-RE integration for 24x7 power, and review of Merit Order Dispatch considering the health and pollution cost. These reforms can increase the share of gas in primary energy basket.
- We may also see the adoption of LNG in the transport sector in coming days primarily in long haul vehicle. GAIL has recently set up dedicated Retail LNG team. GAIL board has recently accorded approval for an investment plan of Rs 650 crore and aspire to capture 50 % plus market share in the next 5-6 years. Accelerated adoption of LNG-fueled heavy-duty vehicles could generate an additional gas demand of approximately 2 bcm/yr by 2030. Potential incentives to achieve this include road toll, exemptions for LNG vehicles, reduced fuel taxes on LNG, domestic gas allocation for LNG vehicles and corporate fuel efficiency standards for heavy-duty vehicle (HDV) manufacturers.
- With growing awareness of urban pollution and the ban on polluting fuels like petcoke and furnace oil, we can see gradual adoption of gas among various industrial & commercial customers. It is envisaged that natural gas will be in the ambit of GST in the coming years and which is expected to boost demand from MSMEs/SMEs due to the benefit of an input tax credit.
- With government policy support, industry initiatives, efforts from developers and suppliers, we may achieve the ambitious target set out for moving the country towards gas-based economy.

¹ <https://www.eia.gov/tools/faqs/faq.php?id=427&t=3>

2. What is the present status of TSO (Transmission Service Operator) as announced in Union Budget for FY21? What would be the broad structure and mandate for the new operator? What are the timelines for its incorporation?

- At present, India has approximately ~ 25,124 km of natural gas pipeline operated by various entities and passing through various states and union territories in the country. Another ~ 10,676 km of pipelines are in various stages of implementation. It is expected that in the next 4-5 years, India will have its natural gas grid in place with a cumulative length of around 35,000 km.
- Indian gas sector is still a developing market and reforms in the Indian gas sector need to ensure the twin objectives of both infrastructure creation and market development. Demonstrating transparency in third-party open access to pipeline networks might support natural gas market creation and further boost infrastructure development.
- GAIL has been the first and only gas pipeline transporter in the country, who, since August 2018, has been successfully operating an online "**Open Access Portal**" through which any registered entity/shipper can book common carrier capacity, at any time, from anywhere, even using their mobile phones, in a completely transparent manner.
- Petroleum and Natural Gas Regulatory Board (PNGRB) has also implemented Unified Tariff for natural gas pipelines with a mission of "One Nation, One Grid and One tariff".
- The reform has been benefitting the consumers located in the far-flung areas where the additive tariff was earlier applicable. This will facilitate development of gas markets and vision of government to increase the gas utilization in the country.
- Ministry of Petroleum and Natural Gas has examined various options for creation of an independent Transport System Operator (TSO). It may be noted that GAIL already has system/portal in place since Aug' 2018 for providing open access to common carrier capacities. Establishment of TSO will be completed after government/regulatory approvals. At present, there is no such proposal of TSO, and any upcoming considerations will be taken up after requisite government and regulatory approvals.

3. What is the Status of unbundling of Gas Transmission segment and government strategies?

- In recent years, the government has placed significant emphasis on developing a gas-based economy. and the country is working towards increasing the share of gas from 6% to 15% in the energy mix by 2030. GAIL has been working relentlessly to improve availability of natural gas across the country, construction of National Gas Grid (NGG) in a time bound manner.
- Indian gas sector is still on a developing phase and it will take some more time for the market to mature. India needs lot of infrastructure development in the natural gas space and need timely completion of NGG without derailing the existing momentum. It has also been observed globally that in less matured gas markets, traditionally vertically integrated entities have played the role of infrastructure development and market development as cash flow from both businesses complement each other making it viable.
- Petroleum and Natural Gas Regulatory Board (PNGRB) has also implemented Unified Tariff for natural gas pipelines with a mission of "One Nation, One Grid and One tariff".

- The reform has been benefitting the consumers located in the far-flung areas where the additive tariff was earlier applicable. This will facilitate development of gas markets and vision of government to increase the gas utilization in the country.
- At present, there is no proposal for unbundling of GAIL's gas transmission business. Any consideration in this matter will be made with a focus on enhancing shareholder value and promoting the development of the natural gas sector in the national interest.

4. What is the status of GAIL's asset monetization for two pipelines under InvIT Model? How many more pipelines/assets are expected to go under InvIT Model?

- Based on the advice of NITI Aayog and MoP&NG, Dahej-Uran-Panvel-Dabhol Pipeline (DUPL-DPPL) and Dabhol-Bangalore Pipeline (DBPL) were identified to test the market for Asset Monetization (AM) through InvIT route.
- 105th meeting of PPPAC was held in Aug'21 for approval of GAIL's monetization of assets. In the meeting, it was advised to study and submit a report through GAIL's Transaction Advisor (TA), inter-alia capturing various options available with GAIL for raising additional resources (Debt, Equity, InvIT, etc.) and conducting a comparative analysis of such costs of financing avenues.
- In this regard, GAIL through its Financial & Technical Advisor (FTA) has carried out the desired study and the report has been submitted to PPPAC. Report of the FTA is under examination.
- Further, identification of assets for monetization is carried out on the basis of a number of criteria such as the nature and location of assets, viable transaction structure, future business potential, strength of their balance sheet, debt equity ratio, comparative cost analysis of all avenues available for raising additional resources etc.
- At present, the projects of GAIL are generally financed through internal generation and borrowing raised on its balance sheet. GAIL has a strong balance sheet and is conservatively leveraged with very low Debt – Equity (D/E) ratio. GAIL has enough headroom available to raise additional debt at cost-effective & competitive levels to fund its Capex requirement.
- Recently, MoP&NG has constituted a 07-member Asset Monetization Industry Group (AMIG), with representation from all major Oil CPSEs for examination and recommendation of asset monetization proposals for MoP&NG. Representative from GAIL is also a member of this committee. AMIG is identifying assets to be monetized and is preparing annual monetization plans for Oil CPSEs over next 4 years, starting from FY 2024-25, so as to meet the monetization targets set by Government of India for MoP&NG.

5. Kindly provide brief write-up on IGX and rationale for GAIL's equity investment in IGX.

- Gas trading hubs are well-advanced and mature in northwest Europe and North America but have just started in emerging markets. In June 2020, India's first gas trading platform, Indian Gas Exchange (IGX) was launched and in December 2020, PNRGB granted authorization to IGX to

set up and operate the exchange as per PNGRB (Gas Exchange) Regulations notified in October 2020.

- The online delivery-based gas trading platform is expected to facilitate market transparency, stimulate demand, and incentivize domestic investment in the natural gas infrastructure by enabling market-based price discovery of natural gas. Further, the gas trading platform will also minimize sellers' counterparty default risk across the cross-spectrum of industries and drive competition across the value chain, supporting the overall affordability of gas and increasing its outreach.
- Gas exchange facilitates spot gas sourcing for various consuming sectors for meeting their day to day additional demand. The exchange market provides many flexible options for sourcing — ranging from day-ahead to six months — that can be optimally utilized by customers, suppliers and marketers.
- GAIL is India's largest gas pipeline, marketing, and processing company. It has diversified interests across the natural gas value chain of trading, transmission, LPG production & transmission, LNG re-gasification, petrochemicals and city gas, among others. GAIL is committed towards development of gas-based economy in the country. The partnership between GAIL and IGX will benefit the gas sector as a whole and pave way for further development of the gas market, more trading through gas exchange and increase in capacity utilization of pipelines.

6. Divestment of Govt. Share in GAIL and its implications?

GoI has 51.52% share in GAIL. The ongoing policy of strategic disinvestment of central public sector enterprises (CPSEs) is not likely to affect GAIL. Government of India has set a target of more than doubling the share of natural gas in the primary energy basket to 15% by 2030. GAIL is working relentlessly towards laying of natural gas pipelines to complete the national gas grid as well as expand city gas distribution network for achieving the same. In view of this, any disinvestment plan for GAIL is not expected.

7. How real is the threat from EV to GAIL's CNG business? Kindly provide the brief in view of various governments coming up with comprehensive EV policies to counter the pollution.

- The world is fast moving towards electrification of automobiles and India is trying to catch up to the speed of countries like China and many EU nations who have fast adopted the electric vehicles. In India, though, the government is focusing on electrification of vehicles, CNG fueled vehicle is here to stay. Of course, there will be competition from newly-launched EVs which may seem greener but the entire ecosystem is so big that every player is going to stay in the medium run.
- The economics and infrastructure are considered to be important decision drivers in vehicle acquisition in India. The economics of CNG vehicles in terms of upfront costs, availability of

finance, resale value, choice of models and infrastructure in terms of refueling, repair and maintenance is considered superior to EVs in the medium term.

- While electric vehicles are low on cost of running, they have a high initial cost of acquisition. CNG vehicles, on the other hand, have a low acquisition cost, low running cost and this makes CNG vehicles a replacement vehicle in the immediate run.
- Government has been implementing several policy push for EV ecosystem, such as- PM E-DRIVE scheme, Scheme for Manufacturing of Electric Cars (SMEC), FAME scheme, Critical Mineral Mission etc. As per the market forecast, passenger vehicle (PV) segment is likely to grow at 10% CAGR. This will result in to PV segment market sale reaching to ~ 62 lakh per year market from ~ 39 lakh per year today in next 5 years. Even at an optimistic 30% EV penetration by 2030, there will still be market for more than 42 lakh ICE vehicles. CNG will, therefore, play a major role in the future of ICE-technology vehicles.
- Another factor that will drive the demand for CNG vehicles is the fast pace at which newly bided GAs are being developed along with setup of CNG/LNG fueling stations. No of CNG stations has increased from around 6,861 fuel stations in FY23 to 7,594 in FY25. These numbers are going to further increase with the creation of infrastructure and government push towards adoption of natural gas as a more sustainable fuel. As the CNG infrastructure increases, growth of CNG segment is inevitable.
- The new domestic gas pricing guidelines will provide a stable pricing regime for domestic gas consumers while at the same time providing adequate protection to producers from adverse market fluctuation. This is going to be a big positive for the CGD sector as a whole.

8. What are GAIL's plan to extend its footprints into renewable business? Comments over Renewable energy sources and their competitive position with NG in terms of costs and in production potential.

- GAIL is actively working towards diversifying the business portfolio to include renewable and alternative energy. GAIL has a total installed capacity of ~ 135 MW of renewable energy (out of which ~ 118 MW is wind and ~ 17 MW are solar energy projects).
- GAIL is also setting up 170 MW captive Wind project in Maharashtra and 100 MW captive Solar project in Uttar Pradesh.
- As a part of GAIL's commitment to reduce its carbon footprint, GAIL aims to increase its renewable energy capacity to 3 GW in 2030.
- In addition, GAIL is also exploring various business opportunities in bio-fuels, alternate energy such as CBG, Ethanol, Hydrogen etc. GAIL is setting up first-generation (1G) 500 KLPD capacity grain-based bio-ethanol blending plant. GAIL has set up a CBG Unit at Ranchi which will produce 5 Tons of Compressed Biogas (CBG) per day and approximately 25 Tons of Fermented Bio Manure per day. GAIL is also in the process of setting up of 26 CBG plant in line with the mandate from MoPNG for promoting Atmanirbhar Bharat.

- Currently, India's total installed power capacity is ~ 475 GW (as on 31.03.2025) with coal having highest share of 46.7%, followed by Renewables at 36.3%, Hydro at 10%, gas at 5.2% and nuclear at 1.7%. In 2010-11, installed RE power capacity was only 18 GW. Since then, with a CAGR of 20%, it has reached 220 GW. The increase in capacity addition is primarily attributed to the reduction in solar panel prices and incentives given by Govt.
- Currently around 25 GW of Gas based power capacity is installed in India which are operating at very low utilization levels due to issues of affordable fuel supply. Therefore, to derive the maximum of the existing gas-based power assets, Govt. needs to provide suitable incentives/schemes for the sector:
- Promoting “Natural Gas” as a source of fuel under the auctions conducted by SECI for Round the Clock power supply
- Considering natural gas-based power plants are much cleaner in respect of PM, SO_x, NO_x & CO₂ emission compared to a coal-based power plant, and almost 12 GW gas-based power plants are being stranded, it is recommended that natural gas should be used in order to reduce the impact of pollution in the environment and health. The cost of externalities of coal-based power plant needs to be considered in merit order dispatch.
- Gas-based power can also be used to satisfy the needs of open customers, and in order to make the power more affordable, the cross-subsidy surcharge for purchase of power from gas-based projects can be relaxed.
- Further, being a polluting fuel, there are no penalties presently applicable for Petcoke and FO in India. However, the applicable GST also fully available for input credit to the consumers, therefore indirectly incentivizing its use. Natural Gas needs to be considered under the ambit of GST. This will help gas to compete with other fuels.
- Waiver of cross-subsidy surcharges for gas-based power in line with the existing Renewable Purchase Obligation which was set up to promote cleaner renewable energy, a policy initiative such as Clean power purchase obligation (CPO) may be framed. The policy will help meet India's INDC commitment as well as the share of the gas-based power will increase in the present electricity generation mix.
- Banning of use of diesel generator in CGD areas
- Through 'Peaking power policy', gas-based power can be promoted in the economy.
- Only renewable and gas-based power should be allowed to export to neighboring countries considering the pollution impact on India.

9. Write-up on the green hydrogen production and sale strategy of the company along with GoI's vision towards green Hydrogen and future expansion plans.

- National Hydrogen Mission documents has been launched by Hon. PM in January, 2023, and it aims to aid the government in meeting its climate targets and making India a green hydrogen hub, with an annual production capacity of 5 million tons, with an associated addition of approximately 125 GW in renewable energy capacity.

- GAIL has started hydrogen blending as a pilot project, through its JV Avantika Gas Limited (AGL) in City Gas Station (CGS) at Indore, to establish the techno-commercial feasibility of blending hydrogen in the CGD network. Initially the pilot project was done with 2 % v/v blending Hydrogen in Natural gas. Currently, 5 v% v/v hydrogen is being blended in AGL's domestic, commercial and industrial PNG network.
- GAIL has engaged M/s EIL to undertake a study to assess impact on CGD pipeline Network/NG pipeline due to various levels of Hydrogen blending. laboratory testing for higher grade of pipelines are undergoing at IIT- Kanpur. Based on the outcome of the pilot projects and recommendation of study by EIL & IIT Kanpur, further decision to scale up the pilot projects for blending/transportation of hydrogen will be taken.
- Moreover, GAIL has set up a 10 MW electrolyser at GAIL, Vijaipur which is of 4.3 TPD/1.4 KTA capacity of Green Hydrogen.
- GAIL has signed MoU with Industrial Promotion and Investment Corporation of Odisha Limited (IPICOL) for the setting up of green Hydrogen/green ammonia plants along with the possibility of Renewable unit.
- GAIL has also an MoU with Pipeline Infrastructure Limited (PIL) to collaborate on potential business areas in transportation and use of hydrogen and to share best practices.
- GAIL is also evaluating setting up of a Centre For Excellence in GAIL, under R&D scheme of National Green Hydrogen Mission.
- Additionally, GAIL is currently developing a comprehensive Hydrogen Roadmap and strategy, while assessing cost competitiveness, conducting market analysis, and exploring viable business models.

10. Write up on “Net Zero”. GAIL’s Strategy along with GoI vision.

- GAIL has developed Net Zero GHG Emissions target to achieve a 100% reduction in Scope 1 and Scope 2 emissions and a 35% reduction in Scope 3 emissions by 2035. GAIL has made its Net Zero strategy based on 4 strategic pillars: Operational Decarbonization, Energy transition, Carbon Capture and Utilization (CCUS), and Offsetting.
- The company has further opened up its energy basket with forays into Wind, Solar, Ethanol Blending, CBG, and Green Hydrogen towards shaping a Net Zero future.
- GAIL is currently undertaking various initiatives while exploring possibilities of investing in projects for utilizing and storing carbon.
- GAIL and LanzaTech, USA signed MoU on 18th July 2023. This is a step towards Circular & Green Economy. his MOU is a landmark step towards Net Zero journey of GAIL. The pioneering technology of LanzaTech converts CO₂ emissions into Biofuel conversion technology will help GAIL in reducing its carbon footprint.
- GAIL is also evaluating possible opportunities for valorization of CO₂ emitted from its Gas processing plant at Vijaipur, Madhya Pradesh and Petrochemical plant at Pata, Uttar Pradesh.

11. Direct LNG sales and LNG transportation potential in India long-term, and what is the GAIL strategy in this regard.

- India has been the world's fourth-largest importer of liquefied natural gas (LNG). The LNG import gas grown by a CAGR of 6.3% during 2015 and this is going to further increase on account of limited domestic natural gas production and government aim of increasing the natural gas share in the primary energy mix. LNG import in the country during FY24 was 37 BCM (~101 MMSCMD), a 15.4% increase compared to the previous fiscal year

LNG in Transport:

- Switching heavy-duty vehicles to LNG reduces harmful air pollution and also diversifies the fuel mix. LNG has the potential to offer fuel cost savings when compared to conventional diesel. LNG as a transportation fuel, particularly for heavy-duty transport and inter-state bus transport, can help lower emissions and has been identified as one of the new generation fuels by the Government of India.
- The Indian government has identified the use of LNG as a transport fuel as a priority area, considering the potential of manifold benefits in terms of reducing vehicular pollution, saving in terms of import bill of the country, and wide-ranging benefits that may accrue to fleet operators, vehicle manufacturers and other entities in the gas sector. LNG offers a cleaner and economical solution to the commercial transport industry.
- GAIL is committed to increasing Natural Gas consumption in the country by exploring various new areas for the use of Natural Gas / LNG. LNG as Transport Fuel in Heavy Duty Trucks and Buses is one of such areas. GAIL has recently set up dedicated Retail LNG team. GAIL board has recently accorded approval for an investment plan of Rs 650 crore and aspire to capture 50% plus market share in the next 5-6 years. Accelerated adoption of LNG-fueled heavy-duty vehicles could generate an additional gas demand of approximately 2 bcm/yr by 2030. Potential incentives to achieve this include road toll, exemptions for LNG vehicles, reduced fuel taxes on LNG, domestic gas allocation for LNG vehicles and corporate fuel efficiency standards for heavy-duty vehicle (HDV) manufacturers.

12. Sector wise demand for Natural Gas in World as well as in India for next five years? How the world has changed for Natural Gas consumption after geopolitical issue of Russia-Ukraine war?

- Sector wise Indian Natural Gas Consumption in India.

Total (MMSCMD)	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
Fertilizer	44.15	48.72	49.53	53.15	57.66	56.30
City Gas Distribution	29.82	25.29	33.36	32.95	39.96	41.40
Power	30.36	29.69	27.83	22.34	24.88	24.80
Refinery	21.33	21.67	14.56	10.71	15.99	16.00

Total (MMSCMD)	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
Petrochemicals	9.78	8.42	7.85	5.37	7.31	9.20
Others	19.34	19.96	35.35	36.31	45.59	48.57
Total	154.78	153.74	168.47	160.83	188.39	196.27

- Global demand for natural gas held up better than demand for other fossil fuels during the first year of the Covid-19 pandemic, and then increased by 5% in 2021, double its average growth rate over the past decade. A dearth of new projects, weather-related increases in demand, LNG outages and reduced Russian exports tightened the global gas supply balance from mid-2021 and put upward pressure on prices, especially in Europe where the Title Transfer Facility (TTF) benchmark rose from less than USD 10 per million metric British thermal units (MMBTU) in the first-half of 2021 to over 30 USD/MMBTU by December 2021.
- Russia's invasion of Ukraine in February 2022 had a huge impact on an already fragile global gas balance. European Union efforts to fill gas storage ahead of the winter have run up against Russia's strategic withholding of gas supply and the prospect of possible supply shortages bringing high levels of market volatility and exceptionally high prices. Europe's TTF benchmark price saw peaks exceeding USD 90/MMBTU in 2022, even as the European Union and its partners debate ways to reduce reliance on Russian gas and curtail its revenue from energy sales.
- The strains on gas supply have also led to energy shortages in several parts of the developing world that rely on imported gas, notably Pakistan and Bangladesh. Major growth markets for gas such as India and China have meanwhile sharply reduced their LNG imports in 2022.
- Gas Demand in the world by region

Demand (BCM)	2010	2022	2023	2030	2035	2050
North America	838	1167	1175	1121	1028	811
United States	678	939	940	883	791	578
Central & South America	150	156	156	168	180	183
Brazil	28	32	30	34	35	32
Europe	697	544	507	462	407	301
European Union	446	358	331	296	257	166
Africa	107	176	182	203	219	290
Middle East	370	566	578	702	750	880
Eurasia	573	652	660	671	671	669
Russia	467	515	523	521	513	485
Asia Pacific	577	904	928	1075	1131	1191
China	110	370	398	499	522	523
India	64	60	64	99	125	172
Japan	95	94	85	61	49	41
South East Asia	149	167	175	209	231	265
International Bunkers	-	-	-	27	36	51
Total	3312	4166	4186	4430	4422	4377

Source: IEA energy outlook 2024. 2030 & 2050 forecast is as per Stated Policies Scenario. The Stated Policies Scenario (STEPS) shows the trajectory implied by today's policy settings

- In India, natural gas demand in the STEPS scenario² is projected to reach 99 bcm by 2030. Most of the growth to come from manufacturing and other industry, helped by the expansion of city gas distribution networks. Heavy industrial and manufacturing sectors, such as iron and steel production, are also driving demand. Gas use in oil refining is expected to increase by more than 4 bcm/yr as more refineries get connected to the grid. Gas demand in the power generation sector is projected to reach nearly 15 bcm/yr by 2030, driven by a recovery in large gas-fired power plants and the rapid expansion of gas use in captive power plants³. Gas consumption in the sponge iron and steel segment is projected to reach close to 2 bcm by 2030. Accelerated adoption of LNG-fueled heavy-duty vehicles could generate an additional gas demand of approximately 2 bcm by 2030.

13. Views on regulatory push towards restriction of use of polluting fuels like FO, Petcoke and coal among industries.

- Though industries understand that natural gas is cleaner, efficient, require less maintenance, and importantly reduces their regulatory cost of implementing pollution control measures, the cost of natural gas renders industries uncompetitive vis-a-vis competitors using polluting fuel. Downstream industries are not getting the benefit of the tax credit of VAT paid on purchases of gas which is available for alternate fuels/feedstock. Domestic LPG is under GST regime and is taxed at a lower rate of 5%. GST in the case of petcoke / fuel oil is 18 % and 5% on coal with full input credit.
- All the possible positive outcomes of rolling out CGD units, expanding National Gas Grid (NGG), introducing BS-VI standard etc. may not fructify if India continues to allow highly polluting fuels (F.O. & Petcoke) in industries. Further, the development and growth of gas trading in exchanges may not be successful due to tax arbitrage in various states resulting in incorrect price discovery.
- States may be persuaded to accept a change in the extant tax system on natural gas and bring natural gas under lower bracket of GST considering overall health benefits and reduction of regulatory pollution control cost to industries.

14. What are the threats for increase in Gas as primary energy source in economy?

- According to various long-term estimates on energy consumption mix, natural gas will remain as the highest growing fossil fuel in the long-term future as coal and oil are expected to peak in short to mid-term.

² Stated Policy Scenarios (STEPS)

³ India Gas Market Report- Outlook to 2030 (Page-6)

- According to B.P. statistical review 2023, natural gas is only fossil fuel that shows growth in levels throughout to 2050 in all scenarios (Accelerated, Net Zero and New Momentum scenarios).
- Historically as well, the natural gas sector has shown a decent growth rate compared to other fossil fuels as shown in the table below:
- India historical Primary Energy Mix4, BCM

India Primary Energy Mix							
BCM	Oil	Natural Gas	Coal	Nuclear energy	Hydro electric	Renewables	Total
2018	278.6	58.6	519.6	9.8	35.0	30.9	932.5
2019	279.7	59.7	520.7	11.3	40.4	37.1	949.0
2020	254.1	61.0	487.2	11.3	43.3	44.4	901.3
2021	259.0	62.7	540.4	11.2	42.3	51.0	966.6
2022	281.4	58.5	562.5	11.8	45.9	60.2	1020.3
2023	293.6	62.6	610.4	12.0	38.7	66.1	1083.5
CAGR	1.05%	1.33%	3.27%	4.13%	2.03%	16.43%	3.05%

- In view of the above, the natural gas sector is well placed for growth in future. However, a few threats, along with measures to address these threats are given as follows:

	Threat	Measures to address the threat
1	Decreasing domestic natural gas production	Attract investment in upstream sector through favourable policies and pricing
2	Volatility in imported LNG price	LNG price in India have to be competitive compared to substitutes. In the last two years we have seen volatility in LNG prices which from the past 6-7 months have reduced. This was mainly attributed to ongoing conflict, geopolitical reason, increased LNG requirement by Europe etc. However, lot of investment are undergoing in upstream sector and many FIDs have taken place. The gas prices are going to come down in medium term. Sourcing of gas on long-term basis from diversified sources will bring stability in imported LNG prices.
3	Declining usage of natural gas in power sector	Incentivise gas usage in power sector by giving incentives similar to RE power sector like purchase obligations, must run status, tax and cost incentives, including Gas under GST, reintroduction of schemes like PSDF scheme. Carbon tax or equivalent may be levied on coal-based generation

	Threat	Measures to address the threat
		to move from the current Merit order dispatch to “Emission Based –Merit Order Dispatch”
4	Competition from alternate polluting fuels like coal, petcoke, fuel oil etc.	Similar to ban on usage of polluting fuel in NCR and Morbi cluster in Gujarat; fuels like petcoke, coal, fuel oil needs to be dis-incentivized in view of cleaner combustion properties of gas compared to these fuels.
5	Competition from new and disruptive technologies/Mobility solution	Green Hydrogen: Currently cost of green hydrogen for use in refinery, fertilizer etc. is not competitive with natural gas price. Electric Vehicle: As per the market forecast, passenger vehicle (PV) segment is likely to grow at 10% CAGR. This will result in to PV segment market sale reaching to ~ 62 lakh per year market from ~ 39 lakh per year today in next 5 years. Even at an optimistic 30% EV penetration by 2030, there will still be market for more than 43 lakh million ICE vehicles. CNG will, therefore, play a major role in the future of ICE-technology vehicles
6	Complicated tax structure across country	Bring natural Gas under lower bracket of GST

15. What Policy support required for India’s vision for increasing share of gas in its economy?

- Natural Gas is to be brought under the ambit of GST under its lowest bracket.
- Emission factored merit order dispatch to be considered for Gas based power
- Gas based power in “Must Run” category as is the case in RE power with 10% Gas purchase obligation (GPO) w.r.t. thermal power.
- Restrictions may be imposed on use of high polluting fuels like coal, pet-coke, Fuel Oil (to be defined in terms of non-permissible emission content) in industrial clusters close to urban centers
- Policy guidelines to encourage the use of LNG in heavy-duty vehicles via tax rationalization, the PLI scheme, and other means.
- Nationwide Unified CGD Policy, mandating state governments/ authorities to allocate land parcels in city development plans and highways for City Gate Stations, CNG/PNG Infrastructure, LNG stations, District Regulating Stations; Single window clearances for the development of CGD/LNG infrastructure
- Declaration of cities covered under PNG as LPG free zone.
- Perpetual exclusivity for marketing and infrastructure for CGDs

- Continued preference for allocation of APM gas to CGDs
- CGD needs supports such as Exemption of CNG from excise duty, Single window permission, Restrictions on polluting fuels in CGD areas, Tax reduction in CNG and LNG vehicles in line with Electric Vehicle, Rationalization of permission and road restoration charges for laying PNG pipelines etc.
- Input tax credit for pipeline laying may be allowed to incentivize the investment in pipeline infrastructure. GST on LNG regasification and Natural Gas compression may be reduced @ 5% with the benefit of Input Tax Credit.
- Single window permission may be implemented in all states for CGD projects with waving off Restoration/Permission charges for one year. Creation of dedicated “Task Force” comprising of member of CGD industry, State Government and Central Government to implement the well drafted “CGD policy”. Soft financing arrangement, including a grant towards switching to PNG may be provided to MSMEs.
- Creation of “Infrastructure Fund” to provide Viability Gap funding for creation of pipeline/ pipeline connectivity to the proposed industrial corridors under National Industrial Corridor Development Corporation.
- Financial support for promoting Gas-based DRI production facilities and adoption of natural gas for conventional iron and steel making industry.

16. Brief details on GAIL’s plan under /for Government of India’s vision Viksit Bharat. Also indicate timelines, action plan and other relevant details thereof.

- Viksit Bharat 2047 is a visionary initiative aimed at transforming India into a developed nation by 2047, the 100th year of independence. The vision emphasizes sustainable development, innovation-led growth, and equitable prosperity for all citizens, aligning with India’s aspirations to be a global leader. GAIL (India) Limited, as a leading Public Sector Undertaking in the energy sector, is strongly aligned with this vision and is strategically positioned to contribute meaningfully to India’s development agenda.
- As per projections by the International Energy Agency (IEA), India’s primary energy demand is expected to reach 2,222 Billion Cubic Metres (bcm) (80 EJ, 1910 mmtoe) by 2050. With Natural Gas setting to have 15% share in India’s primary energy mix- India is looking at a Gas consumption of 333 bcm (12 EJ, 287 mmtoe) from current 64 bcm by 2050.
- Moreover, India’s current per capita petrochemical consumption stands at approximately 12 kilograms, which is just one-third of the global average. By 2047, India aims to reach consumption levels at par with global standards, translating into a projected total demand of 60 million metric tonnes. GAIL intends to play a catalytic role in meeting this demand by augmenting its petrochemical production capacity and developing integrated value chains.
- GAIL is committed to making substantial investments across the Natural Gas and Petrochemical value chains, including pipeline infrastructure, LNG sourcing and regasification, and

downstream petrochemical projects. These investments will be instrumental in ensuring access to clean energy, supporting industrialization, and creating employment across regions.

- In line with Viksit Bharat's emphasis on sustainability, GAIL is also investing in renewable and green energy. The Company is integrating ESG principles into its operations and is working towards achieving net-zero goals in alignment with India's climate commitments.
- Through robust planning, innovation, and partnership with stakeholders, GAIL aims to be a pillar of India's journey towards becoming a sustainable and developed nation by 2047.

17. In view of fast changing energy scenarios, should GAIL make its way in to Renewable energy on a major scale in addition to "Net Zero" investments? Kindly elaborate your view on this.

- GAIL (India) Limited was established in 1984 with the main objective to create necessary infrastructure for transportation, processing and distribution of natural gas. GAIL, since its inception in 1984, has played a pivotal role in building the natural gas infrastructure of the country. Over the years, the company has strategically evolved from a gas transmission entity to an integrated energy company with a strong emphasis on sustainability and clean energy across its operational framework.
- GAIL's mission- "Enhancing quality of life through clean energy and beyond"- reflects its deep-rooted commitment to environmental stewardship. This vision naturally aligns with global and national efforts toward a sustainable energy future.
- GAIL has declared an ambitious Net Zero target by 2035. A robust renewable energy portfolio is instrumental in realising these goals and reflects GAIL's forward-looking approach in supporting national climate objectives and the international transition to a low-carbon future.
- GAIL is actively exploring and investing in a wide array of renewable and alternative energy segments, which include:
- GAIL has set up a CBG Unit at Ranchi which will produce 5 Tons of Compressed Biogas (CBG) per day and approximately 25 Tons of Fermented Bio Manure per day. Setting-up of similar plant at Bengaluru is underway.
- GAIL is setting up first-generation (1G) 500 KLPD capacity grain-based bio-ethanol blending plant
- GAIL is also in the process of setting up of 26 CBG plant in line with the mandate from MoPNG for promoting Atmanirbhar Bharat
- GAIL has established India's first green Hydrogen plant of 10 MW capacity producing 4.3 TPD of 99.999% pure hydrogen. GAIL is also setting up around 20 MW Solar power plants at Vijaipur (both Ground Mounted and Floating) to meet the requirement of green power for the 10 MW PEM Electrolyzer.
- GAIL has also successfully run pilot project of blending Hydrogen in City Gas distribution Network.
- GAIL has set-up a JV with Coal (India) Limited for coal to Synthetic Natural Gas project

- As a part of GAIL's commitment to reduce its carbon footprint, GAIL aims to increase its renewable energy capacity to 3 GW in 2030.
- GAIL is committed to explore the potential of various clean and green energy solutions along with expanding and diversifying its core business portfolio.

Compressed Bio Gas (CBG)

1. Kindly brief about the potential of Compressed Bio Gas (CBG) in India. What is the present status of CBG production in India vs the target?

National Policy on Biofuels - 2018 allows usage of bio fuels in place of (or blend with) diesel, petrol or other fossil fuels. The Biofuels include, Bioethanol, Biodiesel, Bio-CNG (also known as Compressed Bio Gas (CBG)) etc. The National Policy on Biofuels, aims to increase usage of biofuels in the energy and transportation sectors of the country in the coming decade.

Under Swachh Bharat Mission (Grameen), Department of Drinking Water and Sanitation (Ministry of Jal Shakti) has launched **GOBAR-dhan** (Galvanizing Organic Bio-Agro Resources Dhan) initiative in April 2018. GOBARdhan is an umbrella initiative of Government of India based on the whole of Government approach. It covers the entire gamut of schemes / programmes / policies promoting the conversion of organic waste like cattle dung/ agri-residue etc. to biogas / CBG / Bio CNG. **500 new 'waste to wealth' plants** under GOBARdhan initiative will be established for promoting circular economy. These will include **200 Compressed Bio Gas (CBG) plants**, including **75 plants in urban areas**, and **300 community or cluster-based plants at total investment of INR 10,000 crore**.

Under Swachh Bharat Mission - Urban 2.0, the target is to establish 15,000 TPD Bio-CBG plants by 2026. Ministry of Housing and Urban Affairs (MoHUA) has additionally instructed States/Urban Local Bodies (ULBs) to assist Oil and Gas Marketing Companies (OGMCs) in establishing CBG plants. This assistance includes offering long-term concession agreements and land leases, ensuring a continuous supply of segregated organic municipal waste etc.

'**SATAT**' (Sustainable Alternative Towards Affordable Transportation) scheme on Compressed Bio Gas (CBG) was launched by Hon'ble Minister, Petroleum & Natural Gas on 1.10.2018. A total of **403** Letters of Intent (LOIs) have been issued by GAIL against the target of **400** LOIs by **FY 2023-24** to various entrepreneurs for supply of Compressed Bio Gas (CBG) under **SATAT** initiative.

MoP&NG has mandated OGMCs to set up 195 CBG plants out of which GAIL has been mandated to set up 26 CBG plants. Out of these 26 CBG plants, GAIL has already commissioned 1 (one) CBG plant at Ranchi and activities are going on for remaining.

2. Kindly brief about the initiatives of Govt. for development of CBG infrastructure/ecosystem in India.

The following steps have been taken by the **Government of India** to promote SATAT initiative:

I. Financing of CBG projects/ Financial Assistance:

- a) Inclusion of CBG projects under RBI priority Sector lending.
- b) Roll out of standard loan policy for CBG Projects from Financial Institutions/Banks and Canara Bank nominated nodal bank.

- c) MNRE Capital Subsidy for CBG plants: MNRE is providing Central Financial Assistance (CFA) of Rs 4.0 Cr for CBG plant with minimum installed capacity of 4800 kg/day – Max. CFA up to Rs.10 Cr/project (pro rata CFA of Rs 1.0 Cr per 1.2 TPD)

II. Arrangement of Feedstock:

- a) State Governments advised to streamline setting up of raw material supply chain.
- b) Department of Agriculture and Farmers Welfare has notified revised Crop Residue Management Guidelines (CRM 23-24) which inter-alia provides financial assistance to Farmer or group of Farmers or Rural Entrepreneurs or Cooperative Societies of Farmers or Farmers Producer Organizations (FPOs) & Panchayats having agreement with the biomass-based industry to facilitate establishment of crop residue/paddy straw supply chain. Under this scheme government is providing financial support up to 65 % for procurement of set of biomass aggregation equipments having project cost up to Rs. 125 lakhs with 25% of project cost as Industry contribution. Beneficiary have to pay only 10% of project cost.
- c) Government of India has approved the Scheme for providing financial assistance to Compressed Bio Gas Producers (CBG) producers for procurement of biomass aggregation machinery to support collection of biomass with a total financial outlay of Rs. 564.75 crore for the period of FY 2023-24 to FY 2026-27.

III. Off-Take of CBG:

- a) The procurement price of CBG linked with the Retail Selling Price (RSP) of CNG
- b) The Ministry of Petroleum & Natural Gas, Government of India issued policy guidelines dated 09.04.2021, 26.10.2021 and 26.10.2023 for synchronisation of CBG produced by plants in the CGD network. GAIL has been mandated to operationalise the CBG-CGD Synchronisation Scheme and supply Biogas/CBG co-mingled with domestic gas at Uniform Base Price (UBP) to CGD entities for use in CNG (T) and PNG (D) segments of CGD network. Current scheme is valid till 08.04.2034.
- c) The government has mandated blending Compressed Biogas (CBG) into Compressed Natural Gas (CNG) and Piped Natural Gas (PNG), with a goal of 5% CBG blending by FY 2028-29. CBG Blending Obligation (CBO) will be voluntary till FY 2024-2025 and mandatory blending obligation would start from FY 2025-26. CBO shall be kept as 1%, 3% and 4% of total CNG/PNG consumption for FY 2025-26, 2026- 27 and 2027-28 respectively. From 2028-29 onwards CBO will be 5%. A Central Repository Body (CRB) shall monitor and implement the blending mandate based on the operational guidelines approved by Hon'ble Minister, PNG.
- d) The government has approved scheme guidelines for “Development of Pipeline Infrastructure (DPI) for injection of Compressed Bio Gas (CBG) in City Gas Distribution (CGD) network” with a total financial outlay of Rs. 994.50 crore during the period of FY 2023-24 to FY 2025-26. The scheme shall provide financial assistance for creating CBG-CGD grid connectivity for 100 CBG projects and create ecosystem for offtake of CBG with reduced logistic cost.
- e) Exemption of Excise Duty on CNG blended with CBG to the extent of quantity of CBG subject to GST.

IV. Off-Take of FOM/ LFOM:

- a) Inclusion of Fermented Organic Manure (FOM) and Liquid Fermented Organic Manure (LFOM) under Fertilizer Control Order (FCO). Department of Agriculture and Farmers Welfare has waived of requirement of obtaining authorization letter for selling of FOM/LFOM for three years (FY 2022-2025).
- b) Fertilizer Marketing Companies (FMCs) to arrange for mandatory offtake of FOM / bio-fertilizers with chemical fertilizers as a “Basket -Approach”
- c) Market Development Assistance (MDA) in the form of Rs 1500 per MT to support marketing of organic fertilizers (FOM/LFOM/PROM) produced as by-product from Bio- gas /CBG Plants set up under umbrella GOBARDhan initiative.

3. Has there been any country(ies) where CBG has gained considerable share in Energy basket/Natural Gas consumption?

GAIL Corporate CBG Team is not dealing with any matter whatsoever related to CBG outside India. The information below is from online sources

1. Germany

- **Significant Share:** Germany has been a leader in the adoption of biogas and CBG. It has well-established biogas production from agricultural waste, and CBG is used both for electricity generation and as a fuel for vehicles.
- **Infrastructure:** Germany has developed a comprehensive infrastructure for biogas production, distribution, and use, making it one of the largest biogas consumers globally.
- **Regulatory Support:** Policies like the Renewable Energy Sources Act (EEG) have played a key role in incentivizing biogas production and use.

2. Sweden

- **High Utilization:** Sweden is a major user of biogas, and CBG is used extensively for both electricity generation and transportation. The country has integrated biogas into its energy mix, including in the transport sector, where CBG is used as an alternative to natural gas and diesel for public buses and waste collection vehicles.
- **Sustainability Goals:** Sweden's commitment to sustainability and reducing reliance on fossil fuels has driven the growth of biogas production and use.

3. Denmark

- **High Utilization:** Denmark is a global leader in integrating Compressed Biogas (CBG) into its energy mix. As of 2023, biogas supplies around 40% of the country's natural gas consumption, with projections to reach 72% by 2030 and 100% green gas by 2034.
- **Sustainability Goals:** This success is driven by strong policy support, widespread grid integration—over 50 biogas plants are connected to the national gas grid—and environmental benefits such as major CO₂ emission reductions. Denmark's biogas journey highlights its commitment to sustainable energy and sets a global benchmark for renewable gas adoption.

4. United States

- **Growing Adoption:** The U.S. has seen a growth in CBG production, particularly from agricultural waste and landfills. While it's not as large-scale as in Germany or Sweden, CBG is becoming more significant, especially for transportation fuel and power generation.
- **Government Incentives:** Federal and state-level incentives, like the Renewable Fuel Standard (RFS), have encouraged the development of renewable natural gas (RNG) projects, many of which use CBG.

5. Italy

- **Adoption for Transportation:** Italy has been a leader in using CBG for transportation, particularly in cities where there is a strong push to reduce air pollution. The country has also invested in CBG infrastructure for fueling vehicles.

6. Canada

- **Increasing Use:** Canada has also made strides in integrating CBG into its energy mix. Several provinces, including Ontario and British Columbia, have seen growth in CBG projects, particularly related to waste-to-energy and transportation.
- These countries have successfully integrated CBG into their energy systems, showcasing its potential as a viable, sustainable alternative to traditional fossil fuels, particularly in the transport sector and for electricity generation. The key to success in these nations has been strong governmental support, favorable policies, and a clear focus on sustainability and reducing reliance on natural gas and other fossil fuels.

4. What are the major challenges w.r.t. development of CBG ecosystem in India? Kindly brief about the possible resolution to such challenges.

1. Availability of Land

Challenge:

- Suitable land availability for setting up CBG plants is a bottleneck, especially in proximity to feedstock sources and CGD networks.

Resolution:

- **State Government Support:** Fast-track land allotment by district administration and state agencies, especially for projects under national missions like SATAT.
- **Policy Interventions:** States can earmark industrial or waste management zones for CBG projects.
- **Public-Private Models:** Explore land-leasing models from Panchayats or government bodies.

2. Offtake of CBG (due to long plant-to-market distance)

Challenges:

- High logistics cost due to use of cascades (cylindrical transport systems).
- Limited pipeline connectivity for direct offtake into CGD networks.

Resolutions:

- **Enhance Pipeline Infrastructure:** Prioritize CGD expansion in high-CBG-potential areas.
- **Local Usage Incentives:** Encourage local bulk consumers like public transport, industries, and hotels to adopt CBG.
- **Mini-grid Development:** Develop decentralized mini-pipeline networks around large plants.

3. Sale of FOM

Challenges:

- FOM output (~4x of CBG) lacks consistent offtake.
- Fertilizer companies are not mandated by the Government to procure FOM, pricing of FOM is unattractive.

Resolutions:

- Mandate FOM Usage: Involve agriculture departments and make FOM procurement compulsory for fertilizer companies (like ethanol blending mandates).
- Branding & Awareness: Create a government-backed quality certification/branding to boost farmer confidence.
- Incentivize Use: Provide subsidies or promote inclusion in organic farming schemes (Paramparagat Krishi Vikas Yojana (PKVY), Zero Budget Natural Farming (ZBNF)).

4. Funding of CBG projects by banks

Challenges:

- Banks demand high collateral, personal guarantees, and charge heavy processing fees.
- Lack of familiarity with CBG business models.

Resolutions:

- Viability Gap Funding (VGF): Provide capital subsidies or risk guarantees.
- Credit Guarantee Fund: Government or NABARD-backed credit guarantee for green energy projects.

5. Availability of segregated MSW for CBG plants

Challenges:

- Plants struggle to receive segregated waste, critical for MSW-based CBG units.
- Need tipping fees for waste handling and processing.

Resolutions:

- Mandatory Waste Segregation at Source: Strict ULB enforcement with decentralized collection mechanisms.
- Formal Agreements: MSW supply MoUs between Urban Local Bodies and CBG developers.
- Tipping Fees: Include tipping fee reimbursement in plant revenue models like in waste-to-energy projects.

6. Double taxation on CBG (GST on CBG & VAT when CBG blended with CNG)

Challenge:

- GST levied on CBG production and VAT when blended with CNG, leading to cascading tax effect.

Resolutions:

- Tax Harmonization: Exempt VAT when CBG is already taxed under GST.
- Policy Advocacy: Represent the case to GST Council for single-point taxation or concessional tax rate.

8. In absence of the benchmark cost related to CBG projects, project appraisal by banks is difficult

Challenge:

- Banks and financial institutions find it difficult to appraise CBG projects due to lack of standard cost benchmarks.

Resolutions:

- Standardized Cost Models: MNRE or NITI Aayog can publish technology-wise cost norms and IRR models.
- Third-party Validation Mechanism: Certification from empaneled agencies on DPRs for financing institutions.

9. Encashment of Revenue from Carbon Credits

Challenge:

- Developers face challenges in registering and monetizing carbon credits due to high costs and procedural complexity.
- No clarity regarding monetizing/trading mechanisms of Carbon Credit in India

Resolutions:

- Government Support for Carbon Market Access: Help Indian CBG units get registered under global schemes like VERRA or Gold Standard.
- National Carbon Credit Registry: Streamline the verification and encashment of carbon credits within India.
- Capacity Building: Support training and handholding for project developers on carbon trading mechanisms.
- These steps—if taken holistically with strong inter-ministerial coordination (MNRE, MoPNG, MoEFCC, MoA, and state governments)—can accelerate the growth of a robust and self-sustaining CBG ecosystem in India.

5. What is GAIL's vision regarding Compressed Bio Gas? What are targets decided in this regard?

MoP&NG has mandated GAIL to set up a total of **26 CBG plants**.

GAIL's efforts towards formation of JVs for execution of CBG plants:

- a.GAIL signed a Term Sheet for its equity participation in TruAlt's joint venture company, **Leafinti Bioenergy Limited** for setting up 10 CBG plants in Karnataka & Maharashtra (**Press mud**). Approval for JV formation sent to MoP&NG
 - b.GAIL has issued EOI to Consortium of Hindustan Waste Treatment Private Limited (HWTPL) & Vasudha Waste Treatment Private Limited (VWTPL) for setting up MSW based CBG plants through JV mode. JV agreement under discussion
- GAIL has signed MoU with **Verbio India** on 25.10.2024 for setting up **Agri** based CBG plants. JV agreement under discussion

6. Kindly provide the details of GAIL's CBG plants which are operational as on date, under construction and under planning along with location, capacity, type (MSW based, agricultural etc).

MoP&NG has mandated GAIL to set up a total of **26 CBG plants** directly or through its JVs/ Affiliates for which GAIL is undertaking following activities:

1. Operational project: GAIL Ranchi CBG plant:

- a.Plant inaugurated on 02nd Oct., 2024 by Hon'ble PM.

b.Limited biogas production due to inadequate supply of segregated MSW (max. 50 TPD received). Being pursued with Municipal Corporation and matter escalated at all appropriate levels of Govt. for increase in feedstock.

c.Sale of CBG commenced from 17th Jan., 2025. Average sales of nearly 800 kg per day at GAIL Saparom COCO CNG station.

d.MDPE pipeline connectivity expected by end of Apr. 2025 for gas offtake. 12 Kms out of 13 Kms of required length of pipeline laid.

2. Land firmed up (in respect of **upcoming / proposed projects**) for setting up CBG plants at following locations:

Sl.	Location	Feedstock	Capacity (TPD)	DFR Status*	Land Status**	Present Status
1	Kanpur (CUGL)	MSW	20	Prepared	Handed Over	LoI issued to & MOU signed with KEVA Green India Private Limited. Boundary wall construction commenced. Removal of legacy waste by KMC pending.
2	Sultanpur, UP	Paddy Straw	20	Prepared	Confirmed	Lease rent paid. Lease agreement signed on 05.03.2025.
3	Mumbai (MGL)	MSW	35	Prepared	Confirmed	Investment approval obtained. Due to delay in approval of lease rent by Maharashtra Govt., signing of CA & handing over of land from BMC is pending
4	Bengaluru City	MSW	10	Prepared	Confirmed	DFR submitted to KSPCB and request made for considering project under White Category. VGF sought due to high project capex
5	Bengaluru-Rural	MSW	6	Prepared	Confirmed	
6	Amritsar	Paddy Straw	20	Prepared	Confirmed	Price fixation for determining land lease rent completed. Lease agreement pending between
7	SBS Nagar, Punjab		15	Prepared	Confirmed	
8	Moga, Punjab		15	Prepared	Confirmed	

Sl.	Location	Feedstock	Capacity (TPD)	DFR Status*	Land Status**	Present Status
						PEDA & District Administration which is prerequisite for land hand over
9	Fatehgarh Sahib, Punjab	Paddy Straw	20	Prepared	Identified	No progress could be made due to farmer related issues. Being resolved through PEDA & State Govt.
10	Raigarh, Chhattisgarh	Mix of MSW & Paddy Straw	12.5	Prepared	Confirmed	Tripartite MoU signed with the respective Municipal Corporations and CBDA in Jan., 2025. Cabinet approval of Govt. of Chhattisgarh for nominal land lease rental (Rs. 1 per sq. metre per year) is awaited.
11	Korba, Chhattisgarh		12.5	Prepared	Confirmed	
12	Ambikapur, Chhattisgarh		12.5	Prepared	Confirmed	
13/14	Kolhapur, Maharashtra (02 plants)	Press Mud	12	Prepared	Confirmed	
15	Sangli, Maharashtra		12	Prepared	Confirmed	Plants being undertaken through JV which is being formed with M/s LBPL. Approval for JV formation sent to MoP&NG.
16	Solapur, Maharashtra		12	Prepared	Confirmed	
17	Belagaum, Karnataka		12	Prepared	Confirmed	
18	Pandavpura, Karnataka		12	Prepared	Confirmed	

* DFR status “**prepared**” signifies that techno – commercial feasibility report and financial model have been submitted by the consultant and are under finalization. Investment approval shall be sought once process related to land lease is concluded

** Land status “**confirmed**” signifies that land has been assured by the relevant authority (either govt. authority or JV partner) and hand over of same is under progress (to either GAIL or JV company)

** Land status “**identified**” signifies that land has been identified in coordination with relevant authority (either govt. authority or JV partner), however final assurance for availability and further allocation not yet confirmed.

During meeting chaired by Secretary (PNG) in Nov., 2024, GAIL has targeted to commence the project activities of above 18 plants by Sept., 2025. The target can be achieved only if land is handed over to GAIL by March, 2025

3. Apart from above locations, GAIL is also making efforts to set up CBG projects in additional location, the details of which are tabulated below:

Sl.	Location	Feedstock	Capacity (TPD)	DFR Status	Land Status	Remarks
1	Davanagare, Karnataka (MGL)	MSW	TBD	Under Preparation	Confirmed	--
2	Belagavi, Karnataka	MSW	TBD		Confirmed	--
3	Vijayanagara, Karnataka	MSW	TBD		Confirmed	--
4	Tumakuru, Karnataka	MSW	TBD		Confirmed	--
5	Kalaburgi, Karnataka	MSW	TBD		Confirmed	--
6	Kolar, Karnataka	MSW	TBD		Confirmed	--
7	Pathankot, Punjab	Paddy Straw	15	To be prepared	Confirmed	Price fixation for determining land lease rent received vide letter dated 18.03.2025. Lease agreement pending between PEDDA & District Administration which is prerequisite for land hand over
8	Mandaya, Karnataka	MSW	TBD		Identified	Pre-feasibility prepared. Feedstock survey underway
9	Hisar, Haryana	MSW	5		Identified	Site visit done on 24 th Feb., 2025
10	Jaipur, Rajasthan	MSW	5	To be prepared	To be identified	GAIL has submitted its proposal to MC on 05.03.2025.
11	Patiala, Punjab	Paddy Straw	15		Identified	No progress due to farmer related issues. Being resolved through
12	Ludhiana, Punjab		15		Identified	
13	Roop Nagar, Punjab		15		Identified	

Sl.	Location	Feedstock	Capacity (TPD)	DFR Status	Land Status	Remarks
						PEDA & State Govt.
14	Ujjain, Madhya Pradesh	MSW	5		Identified	UMC seeking 50 TPA CBG free of cost and discount on additional CBG qty. Withdrawal of demands required for further action
15	South Goa, Goa	MSW	5		Identified	GWMC is seeking EC from State Govt.

TBD: To be decided

Apart from above locations, 04 additional locations are being identified by LBPL (to be undertaken through JV route)

During meeting chaired by Secretary (PNG) in Nov., 2024, GAIL has set a target to commence the project activities of 07 plants (out of above 15 plants being prospected) by Dec., 2025. The target can be achieved only if land is handed over to GAIL by May/June, 2025.

Retail LNG

1. Kindly brief about the potential of Retail LNG for long haul transport in India. What is the present status of Retail LNG consumption for long haul transport in India vs the target?

The use of LNG as a fuel in transport sector is being accepted as one of the methods for decarbonization worldwide. Same is also a key component of Government of India's ambitious targets to increase the share of Natural Gas in the country's energy basket from presently ~6.5% to 15% by 2030. LNG as a transport fuel is a priority area, considering the potential of various benefits in terms of reducing vehicular pollution and wide-ranging benefits that may accrue to fleet operators, vehicle manufacturers and other entities in the gas sector. As per the draft LNG policy of the MoP&NG, the Government aims to have 1,000 LNG stations across major highways, industrial corridors and mining sectors in the country. In the first phase, GAIL, OMCs (IOCL, BPCL, HPCL), and PLL are setting up 49 LNG stations across the Golden Quadrilateral (GQ) and other major national highways of the country. Moreover, private players such as Cryo Gas, Ultra Gas and Baidyanath LNG have also started developing LNG dispensing infrastructure in the country. Presently, around 20 LNG stations have been commissioned in the country by both Public and Private entities, catering to around 800-900 LNG vehicles. As per various reports potential of Retail LNG for long haul transport in India could be between 15 - 25 MMSCMD by 2030.

Considering the potential of LNG as a transport fuel market in the country, GAIL has set up a separate Business Unit to focus on developing the LNG infrastructure in the country.

2. Kindly brief about the initiatives of Govt. for development of Retail LNG infrastructure/ecosystem in India.

As per MoP&NG's draft LNG policy of 2021, the Government aims to set-up 1,000 LNG stations across major highways, industrial corridors and mining sectors in the country. MoP&NG also came out with a draft scheme for promotion of LNG based mobility, wherein inter-alia, the scheme framework includes commissioning of LNG dispensing stations, LNG Mobile Refueling Units, allocation of domestic gas to this segment etc.

3. Has there been any country(ies) where Retail LNG for long haul transport has been successful? If yes, kindly provide details.

Yes, countries such as China, Italy, Spain, Netherlands etc. have adopted successful models for advancing LNG as a transport fuel. Government support through policy initiatives has been one of the first enablers of LNG as a transport fuel in these countries. Strong support from the Government has sent signals to the private players, thus giving them the confidence to enter in this sector, that has led to further development of the market in such countries.

4. What are the major challenges w.r.t. development of Retail LNG ecosystem in India?

Kindly brief about the possible resolution to such challenges.

Some of the major challenges are highlighted below:

- Availability of LNG vehicles: A crucial factor in the adoption of LNG as a transportation fuel. To bridge the gap between infrastructure development and vehicle scarcity, State Transport Undertakings (STUs) can play a significant role in ensuring the initial demand for LNG vehicles.
- Availability of Cheaper LNG : for transport sector, cheaper LNG compared to Diesel shall pave the way for faster adoption conversion of vehicles using LNG as a transport fuel.
- Availability of LNG retail outlets: The establishment of LNG refueling infrastructure is crucial for the successful adoption of LNG as a transportation fuel. Plans are in place for 49 LNG retail outlets along the Golden Quadrilateral by OGMCS. Further, private players have also starting developing LNG stations.
- Retro-fitment of LNG trucks: There is limited interest on financing of retro-fitment kits, as it can lead to multiple hypothecations of the truck, and thus, create legal issues. In addition, most HDVs stop plying on long distance routes and go to the secondary market once they are more than six years old. Thus, conversion/retro-fitment of trucks does not seem to be a long-term solution for this project.

Taxes: High taxes/duties on LNG vehicles have made it difficult for switching from Diesel to LNG. Government may consider reducing GST rates on LNG vehicles in-line with EVs.

5. What is GAIL's vision regarding Retail LNG? What are targets decided in this regard?

Considering the potential of LNG as a transport fuel market in the country, GAIL has set up a separate Business Unit to focus on developing the LNG infrastructure in the country. Plans are in place to set up 29 LNG stations throughout India at an estimated cost of Rs.650 crores.

6. Kindly provide the details of GAIL's Retail LNG plants/ filling stations which are operational as on date, under construction and under planning along with location, capacity etc.

5 locations have been finalized and tender award for Civil works and Equipment supply is under finalization stage.

7. What is GAIL's expected Retail LNG consumption over upcoming 5 years (till 2030)?

GAIL aspires to capture around 25% of the market share in coming 5 years.

GAIL Subsidiaries Data

1. Physicals and Financials of GGSPL for FY23-24 & 24-25 and estimates for FY25-26?

GAIL GLOBAL (SINGAPORE) PTE. LTD.				
PROFITABILITY				
(in USD)				
PARTICULARS	Budgeted	Actual	Actual	Actual
	2025-26	2024-25	2023-24	2022-23
SALES	1,150,395,000	1,079,223,449	754,710,503	1,727,919,535
PURCHASE	1,146,245,000	1,075,477,534	751,908,753	1,726,092,839
GROSS TRADING PROFIT	4,150,000	3,745,915	2,801,750	1,826,695
OTHER INCOME	282,630	482,986	361,042	201,268
TOTAL EXPENDITURE	2,454,000	1,728,288	1,611,934	1,587,712
PROFIT BEFORE TAX	1,978,630	2,500,614	1,550,858	440,252
INCOME TAX	336,000	381,510	248,532	49,593
PROFIT AFTER TAX (Before OCI)	1,642,630	2,119,105	1,302,325	390,659
PHYSICAL				
LNG Cargo Sales (No. of FOB/DES Cargoes)	28	27.08	20	27
TBTU Volume (DES & FOB)	92.40	88.22	65.85	93.48

2. Kindly provide BCPLs financials, Production and Sales for last 3 years and future outlook for next 3 years. What are the challenges being faced to operate the plant in northeast and what is the market for its product range? Kindly provide key highlights of recent agreement between GAIL & BCPL for marketing its products.

BCPL Statistics for last 3 years:

	2022-23	2023-24	2024-25
Revenue from Operation (Rs. Cr)	3,487	2,979	3,579 (Unaudited)
Polymer Production (MT)	2,52,021	2,46,066	3,05,875
Sale (MT)	2,45,953	2,48,860	2,96,196
PAT (Rs. Crores)	134	-160	15.64

Given the constrained projections for growth of Global Economy, capacity additions in Petrochemical Plants in the world/ India etc, the supply of Polymers (PE/PP), produced domestically as well as imports, is expected to be on the higher side vis-à-vis expected demand projections. Thus, the outlook for next 3 years is expected to remain competitive for all Polymer producers including BCPL, thereby putting pressure on margins.

Key Challenges

1. Feedstock Availability and Plant Utilization

- Persistent shortage of feedstock has hindered smooth operations.
- Resultant sub-optimal plant capacity utilization is affecting overall efficiency and profitability.

2. Increased Logistics Costs

- The non-renewal of the Freight Subsidy Scheme has significantly increased transportation costs.

- This has created a competitive disadvantage in both feedstock procurement and finished goods distribution, especially compared to competitor operating in mainland region.

3. Market Limitations

- Given the availability vs demand scenario projections, the Petrochemical Business Outlook is expected to remain extremely competitive in India.
- Limited demand for finished products within the North-East region restricts local sales and affects economies of scale.
- Export market prices in adjacent countries like Bangladesh remain unviable and lower than domestic, making it economically unfeasible to pursue aggressive export strategies. However, BCPL is exporting some quantities to Bhutan, which is fetching realization equivalent to Domestic market.

Key Highlights of GAIL–BCPL Marketing Agreement

The Marketing Agreement has been signed for Sale of BCPL Polymers for another 10 years with revised Marketing Margin rate of 1.45% of Net Sales Value.

3. GGUI financials and Physical Performance for CY 24, Q1 CY 25 and future outlook in view of volatility in crude price and impact of US tariff.

Performance GAIL Global USA Inc (GGUI):

	Physical Performance		
	CY 2023	CY 2024	Q1 CY2025
Gross Production (in Boe)	1,79,193	1,92,519	38,137
New wells drilled (in Nos)	3	4	0
	Financial Performance		
Turnover (\$ in Million)	10.65	11.60	2.24
PBT (\$ in Million)	(1.31)	(2.87)	(0.50)
Major activities during the year and future business plan	➤ A Promoters Audit by parent company, GAIL was conducted in March 2025. ➤ Ridgemar, one of the Operator of GGUI's assets, sold its interest to M/s Crescent Energy Company with effect from January 31, 2025. Consequently, now there are two operators Texas American Resources (TAR) and Crescent Energy Company (Crescent). ➤ In 107th Board meeting of GGUI held on March 31, 2025, GGUI Board recommendation for closure of sale/divestiture process for GGUI's Eagle Ford Shale assets and requested further guidance from GAIL. In response, GAIL vide email dated April 17, 2025 informed that, the GAIL Board, in its 478th meeting of Board of Directors held on April 9, 2025, has noted the information regarding cancellation of sale/divestiture process of GGUI's entire Eagle Ford shale assets.		

Performance GAIL Global USA LNG LLC (GGULL):

	Physical Performance		
	CY 2023	CY 2024	Q1 CY2025
LNG Sold (in TBTU)	124.49	127.58	34.66
LNG cargoes sold (Nos)	34	34	9
	Financial Performance		

Physical Performance			
	CY 2023	CY 2024	Q1 CY2025
Turnover (\$ in Million)	776.87	735.10	253.02
PBT(\$ Million)	0.16	0.17	0.05
Major activities during the year and future business plan	➤ Refinancing of Credit Agreement for Revolving Credit Facility of US\$75 Million for GGULL based on Corporate Guarantee from parent Company GAIL, has been executed with the existing Banker i.e. Bank of Baroda with effect from April 18, 2025. The facility carries an interest rate of 1-month Term SOFR plus 105 basis points per annum, representing a 20 basis point reduction from the earlier spread of 125 basis point under the previous agreement.		

Strategic Outlook and Implications:

Crude Price Volatility

Crude oil prices have experienced sharp fluctuations in recent years, driven by a volatile mix of factors:

- OPEC+ production decisions
- Geopolitical tensions (e.g., Russia-Ukraine, Middle East conflicts)
- Global economic uncertainty, especially inflationary trends in the U.S.
- Supply chain constraints (i.e. drill rigs, frac sand, steel casing, etc.)
- Rising interest rates impacting investment in energy infrastructure

WTI Crude, the U.S. benchmark, has swung between ~\$65 to \$90 per barrel over the last 12–18 months, making it difficult for shale producers to forecast cash flows or sustain capital expenditures confidently.

Impact of U.S. Tariffs

Recent U.S. tariff escalations, including 25% duties on Canadian and Mexican oil imports and xx% on certain Chinese products, have introduced further market unpredictability. These tariffs could increase costs for U.S. refiners and may suppress demand for imported heavier crude blends, thereby indirectly affecting regional asset valuations, including shale properties.

Why this Hurts U.S. Shale in Particular:

U.S. shale production, including Eagle Ford, is especially sensitive to price volatility because of its unique characteristics:

- a. High Capital Intensity & Rapid Decline Rates
 - Shale wells have high upfront drilling costs and decline rapidly, requiring continuous reinvestment just to maintain output.
 - Operators depend heavily on predictable, steady prices to plan drilling programs. Volatility leads to underinvestment or forced shut-ins.
- b. Price Thresholds for Profitability
 - Many shale plays become uneconomic below \$50 per barrel, depending on well productivity.
 - Smaller operators and non-integrated players like GGUI face tighter margins and less ability to hedge long-term risks.
- c. Investor Pressure

- U.S. shale firms face increasing investor pressure to prioritize capital discipline and free cash flow over production growth.
- This has led to a slowdown in drilling, even when prices rebound briefly, further contributing to output instability.

Future Direction for GAIL and GGUI:

The recent tender to fully divest the asset by February 2025 represents a strategic retreat from upstream exposure in the U.S.—a market now characterized more by uncertainty than opportunity, especially for foreign state-owned entities with less operational flexibility.

The current process has been discontinued. However, wait and watch for the opportune time for divestment!

4. KLL - Brief on Physical and Financial Performance during previous three years and projection for next two years.

Financial Year	2022-23 (Audited)	2023-24 (Audited)	2024-25 (Unaudited)	2025-26 (Projected)	2026-27 (Projected)
Number of Cargoes	19	17	21	34	39
Revenue from Operation (Rs. Cr)	400.86	406.87	478.24*	800.96	963.23
PBT (Rs. Cr)	-180.61	-327.17	-172.70	-145.14	-76.33
PAT (Rs. Cr)	-450.84	-246.57	-125.11	-303.87	-94.18

*It does not include sale of LNG of Rs 300.00 Crores

Energy Conversion Matrix

Frequently used General Conversions in the Gas Industry

1 SCM (Standard Cubic Meter)	= 1 cubic metre @ 1 atmosphere pressure and 15.56 ° C	
1 Cubic Metre	= 35.31 Cubic feet	
1 BCM(Billion Cubic Metre) / Year of gas (consumption or production)	= 2.74 MMSCMD	365 Days a Year
1 TCF (Trillion Cubic Feet) of Gas Reserve	= 3.88 MMSCMD	100% Recoverable for 20 years @ 365 days / Annum)
1 MMTPA of LNG	=3.60 MMSCMD	Mol.Weight of 18 @ 365 days/Annum)
1 MT of LNG	=1314 SCM	Mol.Weight of 18
Gross Calorific Value (GCV)	10000 Kcal/ SCM	
Net Calorific Value (NCV)	90% of GCV	
1 Million BTU (MMBTU)	= 25.2 SCM	@10000 Kcal/SCM; 1 MMBTU= 252,000 Kcal)
Specific Gravity of Gas	=0.62	Molecular Weight of Dry Air=28.964 gm/mole)
Density of Gas	=0.76 Kg/SCM	Mol.Weight of Gas 18 gm/mol
Gas required for 1 MW of Power generation	=4541 SCM per Day	Station Heat Rate (SHR); ~ 1720 Kcal/Kwh-NCV (50% Thermal Efficiency); N.Gas GCV- @10000 Kcal/SCM
Power Generation from 1 MMSCMD Gas	=220 MWH	Station Heat Rate (SHR); ~ 1720 Kcal/Kwh-NCV (50% Thermal Efficiency); N.Gas GCV- @10000 Kcal/SCM